Algorithmic Game Theory

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Lecture 1: Introduction and Examples

The course has three overarching goals, one for each part: design systems with strategic participants that have good performance guarantees; understand when selfish behavior is essentially benign; and understand how strategic players reach an equilibrium, if they do.

1.1 Mechanism design: the science of rule-making

The women's badminton tournament of the 2012 London Olympics shows what goes wrong when rules ignore incentives. Four groups of four teams each played a round-robin; the top two of each group advanced to a knockout tournament (quarterfinals, semifinals, a bronze match, and a final). The winner of group A met the second-best team of group C in the quarterfinals, and so on.

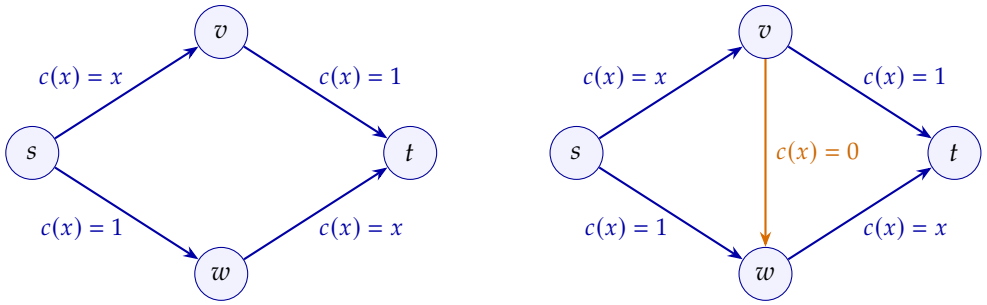
After an upset on the last day of round-robin play, the strong Chinese team QW finished only second in group D. Two teams in group A, both already qualified with a 2-0 record, then faced the following choice: the group winner would meet QW in the semifinals, where a loss means at best bronze, while the runner-up would avoid QW until the final, where silver is guaranteed. Both teams tried to lose their match, as did two teams in group C for the same reason, and all four were disqualified.

The point is that in systems with strategic participants *the rules matter*. The burden lies on the designer to anticipate strategic behavior, not on participants to act against their own interests. Mechanism design is the science of designing rules so that strategic behavior by the participants leads to a desirable outcome. Its major applications include Internet search auctions, wireless spectrum auctions, matching medical residents to hospitals and children to schools, and kidney exchange.

1.2 The price of anarchy and Braess's paradox

Often the rules of a game cannot be designed from scratch, and the task is to analyze a game that occurs "in the wild," such as a road network or the Internet.

In Braess's paradox (Figure 1), a fixed amount of traffic travels from a suburb s to a train station t. Each edge has a cost function $c(x)$ giving the travel time in hours when a fraction x of the traffic uses it. In the initial network there are two disjoint routes, each consisting of a long wide road with $c(x) = 1$ and a short narrow road with $c(x) = x$, so a route used by a fraction x of the traffic costs $1 + x$. The routes are identical, traffic splits evenly, and every driver travels for $1 + \frac{1}{2} = \frac{3}{2}$ hours.



(a) Initial network

(b) Augmented network

Figure 1: Braess's paradox. Adding the zero-cost edge (v, w) to the initial network (a) raises everyone's travel time from $\frac{3}{2}$ to 2 hours in the augmented network (b).

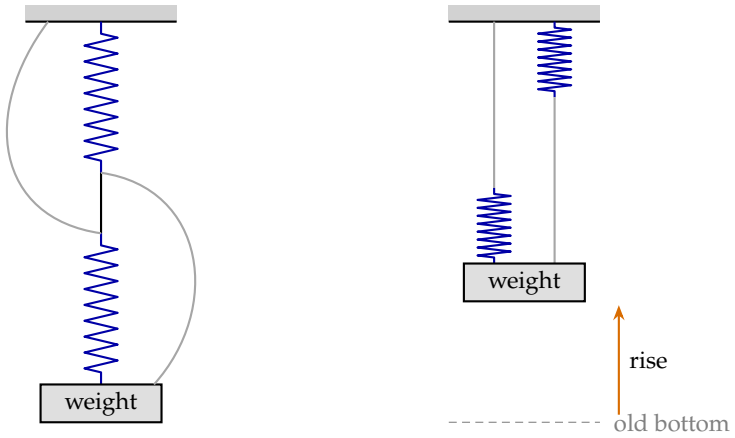
Now add a teleporter, an edge (v, w) with constant cost $c(x) = 0$. The route $s \to v \to w \to t$ costs $x_{sv} + x_{wt} \leq 2$, which is never worse than either original route (each costs $1 + x_{sv}$ or $1 + x_{wt}$, with $x_{sv}, x_{wt} \leq 1$), and strictly better whenever some traffic avoids it. So all drivers switch to the new route. Then (s, v) and (w, t) each carry all the traffic, and every driver now travels for 2 hours. Adding an intuitively helpful zero-cost link has made *all* of the traffic worse off.

Selfish routing therefore does not minimize commute time: in the augmented network, a dictator could route the traffic evenly over the two original paths (ignoring the teleporter) and improve everyone's commute by 25%.

Definition 1.1 [Price of anarchy]. The price of anarchy (POA) is the ratio between the system performance with strategic players and the best possible system performance.

For the augmented Braess network, the POA is $2/\frac{3}{2} = \frac{4}{3}$. The goal for this part of the course is to identify domains and conditions under which the POA is close to 1, so that selfish behavior leads to a near-optimal outcome. Applications include network routing, scheduling, resource allocation, and simple auction formats. For example, modest overprovisioning of network capacity guarantees that the POA of selfish routing is close to 1.

Strings and springs. Braess's paradox has a mechanical analogue (Figure 2). One end of a spring hangs from a fixed support and its other end holds a taut string; a second identical spring hangs from the string and carries a heavy weight. Two slack strings run from the support to the top of the second spring and from the bottom of the first spring to the weight. For ideally elastic springs, the stretched length is linear in the applied force, so the device is a traffic network with force as traffic and distance as cost. Initially the springs are in series, and each bears the full weight. Cutting the taut string leaves them in parallel through the formerly slack strings, so each bears half the weight and stretches half as far, and the weight *rises*. The rise corresponds exactly to the improvement obtained by deleting the zero-cost edge of Figure 1(b).



(a) Before (b) After

Figure 2: Schematic of the strings-and-springs device. (a) The springs are in series through the taut (black) string; the two gray strings are slack. (b) After the taut string is cut, the springs act in parallel and the weight rises. Lengths are not to scale.

1.3 Complexity of equilibria

Some games are easy to play: in the augmented Braess network, using the teleporter is the best route no matter what other drivers do. In most games, the best action depends on what the others do. Informally, an EQUILIBRIUM is a steady state in which each participant, assuming everything else stays the same, wants to remain as it is.

A canonical example is Rock-Paper-Scissors in BIMATRIX form: the row player chooses a row, the column player a column, and the entry lists the payoffs of the row and column player, respectively.

	Rock	Paper	Scissors
Rock	$0, 0$	$-1, 1$	$1, -1$
Paper	$1, -1$	$0, 0$	$-1, 1$
Scissors	$-1, 1$	$1, -1$	$0, 0$

There is no deterministic equilibrium: in every outcome, at least one player benefits from a unilateral move. The key idea, developed largely by von Neumann, is to allow MIXED (randomized) strategies. If both players randomize uniformly, every unilateral deviation yields the deviator expected payoff zero, so neither can gain. A pair of probability distributions with this property is a MIXED-STRATEGY NASH EQUILIBRIUM.

Theorem 1.2 [NASH 1951**].** *Every bimatrix game has a Nash equilibrium. More generally, so does every game with a finite number of players.*

Two computational facts bear on whether equilibria are credible predictions.

- In a ZERO-SUM bimatrix game, where the payoffs in each entry sum to zero (as in Rock-Paper-Scissors), a Nash equilibrium can be computed in polynomial time, by linear programming or, up to a small error, by simple iterative learning algorithms. This supports the Nash equilibrium as a prediction in zero-sum games.
- Under suitable complexity assumptions, there is no polynomial-time algorithm for computing a Nash equilibrium of a general (non-zero-sum) game. The problem is not NP-hard unless NP = coNP, but it is PPAD-hard.

Technically, this makes computing a Nash equilibrium a fundamental problem of intermediate difficulty, like factoring and graph isomorphism. Conceptually, if all parties are boundedly rational, an equilibrium is a credible prediction only if it can be computed with reasonable effort. Intractability thus casts doubt on the predictive power of the Nash equilibrium and motivates easier equilibrium concepts, such as correlated and coarse correlated equilibria.

1.4 What computer science brings

The traditional economic approach to these topics tends to emphasize Bayesian (average-case) analysis and exact solutions and characterizations, and usually ignores computation. Computer science adds a language for computational complexity, approximation bounds for models in which exact solutions are unrealistic or unknowable, and more robust performance guarantees.

Source. Tim Roughgarden, Algorithmic Game Theory, Lecture 1: Introduction and Examples.

Lecture 2: Mechanism Design Basics

2.1 Single-item auctions and quasilinear utility

A seller has one good and there are n strategic bidders. Each bidder i has a valuation v_i, its maximum willingness-to-pay for the item. The valuation is private: unknown to the seller and to the other bidders. In the quasilinear utility model, a bidder who loses has utility 0, and a bidder who wins at price p has utility $v_i - p$.

A sealed-bid auction proceeds in three steps:

1. each bidder i privately communicates a bid b_i to the auctioneer;
2. the auctioneer decides who gets the good, if anyone;
3. the auctioneer decides on a selling price.

In this lecture step 2 always gives the good to the highest bidder. The choice of price in step 3 strongly affects behavior. Charging the winner nothing turns the auction into a contest to name the highest number.

In a first-price auction, the winner pays its bid. Such auctions are common but hard to reason about: a bidder does not know how to bid, and the designer cannot easily predict what will happen. Bidding one's valuation guarantees zero utility, and the right amount to underbid depends on the other bids.

2.2 Second-price auctions

On eBay, a proxy raises a bidder's bid on its behalf until it is the highest bidder or its maximum is reached, so a winner pays the highest other bid plus a small increment. This motivates the following format.

Definition 2.1 [Second-price auction]. A second-price or Vickrey auction is a sealed-bid auction in which the highest bidder wins and pays the second-highest bid.

Write $\mathbf{b}$ for the bid vector and $\mathbf{b}_{-i}$ for the vector of all bids except bidder i's.

Proposition 2.2 [Truthful bidding is dominant]. *In a second-price auction, every bidder i has a dominant strategy: bid $b_i = v_i$. That is, this bid maximizes i's utility no matter what the other bidders do.*

Proof. Fix i, v_i, and $\mathbf{b}_{-i}$, and let $B = \max_{j \neq i} b_j$. Although i can submit infinitely many bids, only two outcomes are possible: if $b_i < B$, then i loses and gets utility 0; if $b_i \geq B$, then i wins at price B and gets $v_i - B$. (Ties are broken in favor of i here; the claim holds for any tie-breaking rule.) So i's utility is at most $\max\{0, v_i - B\}$. If $v_i < B$, this maximum is 0, achieved by bidding truthfully and losing. If $v_i \geq B$, it is $v_i - B$, achieved by bidding truthfully and winning. □

Proposition 2.3 [Nonnegative utility]. *In a second-price auction, every truthful bidder gets nonnegative utility.*

Proof. Losers get 0. The winner i is the highest bidder and bid v_i, so the price p, the second-highest bid, satisfies $p \leq v_i$, and $v_i - p \geq 0$. □

Thus a bidder need not reason about the others (how many there are, their valuations, or whether they bid truthfully) to decide how to bid. Truthful bidding is in fact the unique dominant strategy.

2.3 Awesome auctions

Theorem 2.4 [Vickrey]. *The Vickrey auction is awesome, meaning it has three properties.*

1. Strong incentive guarantees. *It is dominant-strategy incentive-compatible (DSIC): truthful bidding is a dominant strategy and never leads to negative utility (Propositions 2.2 and 2.3).*
2. Strong performance guarantees. *If bidders bid truthfully, it maximizes the social surplus $\sum_{i=1}^{n} v_i x_i$, where $x_i = 1$ if i wins and $x_i = 0$ otherwise, subject to the feasibility constraint $\sum_{i=1}^{n} x_i \leq 1$.*
3. Computational efficiency. *It runs in polynomial, indeed linear, time.*

The selling price does not appear in the surplus: the auctioneer is a player whose utility is its revenue, which cancels the price paid by the winner.

Each property matters. DSIC makes bidding easy, and it lets the designer predict the outcome assuming only that a bidder with an obvious dominant strategy plays it, which is about as weak as a behavioral assumption gets. DSIC alone is not enough: giving the item away free to a random bidder is DSIC but ignores who wants it. Surplus maximization says that, although the valuations are unknown a priori, the auction identifies the bidder with the highest valuation, solving the optimization problem as well as if the data were known. Polynomial running time is needed for an auction to be usable, sometimes in real time.

2.4 Sponsored search auctions

A search results page shows organic results and sponsored links paid for by advertisers. On each query, an auction decides which links are shown, in what order, and what they are charged. Around

2006, these auctions generated roughly 98% of Google's revenue.

Model. The goods are k SLOTS for sponsored links, and the bidders are the advertisers with standing bids on the searched keyword. Unlike a single-item auction, there are several goods and they are not identical. Slot j has CLICK-THROUGH RATE (CTR) α_j, the probability that the user clicks on it, with

$$\alpha_1 \geq \alpha_2 \geq \cdots \geq \alpha_k,$$

and, as a simplification, the CTR of a slot does not depend on its occupant. Advertiser i has a private valuation v_i per click, so its value for slot j is $v_i \alpha_j$.

Goals. An awesome sponsored search auction should be

1. DSIC;
2. surplus-maximizing: the assignment should maximize $\sum_{i=1}^{n} v_i x_i$, where x_i is the CTR of i's slot (0 if i gets no slot), each slot gets at most one bidder and each bidder at most one slot;
3. computable in polynomial time.

Two-step design approach. A mechanism must decide jointly who wins what and who pays what; even for a single item, the correct winner is not enough if the payments let bidders game the system. In many applications the two decisions can be made one at a time. **Step 1**: assume, without justification, that bids are truthful, and choose the allocation so that properties 2 and 3 hold. **Step 2**: given Step 1, choose payments so that property 1 holds. If both succeed, the auction is awesome: Step 2 makes truthful bidding dominant, which justifies the assumption of Step 1, so the outcome maximizes surplus and is computed in polynomial time.

For sponsored search, Step 1 is solved by the greedy rule: assign the jth highest bidder to the jth slot for $j = 1, \ldots, k$. It runs in $O(n \log n)$ time by sorting. It is optimal: treat an unassigned bidder as occupying a slot with CTR 0. If a bidder with a higher value sits in a lower slot than one with a lower value, swapping them changes the surplus by $(v_i - v_{i'})(\alpha_j - \alpha_{j'}) \geq 0$, where $v_i \geq v_{i'}$ and $\alpha_j \geq \alpha_{j'}$; repeated swaps reach the sorted assignment without decreasing surplus. Whether Step 2 can be carried out, that is, whether there is an analogue of the second-price rule here, is answered by Myerson's lemma in Lecture 3.

Source. Tim Roughgarden, Algorithmic Game Theory, Lecture 2: Mechanism Design Basics.

Lecture 3: Myerson's Lemma

Lecture 2 proposed a two-step design approach: first choose an allocation rule that maximizes surplus in polynomial time assuming truthful bids, then choose payments that make the auction DSIC. For sponsored search, Step 1 is the greedy assignment. Myerson's lemma is a general tool for Step 2.

3.1 Single-parameter environments

Definition 3.1 [Single-parameter environment]. A single-parameter environment has n bidders, where bidder i has a private valuation v_i, its value per unit of "stuff," and a feasible set X of vectors $(x_1, \ldots, x_n)$, where x_i is the amount of stuff given to bidder i.

- Single-item auction: X is the set of 0-1 vectors with $\sum_{i=1}^{n} x_i \leq 1$.
- k identical goods, at most one per bidder: 0-1 vectors with $\sum_{i=1}^{n} x_i \leq k$.
- Sponsored search: X corresponds to assignments of bidders to slots, each slot to at most one bidder and each bidder to at most one slot; if i gets slot j, then $x_i = \alpha_j$, the CTR of slot j.

A sealed-bid auction collects bids $\mathbf{b} = (b_1, \ldots, b_n)$, chooses a feasible allocation $\mathbf{x}(\mathbf{b}) \in X \subseteq \mathbb{R}^n$ by an allocation rule, and chooses payments $\mathbf{p}(\mathbf{b}) \in \mathbb{R}^n$ by a payment rule. With quasilinear utilities, bidder i's utility on bid profile $\mathbf{b}$ is

$$u_i(\mathbf{b}) = v_i \cdot x_i(\mathbf{b}) - p_i(\mathbf{b}).$$

Throughout, payments satisfy

$$p_i(\mathbf{b}) \in [0,\ b_i \cdot x_i(\mathbf{b})] \qquad \text{for every } i \text{ and } \mathbf{b}. \tag{3.1}$$

The lower bound forbids the seller from paying bidders. The upper bound guarantees a truthful bidder nonnegative utility, since with $b_i = v_i$ it gives $u_i \geq v_i x_i - v_i x_i = 0$.

3.2 Statement of Myerson's lemma

Definition 3.2 [Implementable allocation rule]. An allocation rule $\mathbf{x}$ for a single-parameter environment is implementable if there is a payment rule $\mathbf{p}$ such that the sealed-bid auction $(\mathbf{x}, \mathbf{p})$ is DSIC.

The implementable rules are exactly the allocation rules of DSIC mechanisms, so they form the

design space. For a single item, awarding the good to the highest bidder is implementable via the second-price rule. Whether awarding it to the second-highest bidder is implementable is not clear directly from the definition.

Definition 3.3 [Monotone allocation rule]. An allocation rule $\mathbf{x}$ is monotone if for every bidder i and bids $\mathbf{b}_{-i}$ of the others, the allocation $x_i(z, \mathbf{b}_{-i})$ is nondecreasing in i's bid z.

Bidding higher can only get a bidder more stuff. Awarding a single item to the highest bidder is monotone: a winner who raises its bid still wins. Awarding it to the second-highest bidder is not: a winner who raises its bid enough loses. The greedy sponsored search rule is monotone, since raising a bid can only move the bidder up in the sorted order, to a slot with a higher CTR.

Theorem 3.4 [Myerson's lemma]. *Fix a single-parameter environment.*

(a) An allocation rule $\mathbf{x}$ is implementable if and only if it is monotone.

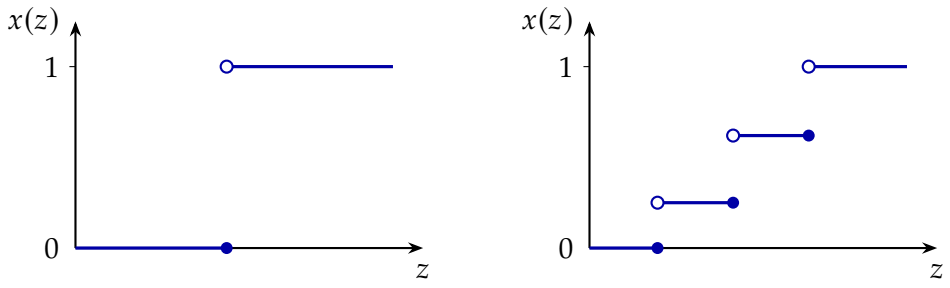
(b) If $\mathbf{x}$ is monotone, there is a unique payment rule $\mathbf{p}$ such that $(\mathbf{x}, \mathbf{p})$ is DSIC, assuming the normalization that $b_i = 0$ implies $p_i(\mathbf{b}) = 0$.

(c) This payment rule is given by the explicit formula (3.3) below (or (3.4) in general).

Part (a) is powerful because implementability describes the design goal but is unwieldy to verify, while monotonicity is usually easy to check. The normalization in (b) follows from (3.1).

3.3 Proof of Myerson's lemma

Let $\mathbf{x}$ be any allocation rule and suppose $(\mathbf{x}, \mathbf{p})$ is DSIC. Fix i and $\mathbf{b}_{-i}$, and write $x(z)$ and $p(z)$ for $x_i(z, \mathbf{b}_{-i})$ and $p_i(z, \mathbf{b}_{-i})$. Figure 3 shows two possible allocation curves.



(a) 0-1 monotone curve (b) Piecewise constant monotone curve

Figure 3: Two monotone allocation curves $x(\cdot)$: (a) a 0-1 curve with one jump; (b) a piecewise constant curve with several jumps. At each breakpoint, the filled dot marks the value of x.

The swapping trick. Take any bids $y, z \geq 0$. A bidder with valuation z could bid y, and one with valuation y could bid z, so DSIC requires

$$\underbrace{z \cdot x(z) - p(z)}_{\text{utility of bidding } z} \geq \underbrace{z \cdot x(y) - p(y)}_{\text{utility of bidding } y} \qquad \text{and} \qquad \underbrace{y \cdot x(y) - p(y)}_{\text{utility of bidding } y} \geq \underbrace{y \cdot x(z) - p(z)}_{\text{utility of bidding } z}.$$

Rearranging gives the PAYMENT DIFFERENCE SANDWICH

$$z \cdot [x(y) - x(z)] \;\leq\; p(y) - p(z) \;\leq\; y \cdot [x(y) - x(z)]. \tag{3.2}$$

Necessity of monotonicity. Comparing the outer terms of (3.2) gives $(y - z)[x(y) - x(z)] \geq 0$, so $y > z$ implies $x(y) \geq x(z)$. Thus every implementable rule is monotone. From now on, assume $\mathbf{x}$ is monotone.

Uniqueness of payments. Suppose first that x is piecewise constant: flat except for finitely many jumps. Fix z and let $y \downarrow z$ in (3.2). If x has no jump at z, both bounds tend to 0. If x jumps by h at z, both tend to $z \cdot h$. Hence

$$\text{jump in } p \text{ at } z = z \cdot \text{jump in } x \text{ at } z,$$

and p is constant where x is. With $p(0) = 0$, this forces the PAYMENT FORMULA

$$p_i(b_i, \mathbf{b}_{-i}) = \sum_{j=1}^{\ell} z_j \cdot \big(\text{jump in } x_i(\cdot, \mathbf{b}_{-i}) \text{ at } z_j\big), \tag{3.3}$$

where $z_1, \ldots, z_\ell$ are the breakpoints of $x_i(\cdot, \mathbf{b}_{-i})$ in $[0, b_i]$ whose jump is already included in $x_i(b_i, \mathbf{b}_{-i})$. If instead x is differentiable, dividing (3.2) by $y - z$ and letting $y \downarrow z$ gives $p'(z) = z \cdot x'(z)$, and with $p(0) = 0$,

$$p_i(b_i, \mathbf{b}_{-i}) = \int_0^{b_i} z \cdot \frac{d}{dz} x_i(z, \mathbf{b}_{-i})\, dz. \tag{3.4}$$

So for every allocation rule there is at most one DSIC payment rule, which proves the uniqueness in (b). It remains to show that this rule works when $\mathbf{x}$ is monotone.

In both cases, integrating by parts (or summing the rectangles to the left of the curve) gives the equivalent form

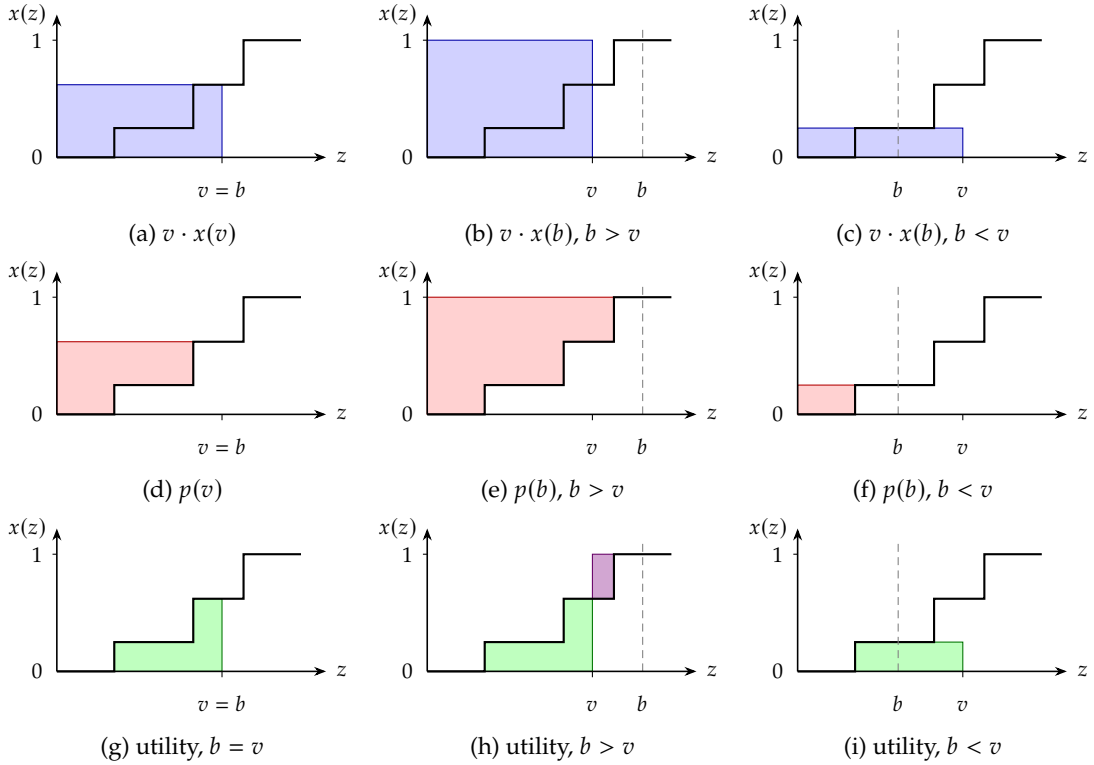
$$p(b) = b \cdot x(b) - \int_0^b x(z)\, dz, \tag{3.5}$$

which is the area to the left of the allocation curve, between heights 0 and $x(b)$. Since $0 \leq \int_0^b x(z)\, dz \leq b \cdot x(b)$ for monotone $x \geq 0$, these payments satisfy (3.1).

Sufficiency. Let the bidder have valuation v and bid b. Its utility $v \cdot x(b) - p(b)$ is the rectangle of width v and height $x(b)$ minus the payment area (Figure 4). By (3.5),

$$u(v) - u(b) = \begin{cases} (b - v)\, x(b) - \displaystyle\int_v^b x(z)\, dz \;\geq 0, & b > v, \\ \displaystyle\int_b^v x(z)\, dz - (v - b)\, x(b) \;\geq 0, & b < v, \end{cases}$$

because $x(z) \leq x(b)$ on $[v, b]$ in the first case and $x(z) \geq x(b)$ on $[b, v]$ in the second, by monotonicity. In pictures: a truthful bidder's utility is the area under the allocation curve on $[0, v]$; overbidding subtracts the area above the curve on $[v, b]$ up to height $x(b)$; and underbidding yields only part of the area under the curve on $[0, v]$. So truthful bidding is a dominant strategy, $(\mathbf{x}, \mathbf{p})$ is DSIC, and all three parts are proved.



(a) $v \cdot x(v)$ (b) $v \cdot x(b)$, $b > v$ (c) $v \cdot x(b)$, $b < v$

(d) $p(v)$ (e) $p(b)$, $b > v$ (f) $p(b)$, $b < v$

(g) utility, $b = v$ (h) utility, $b > v$ (i) utility, $b < v$

Figure 4: Proof by picture for a monotone piecewise constant allocation curve. Columns: truthful bidding ($b = v$), overbidding ($b > v$), underbidding ($b < v$). Rows: the surplus $v \cdot x(b)$, the payment $p(b)$, and the utility $v \cdot x(b) - p(b)$. In (h), the purple region is negative utility.

For a truthful bidder, the surplus $v \cdot x(v)$ splits into its utility (the area under the curve on $[0, v]$) and the seller's revenue (the area above it, within the rectangle).

3.4 Applying the payment formula

Single item. For the highest-bidder rule, fix i and $\mathbf{b}_{-i}$ and let $B = \max_{j \neq i} b_j$. The curve $x_i(\cdot, \mathbf{b}_{-i})$ is 0 up to B and 1 after, so (3.3) gives payment 0 if $b_i < B$ and B if $b_i > B$. Myerson's lemma regenerates the Vickrey auction.

Sponsored search. Let $\mathbf{x}$ assign the jth highest bidder to slot j, for slots with CTRs $\alpha_1 \geq \cdots \geq \alpha_k$. It is surplus-maximizing and monotone, so by Myerson's lemma it has a unique DSIC payment rule. Re-index so that $b_1 \geq b_2 \geq \cdots \geq b_n$, and set $\alpha_{k+1} = 0$ and $b_j = 0$ for $j > n$. As bidder $i \leq k$ raises its bid z from 0 to b_i, others fixed, it enters slot j when z passes b_{j+1}, the jth highest of the other bids, for $j = k, k-1, \ldots, i$. Its allocation therefore jumps by $\alpha_j - \alpha_{j+1}$ at b_{j+1}, and (3.3) becomes

$$p_i(\mathbf{b}) = \sum_{j=i}^{k} b_{j+1} \, (\alpha_j - \alpha_{j+1}). \tag{3.6}$$

Bidders who get no slot pay nothing. Advertisers value clicks, not impressions, so it is natural to charge per click: dividing by α_i,

$$\frac{p_i(\mathbf{b})}{\alpha_i} = \sum_{j=i}^{k} b_{j+1} \, \frac{\alpha_j - \alpha_{j+1}}{\alpha_i}.$$

The weights are nonnegative and sum to 1, so a clicked advertiser pays a convex combination of the lower bids $b_{i+1}, \ldots, b_{k+1}$.

In practice, sponsored search uses the GENERALIZED SECOND PRICE (GSP) auction, whose per-click payment is simply the next lower bid b_{i+1}. Since the payments above are the unique DSIC ones, GSP is not DSIC.

Source. Tim Roughgarden, Algorithmic Game Theory, Lecture 3: Myerson's Lemma.

Lecture 4: Algorithmic Mechanism Design

4.1 Knapsack auctions

Definition 4.1 [Knapsack auction]. In a knapsack auction, each bidder i has a publicly known size w_i and a private valuation v_i, and the seller has a capacity W. The feasible set X consists of the 0-1 vectors $(x_1, \ldots, x_n)$ with $\sum_{i=1}^{n} w_i x_i \leq W$, where $x_i = 1$ means that i wins.

Examples are TV ads of different lengths in a commercial break, files on a shared server, data streams on a shared channel, or jobs on a shared supercomputer: any shared resource with limited capacity. The k-item auction is the special case $w_i = 1$ for all i and $W = k$.

This is a single-parameter environment, so the two-step approach applies. Assuming truthful bids, surplus maximization dictates the allocation rule

$$\mathbf{x}(\mathbf{b}) = \arg\max_{\mathbf{x} \in X} \sum_{i=1}^{n} b_i x_i, \tag{4.1}$$

which solves the Knapsack problem with item values $b_1, \ldots, b_n$, sizes $w_1, \ldots, w_n$, and capacity W. When bids are truthful, it maximizes the social surplus.

Critical bids. This rule is monotone: raising a winner's bid increases the value of every feasible set containing it and leaves the others unchanged, so the winning set stays optimal. Hence Myerson's lemma gives a DSIC payment rule. Fix i and $\mathbf{b}_{-i}$. Since the rule is monotone and 0-1, the curve $x_i(\cdot, \mathbf{b}_{-i})$ is 0 up to a single breakpoint z and 1 after it. By Myerson's payment formula, i pays 0 if it bids below z and $z \cdot (1 - 0) = z$ if it bids above. So a winner pays its critical bid, the lowest bid with which it would still win, holding $\mathbf{b}_{-i}$ fixed. The Vickrey auction is the special case of a single item.

Intractability. The mechanism is DSIC and surplus-maximizing, but it is not awesome: the Knapsack problem is NP-hard, so unless P = NP the rule (4.1) has no polynomial-time implementation. Of the three goals (DSIC, surplus maximization, polynomial time), the last two conflict, so relaxing DSIC does not help. One option is to relax polynomial time: (4.1) can be computed in pseudopolynomial time by dynamic programming, and whenever exact surplus maximization is affordable, the rule is monotone and extends to a DSIC mechanism. Computing the payments, however, generally requires solving n further surplus-maximization problems, one per bidder.

4.2 Algorithmic mechanism design

The dominant paradigm of ALGORITHMIC MECHANISM DESIGN relaxes surplus maximization as little as possible, subject to DSIC and polynomial time. In single-parameter environments, by Myerson's lemma this means: design a polynomial-time *monotone* allocation rule that comes as close as possible to maximizing surplus. This is the goal of approximation algorithms with one extra constraint, monotonicity, into which the entire game-theoretic aspect has been compiled.

The "holy grail" is to match, for as many NP-hard problems as possible, the best approximation guarantee of arbitrary polynomial-time algorithms, or even the best possible one assuming P $\neq$ NP, so that DSIC causes no surplus loss beyond what polynomial time already costs. With exact surplus maximization, monotonicity came for free; the question is whether this persists for approximate surplus maximization.

Greedy knapsack auctions. Assume without loss of generality that $w_i \leq W$ for every i, since bidders with $w_i > W$ can be deleted. The greedy rule $\mathbf{x}^G$ orders bidders by "bang per buck," the value per unit of capacity, favoring high bids and small sizes.

Algorithm 4.1: Greedy knapsack allocation rule $\mathbf{x}^G$

Input: Bids $b_1, \ldots, b_n$, sizes $w_1, \ldots, w_n \leq W$, capacity W.
Output: A feasible set S of winners.

Sort and re-index the bidders so that $b_1/w_1 \geq b_2/w_2 \geq \cdots \geq b_n/w_n$
$S_1 \leftarrow \emptyset$
for $i \leftarrow 1$ **to** n **do**
    if $\sum_{j \in S_1} w_j + w_i > W$ **then break**
    $S_1 \leftarrow S_1 \cup \{i\}$
$S_2 \leftarrow \{i^*\}$ for a highest bidder i^*
return whichever of S_1 and S_2 has the larger total bid

Theorem 4.2 [GREEDY IS A $\frac{1}{2}$-APPROXIMATION]. *Assuming truthful bids, the surplus of the greedy allocation rule is at least* 50% *of the maximum possible surplus.*

Proof. Let the bids be the true values $v_1, \ldots, v_n$. Relax the problem so that a bidder can be chosen fractionally, with its value pro-rated: choosing 70% of a bidder with value 10 contributes 7. In this FRACTIONAL KNAPSACK PROBLEM, the greedy algorithm that takes bidders in the sorted order until the capacity is full, taking the last one fractionally, is optimal by an exchange argument: if an optimal fractional solution uses less of some bidder i and more of a later bidder j, shifting capacity from j to i does not decrease the value, since $v_i/w_i \geq v_j/w_j$. If all bidders fit, S_1 is optimal. Otherwise, suppose the optimal fractional solution takes bidders $1, \ldots, k$ fully and bidder $k+1$ fractionally. Then $S_1 = \{1, \ldots, k\}$ has value $v_1 + \cdots + v_k$, and S_2 has

value at least v_{k+1}. Hence

$$2\max\{v(S_1), v(S_2)\} \;\geq\; \sum_{i=1}^{k+1} v_i \;\geq\; \text{fractional optimum} \;\geq\; \text{integral optimum.} \qquad \square$$

The guarantee improves under additional assumptions. If $w_i \leq \alpha W$ for every i, with $\alpha \in (0, \frac{1}{2}]$, then S_1 alone has surplus at least $(1-\alpha)$ times the optimum. Indeed, if S_1 stops at bidder $k+1$, then its size is $w(S_1) > W - w_{k+1} \geq (1-\alpha)W$, and since every bidder in S_1 has density at least v_{k+1}/w_{k+1}, the fractional optimum is at most

$$v(S_1) + \frac{v_{k+1}}{w_{k+1}}\big(W - w(S_1)\big) \leq v(S_1) + \frac{v(S_1)}{w(S_1)}\big(W - w(S_1)\big) = v(S_1)\frac{W}{w(S_1)} < \frac{v(S_1)}{1-\alpha}.$$

Exact surplus maximization is monotone, and so is the greedy rule. If i wins through S_1, raising its bid moves it earlier in the order, and the set of bidders preceding the stopping point does not change, so S_1 is the same set with a larger value. The set S_2 changes only if i becomes the highest bidder, in which case $S_2 = \{i\}$. If i wins as the highest bidder, raising its bid keeps it the highest bidder and increases the value of S_2, while S_1 either still excludes i with unchanged value or now contains i. Either way i still wins. By Myerson's lemma, the greedy rule extends to a DSIC mechanism in which winners pay their critical bids, and this mechanism is a $\frac{1}{2}$-approximation.

Natural allocation rules are not always monotone, however. For every $\epsilon > 0$, the Knapsack problem has a $(1-\epsilon)$-approximation algorithm running in time polynomial in the input and $1/\epsilon$ (an FPTAS). The rule induced by its standard implementation is not monotone, though it can be modified to restore monotonicity without degrading the guarantee. This is typical of the field: take an NP-hard optimization problem, check whether the state-of-the-art approximation algorithm is monotone and, if not, tweak it or design a new one that is, ideally with the same guarantee.

Black-box reductions. State-of-the-art approximation algorithms for single-parameter problems are usually monotone or can be redesigned to be. This raises the question of whether some natural single-parameter problem has a strictly better polynomial-time approximation guarantee than any polynomial-time *monotone* algorithm. A negative answer could come from a BLACK-BOX REDUCTION, a generic way of turning any polynomial-time algorithm into a monotone one with the same guarantee, or even one worse by a constant factor. Chawla, Immorlica, and Lucier showed that no fully general reduction of this type exists for single-parameter environments. Whether one exists for large subclasses, such as DOWNWARD-CLOSED environments, where giving a bidder less stuff never makes an outcome infeasible, is open.

4.3 The revelation principle

DSIC is attractive because a participant just plays its obvious dominant strategy, and the designer can predict the outcome under that weak assumption. Yet non-DSIC mechanisms, such as first-price auctions, are also useful in practice. The DSIC definition combines two separate requirements:

1. every participant has a dominant strategy, whatever its private valuation;
2. this dominant strategy is DIRECT REVELATION: truthfully reporting all of its private information to the mechanism.

Requirement 1 can hold without 2. If a single-item auction runs a Vickrey auction on the doubled bids $2\mathbf{b}$, every bidder's dominant strategy is to bid half its value.

Relaxing requirement 1 needs stronger assumptions to predict behavior, such as a Bayes-Nash equilibrium with respect to a common prior, or a Nash equilibrium in a full-information model. Sometimes this lets non-DSIC mechanisms do better. As a rough rule, for simple problems like those so far, DSIC mechanisms can do anything non-DSIC ones can, while in more complex problems weaker incentive constraints, such as implementation in Bayes-Nash equilibrium, can achieve things provably impossible for DSIC mechanisms. Then the two are incomparable: one has stronger incentive guarantees, the other better performance guarantees. Relaxing requirement 2, by contrast, gains nothing.

Theorem 4.3 [REVELATION PRINCIPLE]. *For every mechanism M in which every participant has a dominant strategy, whatever its private information, there is an equivalent direct-revelation DSIC mechanism M'.*

Proof. For each participant i and private information v_i, let $s_i(v_i)$ be a dominant strategy of i in M. The mechanism M' plays these strategies on the participants' behalf (Figure 5): it accepts sealed bids $b_1, \ldots, b_n$, submits $s_1(b_1), \ldots, s_n(b_n)$ to M, and chooses the same outcome as M. When everyone reports truthfully, M' produces the outcome of M under its dominant strategies. If participant i has private information v_i and reports something else, M' plays a strategy other than $s_i(v_i)$ for i in M, which cannot increase i's utility, whatever the others report. So truthful reporting is dominant in M'. □

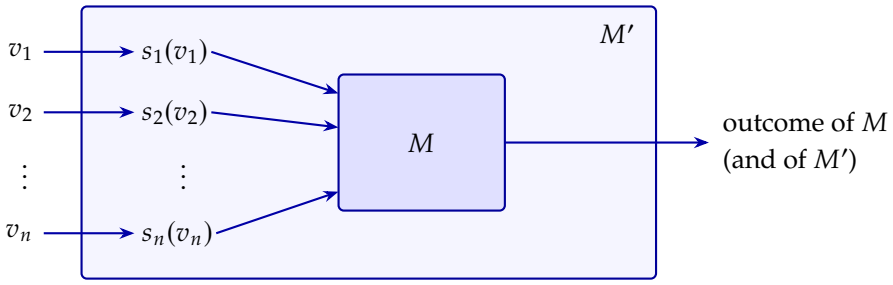


Figure 5: The direct-revelation mechanism M' simulates the dominant strategies $s_1, \ldots, s_n$ of the given mechanism M and outputs the same outcome.

So a designer who wants dominant strategies might as well make direct revelation (in auctions, truthful bidding) the dominant strategy. Other equilibrium concepts, such as Bayes-Nash equilibrium, have their own revelation principles: once the incentive constraints are chosen, direct revelation is without loss of generality. Truthfulness per se is not important. What makes mechanism design hard is requiring a desired outcome, without loss of generality under truthful reporting, in an equilibrium of some type. Stronger equilibrium concepts, like dominant strategies, need weaker behavioral assumptions but have narrower reach than weaker ones, like Bayes-Nash equilibrium.

Source. Tim Roughgarden, Algorithmic Game Theory, Lecture 4: Algorithmic Mechanism Design.

Lecture 5: Revenue-Maximizing Auctions

So far the goal has been the welfare $\sum_{i=1}^{n} v_i x_i$ over feasible outcomes $(x_1, \ldots, x_n) \in X$, with revenue only a side effect of incentivizing truthful reports. Welfare is special: in every single-parameter environment some DSIC mechanism maximizes welfare ex post, as well as if the designer knew every v_i in advance. This lecture designs auctions that maximize revenue, where no such input-independent guarantee is possible.

5.1 The challenge of revenue maximization

With one item and one bidder of private valuation v, the deterministic DSIC auctions are exactly the posted prices, or take-it-or-leave-it offers. A price r earns r if $v \geq r$ and 0 if $v < r$. Welfare is maximized by $r = 0$ whatever v is. For revenue, the ex post best price is $r = v$, which is unavailable since v is private. A posted price of 20 does very well when v is 20 or a little more and terribly when v is smaller. For revenue, different auctions do better on different inputs, so comparing them requires a model of the inputs.

5.2 The Bayesian model

The classical model is average-case, or Bayesian, analysis.

- A single-parameter environment (Lecture 3).
- The private valuation v_i of participant i is drawn from a distribution F_i with density f_i and support contained in $[0, v_{\max}]$, where $F_i(z)$ is the probability of a value at most z. The distributions $F_1, \ldots, F_n$ are independent but not necessarily identical. In practice they come from data, such as past bids.
- The distributions are known to the designer in advance; the realizations $v_1, \ldots, v_n$ are private. Since the auctions considered are DSIC, bidders have dominant strategies and need not know the distributions.

The optimal auction is the DSIC auction with the highest expected revenue, where the expectation is over $\mathbf{v} \sim F_1 \times \cdots \times F_n$ and bidding is truthful.

With one bidder, a posted price r has expected revenue $r \cdot (1 - F(r))$: the revenue of a sale times its probability. The maximizer is the monopoly price of F, and posting it is the optimal auction, since DSIC auctions here are posted prices and randomizations of them. For F uniform on $[0, 1]$, the monopoly price is $\frac{1}{2}$, with expected revenue $\frac{1}{4}$ (Figure 6).

With two bidders, the space of DSIC auctions is richer. For valuations i.i.d. uniform on $[0,1]$, the Vickrey auction earns the expected smaller value, $\frac{1}{3}$. The Vickrey auction with reserve price r awards the item to the highest bidder unless all bids are below r, in which case no one gets it, and charges the winner the larger of the second-highest bid and r. A reserve loses revenue when all bids are below r and gains it when exactly one bid is above r. Here the reserve $\frac{1}{2}$ is a net gain, raising expected revenue from $\frac{1}{3}$ to $\frac{5}{12}$. The rest of the lecture determines the optimal auction in general (Myerson, 1981).

5.3 Expected revenue equals expected virtual welfare

By the Revelation Principle (Lecture 4), every DSIC auction is equivalent to a direct-revelation DSIC mechanism $(\mathbf{x}, \mathbf{p})$ with the same expected revenue, so assume this and truthful bids $\mathbf{b} = \mathbf{v}$ from now on. The expected revenue is $\mathbb{E}_{\mathbf{v}}\left[\sum_{i=1}^{n} p_i(\mathbf{v})\right]$. This is hard to maximize directly, so we rewrite it in terms of the allocation rule alone, using Myerson's payment formula (Lecture 3)

$$p_i(b_i, \mathbf{b}_{-i}) = \int_0^{b_i} z \cdot x_i'(z, \mathbf{b}_{-i})\, dz. \tag{5.1}$$

This was derived for differentiable x_i, but it holds for every monotone $x_i(z, \mathbf{b}_{-i})$, including piecewise constant ones, with a suitable interpretation of the derivative and integral. The calculus below (swapping integrals, integration by parts) is likewise rigorous for bounded monotone functions.

Definition 5.1 [Virtual valuation]. The virtual valuation of bidder i with valuation v_i drawn from F_i is

$$\varphi_i(v_i) = v_i - \frac{1 - F_i(v_i)}{f_i(v_i)}.$$

It depends only on the bidder's own valuation and distribution. For F_i uniform on $[0,1]$, $F_i(z) = z$ and $f_i(z) = 1$, so $\varphi_i(z) = z - (1 - z) = 2z - 1$. A virtual valuation can be negative.

Lemma 5.2 [Expected payment as virtual value]. *For every bidder i and every $\mathbf{v}_{-i}$,*

$$\mathbb{E}_{v_i \sim F_i}[p_i(\mathbf{v})] = \mathbb{E}_{v_i \sim F_i}[\varphi_i(v_i) \cdot x_i(\mathbf{v})].$$

Proof. Fix $\mathbf{v}_{-i}$. By independence, the distribution of v_i is still F_i, so by (5.1)

$$\mathbb{E}_{v_i \sim F_i}[p_i(\mathbf{v})] = \int_0^{v_{\max}} \left[\int_0^{v_i} z \cdot x_i'(z, \mathbf{v}_{-i})\, dz \right] f_i(v_i)\, dv_i.$$

Reversing the order of integration, the inner integral over $v_i \in [z, v_{\max}]$ is $1 - F_i(z)$, giving

$$\int_0^{v_{\max}} (1 - F_i(z))\, z \cdot x_i'(z, \mathbf{v}_{-i})\, dz.$$

Integrate by parts with $(1 - F_i(z))\, z$ and $x_i(z, \mathbf{v}_{-i})$. The boundary term $(1 - F_i(z))\, z\, x_i(z, \mathbf{v}_{-i})\big|_0^{v_{\max}}$ is $0 - 0$, since the factor z vanishes at 0 and $F_i(v_{\max}) = 1$. The derivative of $(1 - F_i(z))z$ is $1 - F_i(z) - z f_i(z)$, so the integral becomes

$$\int_0^{v_{\max}} x_i(z, \mathbf{v}_{-i})\big(z f_i(z) - (1 - F_i(z))\big) dz = \int_0^{v_{\max}} \Big(z - \frac{1 - F_i(z)}{f_i(z)}\Big) x_i(z, \mathbf{v}_{-i}) f_i(z)\, dz,$$

which is $\mathbb{E}_{v_i \sim F_i}[\varphi_i(v_i)\, x_i(\mathbf{v})]$. □

Note. One coarse reading is that v_i is what you would like to charge i and $(1 - F_i(v_i))/f_i(v_i)$ is the INFORMATION RENT, the revenue lost by not knowing v_i in advance. More accurately, $\varphi_i(v_i)$ is the slope at v_i of the REVENUE CURVE, which plots the expected revenue from a bidder with valuation drawn from F_i against the probability of a sale.

Theorem 5.3 [EXPECTED REVENUE EQUALS EXPECTED VIRTUAL WELFARE]. *For every DSIC auction* $(\mathbf{x}, \mathbf{p})$ *in a single-parameter environment with independent valuations,*

$$\mathbb{E}_{\mathbf{v}}\Big[\sum_{i=1}^n p_i(\mathbf{v})\Big] = \mathbb{E}_{\mathbf{v}}\Big[\sum_{i=1}^n \varphi_i(v_i) \cdot x_i(\mathbf{v})\Big]. \tag{5.2}$$

Proof. Take the expectation over $\mathbf{v}_{-i}$ of both sides of Lemma 5.2 to get $\mathbb{E}_{\mathbf{v}}[p_i(\mathbf{v})] = \mathbb{E}_{\mathbf{v}}[\varphi_i(v_i) x_i(\mathbf{v})]$, and sum over i using linearity of expectation. □

The quantity $\sum_i \varphi_i(v_i) x_i(\mathbf{v})$ is the VIRTUAL WELFARE of the auction on $\mathbf{v}$; without the φ_i it would be the welfare. Although only payments matter, maximizing expected revenue over DSIC auctions reduces to maximizing expected virtual welfare, which involves only the allocation rule.

5.4 Bayesian optimal auctions

Since $\mathbf{x}(\mathbf{v})$ can be chosen separately for each input, expected virtual welfare is maximized pointwise: for each $\mathbf{v}$, choose the feasible $\mathbf{x}(\mathbf{v}) \in X$ maximizing $\sum_i \varphi_i(v_i) x_i(\mathbf{v})$. This VIRTUAL WELFARE-MAXIMIZING ALLOCATION RULE maximizes expected virtual welfare over all allocation rules, monotone or not. If it is monotone, Myerson's Lemma extends it to a DSIC auction, which by (5.2) is optimal.

In a single-item auction, where feasibility means $\sum_i x_i(\mathbf{v}) \le 1$, the rule awards the item to the bidder with the highest virtual valuation, unless every virtual valuation is negative, in which case no one gets it (as in the one-bidder case, optimal revenue sometimes means not selling). Monotonicity depends on the distributions.

Definition 5.4 [REGULAR DISTRIBUTION]. A distribution F is REGULAR if its virtual valuation function $v - \frac{1-F(v)}{f(v)}$ is strictly increasing.

For most applications, nondecreasing suffices. Uniform, exponential, and lognormal distributions are regular. Many multimodal distributions and distributions with sufficiently heavy tails are not.

Theorem 5.5 [Optimal single-item auction]. *For a single-item auction whose bidders' valuations are i.i.d. from a regular distribution F with virtual valuation function φ, the optimal auction is the Vickrey auction with reserve price $\varphi^{-1}(0)$.*

Proof. All bidders share the strictly increasing φ, so the bidder with the highest virtual valuation is the bidder with the highest valuation, and its virtual valuation is nonnegative exactly when its valuation is at least $\varphi^{-1}(0)$. So the virtual welfare-maximizing rule is the allocation rule of the Vickrey auction with reserve $\varphi^{-1}(0)$. It is monotone, since raising a bid can only make a bidder the highest and above the reserve, and its critical bid, hence Myerson payment, is the larger of the second-highest bid and the reserve. □

So for i.i.d. bidders and a regular distribution, eBay with a suitable opening bid is the optimal auction format. For the uniform distribution on $[0,1]$, $\varphi^{-1}(0) = \frac{1}{2}$, so the reserve $\frac{1}{2}$ of the two-bidder example is optimal. By (5.2) its revenue is the expected virtual valuation of the maximum M of the two values when $M \geq \frac{1}{2}$; with M of density $2z$,

$$\int_{1/2}^{1} (2z-1)\,2z\,dz = \left[\tfrac{4}{3}z^3 - z^2\right]_{1/2}^{1} = \tfrac{1}{3} + \tfrac{1}{12} = \tfrac{5}{12}.$$

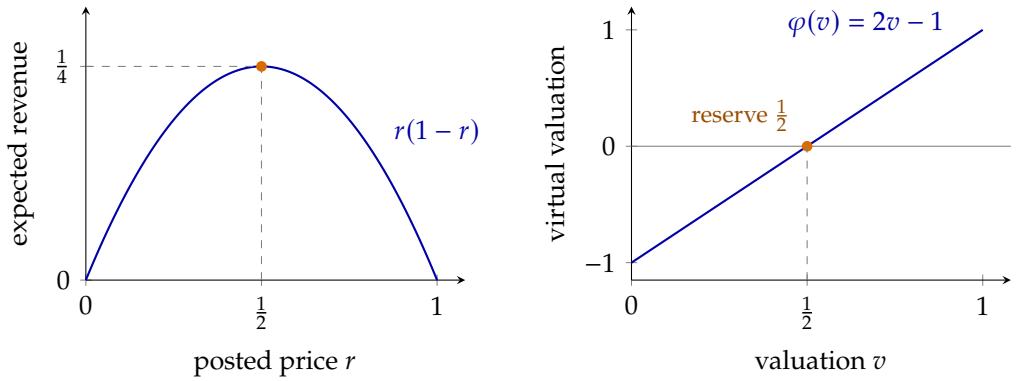


Figure 6: The uniform distribution on $[0,1]$. Left: the expected revenue $r(1-F(r)) = r(1-r)$ of a posted price, maximized by the monopoly price $\frac{1}{2}$ with revenue $\frac{1}{4}$. Right: the virtual valuation $\varphi(v) = 2v - 1$, negative below $\varphi^{-1}(0) = \frac{1}{2}$, the optimal reserve price.

More generally, in an arbitrary single-parameter environment with independent valuations, the virtual welfare-maximizing rule chooses, for each $\mathbf{v}$, the feasible outcome of maximum virtual welfare. If every F_i is regular, it is monotone (with ties broken consistently). Fix $\mathbf{v}_{-i}$, let $W_{-i}(\mathbf{x}) = \sum_{j \neq i} \varphi_j(v_j) x_j$, and let $\mathbf{x}, \mathbf{x}'$ be chosen at valuations $v_i < v_i'$ for i, with $\varphi = \varphi_i(v_i) < \varphi' = \varphi_i(v_i')$. Optimality of each choice gives $\varphi x_i + W_{-i}(\mathbf{x}) \geq \varphi x_i' + W_{-i}(\mathbf{x}')$ and $\varphi' x_i' + W_{-i}(\mathbf{x}') \geq \varphi' x_i + W_{-i}(\mathbf{x})$. Adding, $(\varphi' - \varphi)(x_i' - x_i) \geq 0$, so $x_i' \geq x_i$. With the unique DSIC payment rule, this gives the optimal auction

for every single-parameter environment with regular distributions.

5.5 Extensions

Myerson's theory extends further. For irregular distributions the virtual welfare-maximizing rule is not monotone, and one instead finds the monotone rule of maximum expected virtual welfare by IRONING the virtual valuation functions into monotone ones while preserving the virtual welfare of the relevant auctions (Hartline, *Mechanism Design and Approximation*, Chapter 3). Also, the DSIC optimal auctions above remain optimal among all BAYESIAN INCENTIVE COMPATIBLE mechanisms; for example, a first-price auction cannot earn more revenue at equilibrium than the best DSIC auction. Independence of the distributions, however, is crucial and cannot be relaxed without significantly changing the results.

Source. Tim Roughgarden, Algorithmic Game Theory, Lecture 5: Revenue-Maximizing Auctions.

Lecture 6: Simple Near-Optimal Auctions

Recall from Lecture 5 that for every DSIC auction in a single-parameter environment with independent valuation distributions $F_1, \ldots, F_n$, expected revenue equals expected virtual welfare:

$$\mathbb{E}_{\mathbf{v}}\Big[\sum_{i=1}^{n} p_i(\mathbf{v})\Big] = \mathbb{E}_{\mathbf{v}}\Big[\sum_{i=1}^{n} \varphi_i(v_i) \cdot x_i(\mathbf{v})\Big], \qquad \varphi_i(v_i) = v_i - \frac{1 - F_i(v_i)}{f_i(v_i)}.$$

If every F_i is regular, the allocation rule $\mathbf{x}(\mathbf{v}) = \arg\max_X \sum_i \varphi_i(v_i) x_i(\mathbf{v})$ is monotone and, with suitable payments, is the optimal auction. For a single item with i.i.d. bidders and a regular F, this is simply the Vickrey auction with reserve price $\varphi^{-1}(0)$.

6.1 Optimal auctions can be complex

For a single item with independent regular but non-identical distributions, the optimal auction can be weird and unlike any auction used in practice. Someone other than the highest bidder might win, and the winner's payment is hard to explain without virtual valuations. This is unavoidable if one is fully confident in the F_i and wants every last cent of expected revenue: the allocation and payment rules are then forced. The goal of this lecture is auctions that are simpler, more practical, and more robust, which can then only be approximately optimal. Unlike polynomial running time in Lecture 4, terms like simple and robust are left informal here.

6.2 The prophet inequality

In a game with n stages, stage i offers a nonnegative prize π_i drawn from a distribution G_i. The independent distributions $G_1, \ldots, G_n$ are known in advance, but π_i is revealed only at stage i. After seeing π_i, you either accept it and end the game or discard it and continue. The difficulty is the trade-off between accepting a reasonable prize early and missing a great one later, and having to settle for a lousy prize at the end. A prophet who knows all the realizations earns $\mathbb{E}[\max_i \pi_i]$.

Theorem 6.1 [Prophet inequality; Samuel-Cahn]. *For every sequence $G_1, \ldots, G_n$ of independent distributions, some strategy guarantees expected reward at least $\frac{1}{2}\mathbb{E}_{\boldsymbol{\pi}}[\max_i \pi_i]$. In fact, there is such a* threshold strategy *with a threshold t, which accepts prize i if and only if $\pi_i \geq t$.*

Proof. Write $z^+ = \max\{z, 0\}$, and let $q(t)$ be the probability that the t-threshold strategy accepts no prize. We bound the strategy's payoff from below and the prophet's from above by expressions that are easy to compare.

With probability $1 - q(t)$ the strategy gets at least t. Credit it the extra $\pi_i - t$ only when π_i is the unique prize that is at least t (these events are disjoint), and only the baseline t when two or more prizes reach the threshold. Then

$$\begin{aligned}\mathbb{E}[\text{payoff}] &\geq (1 - q(t))\, t + \sum_{i=1}^{n} \mathbb{E}[\pi_i - t \mid \pi_i \geq t,\ \pi_j < t\ \forall j \neq i]\, \mathbb{P}[\pi_i \geq t]\, \mathbb{P}[\pi_j < t\ \forall j \neq i] \\ &= (1 - q(t))\, t + \sum_{i=1}^{n} \underbrace{\mathbb{E}[\pi_i - t \mid \pi_i \geq t]\, \mathbb{P}[\pi_i \geq t]}_{=\mathbb{E}[(\pi_i - t)^+]}\ \underbrace{\mathbb{P}[\pi_j < t\ \forall j \neq i]}_{\geq q(t)} \\ &\geq (1 - q(t))\, t + q(t) \sum_{i=1}^{n} \mathbb{E}\big[(\pi_i - t)^+\big]. \end{aligned} \tag{6.1}$$

The first step uses independence to factor the probability, the second uses it to drop the conditioning on the other prizes, and the third uses $q(t) = \mathbb{P}[\pi_j < t\ \forall j] \leq \mathbb{P}[\pi_j < t\ \forall j \neq i]$. For the prophet, add and subtract t:

$$\mathbb{E}\Big[\max_i \pi_i\Big] = \mathbb{E}\Big[t + \max_i (\pi_i - t)\Big] \leq t + \mathbb{E}\Big[\max_i (\pi_i - t)^+\Big] \leq t + \sum_{i=1}^{n} \mathbb{E}\big[(\pi_i - t)^+\big]. \tag{6.2}$$

Choose t with $q(t) = \frac{1}{2}$, a 50/50 chance of accepting some prize. Then (6.1) is $\frac{1}{2}\big(t + \sum_i \mathbb{E}[(\pi_i - t)^+]\big)$, at least half of (6.2). If point masses rule out $q(t) = \frac{1}{2}$ exactly, a minor extension of the argument gives the same result. □

The proof shows more. The lower bound (6.1) credits only t whenever several prizes reach the threshold, so the guarantee of $\frac{1}{2}\mathbb{E}[\max_i \pi_i]$ holds even if, among several prizes above the threshold, the strategy somehow picks the smallest.

6.3 Simple single-item auctions

Return to a single-item auction whose bidders have independent regular distributions $F_1, \ldots, F_n$, not necessarily identical. Regard $\varphi_i(v_i)^+$ as the ith prize; its distribution G_i is induced by F_i, so the G_i are independent. The optimal auction gives the item to the highest nonnegative virtual valuation, so its expected revenue is $\mathbb{E}_{\mathbf{v}}[\max_i \varphi_i(v_i)^+]$, exactly the prophet's expected prize. Consider any allocation rule of the following form.

1. Choose t such that $\mathbb{P}[\max_i \varphi_i(v_i)^+ \geq t] = \frac{1}{2}$.
2. Give the item to a bidder i with $\varphi_i(v_i) \geq t$, if any, breaking ties among several candidates arbitrarily (subject to monotonicity).

Since $\mathbb{P}[\max_i \varphi_i(v_i)^+ \geq 0] = 1$, we have $t > 0$, so every winner has a nonnegative virtual valuation. By the strong form of the prophet inequality and expected revenue equals expected virtual welfare,

every such auction satisfies

$$\mathbb{E}_{\mathbf{v}}\left[\sum_{i=1}^{n} \varphi_i(v_i)^+ x_i(\mathbf{v})\right] \geq \frac{1}{2}\,\mathbb{E}_{\mathbf{v}}\left[\max_i \varphi_i(v_i)^+\right],$$

that is, it earns at least half of the optimal expected revenue. One specific such rule:

1. Set a reserve price $r_i = \varphi_i^{-1}(t)$ for each bidder i, with t as above.
2. Give the item to the highest bidder that meets its reserve, if any.

By regularity, $v_i \geq r_i$ exactly when $\varphi_i(v_i) \geq t$, so this rule has the required form, and it is clearly monotone. It is qualitatively simpler than the optimal auction in two ways. The winner pays the larger of its reserve and the highest bid of another bidder that meets its own reserve, so virtual valuations are used only to set reserves. And the highest bidder wins whenever it clears its reserve.

The remaining drawback is that the reserves are bidder-specific. Such non-anonymous reserves appear in some practice (in sponsored search, higher-quality advertisers generally face lower reserves) but are rare; eBay, for instance, allows one opening bid. For the Vickrey auction with an *anonymous* reserve price and non-i.i.d. regular bidders, it is known that some such auction always recovers at least 25% of the optimal expected revenue, and none always recovers more than 50% (Hartline and Roughgarden, 2009); the best constant was open as of 2013.

6.4 Prior-independent auctions and the Bulow–Klemperer theorem

A second critique is that the optimal auction assumes the seller knows the distributions F_i in advance. This is reasonable with plenty of data and slowly changing preferences, but not in thin markets with little data, such as keyword auctions for rare but valuable search queries. The PRIOR-INDEPENDENT approach keeps valuations random but unknown: distributions are used in the analysis of an auction but not in its design. The goal is an auction, described independently of the distributions, that performs almost as well as if they were known.

Theorem 6.2 [BULOW–KLEMPERER]. *Let F be a regular distribution and n a positive integer. Then*

$$\mathbb{E}_{v_1,\ldots,v_{n+1}\sim F}\left[\text{Rev(VA) } (n+1 \textit{ bidders})\right] \geq \mathbb{E}_{v_1,\ldots,v_n\sim F}\left[\text{Rev}(\text{OPT}_F) \text{ } (n \textit{ bidders})\right], \tag{6.3}$$

where VA *is the Vickrey auction and* OPT_F *is the optimal auction for F, the Vickrey auction with monopoly reserve $\varphi^{-1}(0)$.*

The left side of (6.3) is prior-independent, while the right side depends on F through its reserve. So a single auction is simultaneously competitive with infinitely many optimal auctions, across all single-item environments with i.i.d. regular valuations. The usual reading is that extra competition matters more than getting the auction format just right.

Proof. Define a fictitious auction A with $n+1$ bidders: run OPT_F on bidders $1, \ldots, n$, and if it does not allocate the item, give it to bidder $n+1$ for free. Then A is DSIC (bidder $n+1$ cannot affect its outcome, and bidders $1, \ldots, n$ face OPT_F), its expected revenue with $n+1$ bidders equals that of OPT_F with n bidders, and it always allocates the item.
It remains to show that, for i.i.d. regular valuations, the Vickrey auction maximizes expected revenue over all DSIC auctions that always allocate the item. By expected revenue equals expected virtual welfare, the best such auction gives the item to the bidder with the highest virtual valuation, even if it is negative. All bidders share the same increasing φ, so this is the bidder with the highest valuation, which is the Vickrey allocation. Hence the Vickrey auction with $n+1$ bidders earns at least as much as A, and so at least as much as OPT_F with n bidders. □

Corollary 6.3 [Vickrey versus optimal with the same bidders]. *For every $n \geq 2$ i.i.d. bidders with a regular distribution, the expected revenue of the Vickrey auction is at least $\frac{n-1}{n}$ times that of the optimal auction.*

Proof. By Theorem 6.2, Vickrey with n bidders earns at least OPT_F with $n-1$ bidders, whose expected revenue is $\mathbb{E}[\max_{i \leq n-1} \varphi(v_i)^+]$. As the bidders are i.i.d., this is also the expected maximum of $\varphi(v_i)^+$ over a uniformly random set of $n-1$ of the n bidders, chosen independently of the values. That set contains the bidder with the largest $\varphi(v_i)^+$ with probability $\frac{n-1}{n}$, and all terms are nonnegative, so the expectation is at least $\frac{n-1}{n}\mathbb{E}[\max_{i \leq n} \varphi(v_i)^+]$, the optimal revenue with n bidders. □

6.5 Case study: reserve prices in Yahoo! keyword auctions

In the standard keyword auction model (Lecture 2), if valuations per click are i.i.d. from a regular F, the revenue-maximizing auction ranks bidders by bid, from the best slot to the worst, after applying the monopoly reserve $\varphi^{-1}(0)$ to all bidders. Up to 2008, Yahoo! used low reserves (initially \$0.01, later \$0.05, and \$0.10 in 2008), and the same reserve for every keyword, though a query like "divorce lawyer" surely warrants a higher reserve than "pizza".

A 2008 field experiment (Ostrovsky and Schwarz) asked how revenue would change with theoretically optimal reserves, set independently per keyword.

1. A lognormal valuation distribution was fitted for each of about half a million keywords from past bidding data. Yahoo!'s auction is a non-DSIC variant of GSP, so valuations were reverse-engineered from bids, assuming bidders play the equilibrium outcome-equivalent to the dominant-strategy outcome of the DSIC auction. There is no evidence that the conclusions depend on these details.
2. Optimal reserves, computed per keyword, varied widely, with many at \$0.30 or \$0.40, so the uniform \$0.10 was often much too low.

To be conservative, Yahoo! set each new reserve to the average of the old $0.10 and the theoretically optimal reserve. Revenue rose several percent, most of it from this initial change; raising reserves further had little effect. The gains were largest in valuable but thin markets, with fewer than six bidders, and Yahoo!'s president credited better reserve prices as the biggest reason for higher search revenue in its third-quarter 2008 report.

Source. Tim Roughgarden, Algorithmic Game Theory, Lecture 6: Simple Near-Optimal Auctions.

Lecture 7: Multi-Parameter Mechanism Design and the VCG Mechanism

7.1 General mechanism design problems

In single-parameter problems each participant has one private number, its valuation per unit of stuff. Once a participant has different private valuations for different items, so that it is unclear whether it prefers item A to item B, the problem is multi-parameter. A general mechanism design problem has

- n strategic participants, or agents;
- a finite set Ω of outcomes;
- for each agent i, a private valuation $v_i(\omega)$ for each outcome $\omega \in \Omega$.

The set Ω is abstract and may be very large. In a single-item auction, Ω has $n+1$ elements, one for each possible winner and one for no sale. The single-parameter model assumes that an agent's valuation is 0 in the n outcomes where it does not win, leaving one unknown per agent. In the multi-parameter model an agent may value each possible winner differently: in a bidding war over a startup, an agent most wants to acquire it, but if it loses it prefers a company in a different market to a direct competitor.

7.2 The VCG mechanism

Theorem 7.1 [The Vickrey–Clarke–Groves (VCG) mechanism]. *In every general mechanism design environment, there is a DSIC welfare-maximizing mechanism.*

Of the three properties of awesome auctions (Lecture 2), this asserts the first two but not polynomial running time, which cannot always be had even in single-parameter environments unless P = NP. The Revelation Principle holds unchanged in multi-parameter environments, so direct-revelation mechanisms suffice. We follow the usual two steps.

First, assume truthful bids and pick the outcome. Welfare maximization forces the welfare-maximizing outcome with bids as proxies for valuations: given bids $\mathbf{b}_1, \ldots, \mathbf{b}_n$, each indexed by Ω,

$$\mathbf{x}(\mathbf{b}) = \arg\max_{\omega \in \Omega} \sum_{i=1}^{n} b_i(\omega). \tag{7.1}$$

Second, find payments that make this DSIC. Myerson's Lemma does not extend beyond single-

parameter environments: with bids in several dimensions it is not even clear what monotonicity means, and critical bids have no obvious analog. The key idea is instead to charge each agent its EXTERNALITY, the welfare loss its presence inflicts on the other $n-1$ agents. In a single-item auction with truthful bids the winner inflicts a loss equal to the second-highest bid, which is exactly the Vickrey payment. In general, with $\omega^* = \mathbf{x}(\mathbf{b})$,

$$p_i(\mathbf{b}) = \underbrace{\max_{\omega \in \Omega} \sum_{j \neq i} b_j(\omega)}_{\text{without } i} - \underbrace{\sum_{j \neq i} b_j(\omega^*)}_{\text{with } i}. \tag{7.2}$$

This is nonnegative, since ω^* is one of the outcomes in the maximum. The VCG MECHANISM is $(\mathbf{x}, \mathbf{p})$; by definition it maximizes welfare under truthful bids.

Proof that VCG is DSIC. This is proved from scratch, without Myerson's Lemma. Fix i and $\mathbf{b}_{-i}$. When the chosen outcome is ω^*, agent i's quasilinear utility is

$$v_i(\omega^*) - p_i(\mathbf{b}) = \underbrace{\left[v_i(\omega^*) + \sum_{j \neq i} b_j(\omega^*)\right]}_{\text{(A)}} - \underbrace{\left[\max_{\omega \in \Omega} \sum_{j \neq i} b_j(\omega)\right]}_{\text{(B)}}.$$

Term (B) does not depend on $\mathbf{b}_i$, so i wants to maximize term (A). Even if i could choose ω^* directly, it would choose an outcome maximizing (A). Bidding $\mathbf{b}_i = \mathbf{v}_i$ makes the objective (7.1) that the mechanism maximizes identical to (A), so truthful bidding yields an outcome maximizing i's utility, and no other bid can do better. □

Rewriting (7.2),

$$p_i(\mathbf{b}) = \underbrace{b_i(\omega^*)}_{\text{bid}} - \underbrace{\left[\sum_{j=1}^{n} b_j(\omega^*) - \max_{\omega \in \Omega} \sum_{j \neq i} b_j(\omega)\right]}_{\text{rebate}}, \tag{7.3}$$

so agent i pays its bid minus a rebate equal to the increase in welfare attributable to its presence. In the Vickrey auction the highest bidder pays b_1 minus the rebate $b_1 - b_2$. With nonnegative bids the rebate is nonnegative, because the welfare-maximizing ω^* has $\sum_j b_j(\omega^*) \geq \sum_j b_j(\omega) \geq \sum_{j \neq i} b_j(\omega)$ for every ω. Hence $p_i(\mathbf{b}) \leq b_i(\omega^*)$, and truthful agents are guaranteed nonnegative utility.

So DSIC welfare maximization is always possible in principle in general multi-parameter environments. The VCG mechanism can be infeasible in practice, but it is a useful benchmark for more practical approaches.

7.3 Combinatorial auctions

Combinatorial auctions are important in practice, for instance in government spectrum auctions and for take-off and landing slots at airports, and notoriously difficult in theory and practice. A COMBINATORIAL AUCTION has n bidders (say, Verizon, AT&T, and several regional providers) and a set M of m non-identical items (say, licenses to broadcast on a frequency in a given area). An outcome is a vector $(S_1, \ldots, S_n)$ of disjoint BUNDLES, with $S_i \subseteq M$ the items given to bidder i; there are $(n+1)^m$ outcomes. Each bidder i has a private valuation $v_i(S)$ for every bundle $S \subseteq M$, so 2^m private parameters. One generally assumes $v_i(\emptyset) = 0$ and FREE DISPOSAL: $v_i(S) \leq v_i(T)$ whenever $S \subseteq T$. The welfare of $(S_1, \ldots, S_n)$ is $\sum_{i=1}^n v_i(S_i)$.

In principle the VCG mechanism gives a DSIC welfare-maximizing auction, which is useful when valuations are sufficiently structured, as with unit-demand bidders. In general there are four major impediments.

1. Preference elicitation. Each bidder has $2^m - 1$ private parameters, roughly a thousand when $m = 10$ and a million when $m = 20$. No bidder wants to report that many bids and no seller wants to hear them, so the VCG mechanism, and every direct-revelation mechanism, is a nonstarter in practice. This never arises in single-parameter problems, where each bidder reports one number. The alternative is INDIRECT MECHANISMS, which learn about preferences only on a need-to-know basis. The canonical one is the ascending ENGLISH AUCTION. In its analysis-friendly Japanese variant, a publicly displayed price rises steadily from an opening price, each bidder is either in or out and cannot return after dropping out, and the last bidder in wins at the price where the second-to-last dropped out. Each bidder has a dominant strategy: stay in while the price is below its valuation and drop out when the price reaches it. Under these strategies the outcome is that of the sealed-bid Vickrey auction, which is what the Revelation Principle produces from the English auction. Whether indirect mechanisms can guarantee non-trivial welfare while eliciting only, say, polynomially much information in n and m has roughly the answer no for complex valuations and yes for sufficiently simple ones.

2. Computational intractability. Even when direct revelation is practical, as for single-parameter bidders, welfare maximization can be NP-hard, as for Knapsack auctions (Lecture 4). No clever auction format avoids this. Since valuations are unknown and practical auctions are not DSIC, closeness to optimal welfare cannot be checked directly, but sanity checks help: did bidders acquire sensible packages, such as licenses adjacent geographically or in spectrum, and did similar items sell for similar prices?

3. Bad revenue and incentives. The VCG mechanism, essentially the unique DSIC welfare-maximizing mechanism, can have bad revenue and incentive properties even when the first two challenges are absent.

Example: Take items A and B. Bidder 1 wants only both: $v_1(AB) = 1$ and $v_1 = 0$ otherwise. Bidder 2 wants only A: $v_2(A) = v_2(AB) = 1$ and $v_2 = 0$ otherwise. The maximum welfare is 1; if bidder 1 gets AB, it pays the welfare 1 that bidder 2 would obtain without it, and if bidder 2 gets A, it pays bidder 1's 1 for the same reason, so revenue is 1 either way. Now add bidder 3 with $v_3(B) = v_3(AB) = 1$ and 0 otherwise. The maximum welfare rises to 2 (A to bidder 2, B to bidder 3), but bidder 2 pays $\max_\omega \sum_{j \neq 2} v_j(\omega) - v_3(B) = 1 - 1 = 0$, and bidder 3 likewise pays 0. Revenue drops to 0.

Zero revenue in a seemingly competitive environment is a dealbreaker in practice. Revenue non-monotonicity also implies incentive problems, such as vulnerability to collusion and to false-name bids. None of these issues affects the single-item Vickrey auction.

4. Strategic behavior in iterative auctions. Because of the first challenge, almost all practical combinatorial auctions are iterative, proceeding in rounds, and this creates new opportunities for strategizing. In an early, relatively uncompetitive spectrum auction, bidders used the low-order digits of bids to signal (Cramton and Schwartz). USWest and McLeod repeatedly outbid each other on license #378, for Rochester, Minnesota. USWest then bid on several licenses elsewhere on which McLeod was the standing high bidder and USWest had shown no interest. McLeod won them back at higher prices, and every one of USWest's retaliatory bids was a multiple of 1000 plus 378. Requiring bids to be multiples of a large number largely eliminates this kind of signaling, but other opportunities for undesirable strategic behavior remain.

Source. Tim Roughgarden, Algorithmic Game Theory, Lecture 7: Multi-Parameter Mechanism Design and the VCG Mechanism.

Lecture 8: Combinatorial and Wireless Spectrum Auctions

8.1 Selling items separately

A combinatorial auction has n bidders and m non-identical items, and bidder i has a private valuation $v_i(S)$ for every bundle $S \subseteq M$. Reporting 2^m bids is absurd unless m is tiny, so we must use indirect, and especially iterative, mechanisms that query valuations on a need-to-know basis, giving up both the DSIC guarantee and exact welfare maximization. The simplest such mechanisms sell the items separately, with a single-item auction for each, needing only one bid per bidder per item.

Whether this can work depends on a fundamental dichotomy. Items are SUBSTITUTES if they give diminishing returns, so that having one makes the others less valuable; for two items, $v(AB) \leq v(A) + v(B)$. Two licenses for the same area with equal-sized frequency ranges are usually substitutes. Items are COMPLEMENTS if they have synergies, so that having one makes the others more valuable; for two items, $v(AB) > v(A) + v(B)$. Complements arise when bidders want licenses adjacent in geography or frequency.

Substitutes are far easier, in theory and practice. With known valuations, welfare maximization is computationally tractable, and the undesirable properties of the VCG mechanism from Lecture 7 disappear, generalizing the single-item case; selling items separately has a chance to work well. With complements, welfare maximization is very difficult even without incentive constraints, and separate single-item auctions cannot be expected to perform well. Real spectrum auctions mix both: if items are mostly substitutes, properly implemented separate auctions may already do well, and otherwise extra ideas (package bidding) are needed.

8.2 Two rookie mistakes

Rookie mistake 1: hold the single-item auctions sequentially. Take the easy case of identical items where each bidder wants at most one, which a single auction allocating all items solves well. Suppose instead two identical items are sold by back-to-back Vickrey auctions, and you have a very high valuation, so you expect to win any auction you enter. If everyone else bids straightforwardly (entering the next auction with their true valuation until they win), then entering the first auction wins at the second-highest valuation, while skipping it lets the second-highest bidder win and leave, so you win the second auction at the third-highest valuation. Straightforward bidding is therefore not a dominant strategy. Bidders must guess future selling prices, which makes outcomes unpredictable, with possibly low welfare and revenue. In March 2000, Switzerland sold three spectrum blocks by sequential Vickrey auctions. The first two, identical 28 MHz blocks, sold for 121 and 134 million Swiss francs, and the third, a larger 56 MHz block, sold for only 55 million.

Rookie mistake 2: use sealed-bid single-item auctions. In 1990, New Zealand sold essentially identical television licenses by simultaneous sealed-bid Vickrey auctions. With 10 licenses and wanting only one, a bidder can go for one license chosen at random, or bid less aggressively on several hoping for a bargain, trading off the risk of winning too many against the risk of winning too few. Such miscoordination lowers welfare and revenue. For example, with three bidders, two identical items, and unit demand, the natural extension of Vickrey sells to the two highest bidders at the smallest valuation; but if each bidder targets one license in a simultaneous sealed-bid auction, one license likely gets a single bidder and goes for free (or the reserve price). New Zealand raised $36 million against a projected $250 million. On one license the high bid was $100,000 and the selling price $6; on another, $7 million and $5,000. A later switch to first-price sealed bids probably did not fix the miscoordination, but made the losses less visible to the public.

8.3 Simultaneous ascending auctions

Simultaneous ascending auctions (SAAs) underlie most spectrum auctions of the past 20 years. Conceptually, they are parallel single-item English auctions, one auctioneer per item. In each round, each bidder may place new bids on any subset of items, subject to an activity rule, which forces participation from the start; its gist is that the number of items a bidder bids on can only decrease as prices rise. High bids and bidders are generally public, even though this enables signaling. The first round with no new bids ends the auction.

SAAs work better than sequential or sealed-bid auctions mainly because of price discovery: as bidders learn likely selling prices, they make mid-course corrections, abandoning fiercely contested licenses, snapping up bargains, and rethinking which packages to assemble. This typically resolves miscoordination. With two identical items and three bidders, some bidder is losing both auctions in every round, and when it jumps back in it bids on the cheaper item, keeping the two prices close. Bidders also need valuations only on a need-to-know basis: coarse estimates for most bundles and precise ones only for those that matter.

SAAs are believed to achieve good welfare and revenue. This is hard to verify, but sanity checks suggest it: little or no resale after the auction, at prices comparable to the auction's; similar prices for similar items; revenue meeting or exceeding projections; evidence of price discovery, such as midpoint prices and provisional winners highly correlated with the final ones; and sensible packages, such as licenses adjacent geographically or in frequency.

SAAs have two big vulnerabilities.

Demand reduction. A bidder asks for fewer items than it wants, to lower competition and hence prices; this arises even with substitutes. Take two identical items. Bidder 1 values one item at 10 and both at 20; bidder 2 values one item at 8 and does not want a second (also 8 for both). Giving both items to bidder 1 maximizes welfare at 20, and VCG would earn revenue 8. In an SAA, bidder 2 happily takes either item at any price below 8, so it drops out only once both prices reach 8. If

bidder 1 insists on both items, its utility is $20 - 16 = 4$. If it targets just one, each bidder gets one item at a near-zero price, and bidder 1's utility is close to 10. Demand reduction loses welfare and revenue relative to VCG, and there is ample evidence of it in spectrum auctions.

The exposure problem. This arises with complements. Take two non-identical items. Bidder 1 wants only both, valuing them at 100 (and 0 otherwise); bidder 2 will pay 75 for either item. VCG gives both items to bidder 1, for welfare 100, with revenue 75. In an SAA, bidder 2 drops out only when each price reaches 75, so bidder 1 would pay 150 for both, more than its value, and winning only one item at a non-trivial price is worse still. If bidder 2's value per item were only 40, bidder 1 should simply go for both, but it cannot know which case holds. The exposure problem makes bidding hard for bidders with complementary items, and often leads to risk-averse, tentative bidding.

8.4 Package bidding

Whether and how to add PACKAGE BIDDING, bids on sets of items as well as individual items, is difficult and controversial. Package bids mainly alleviate the exposure problem, freeing bidders who want bundles to bid aggressively, and can sometimes remove the incentive for demand reduction. The conservative view, which long dominated practice, is that they add complexity to a functional format and may do more harm than good; limited forms have been used only in the last 5 to 10 years.

- One approach adds a single PROXY ROUND after the SAA (Ausubel and Milgrom), in which bidders submit package bids on any subsets, subject to an activity rule. These compete with each other and with the winning individual bids of the SAA phase, and the final allocation maximizes welfare treating bids as true values. Computing prices is tricky: VCG payments are avoided because of their poor revenue and incentive properties, and a more aggressive, non-DSIC payment rule with other good incentive properties is used, whose effect on bidding is not well understood.
- Another approach predefines a limited set of allowed packages, ideally matching what bidders want while structured enough for simple allocation and payment rules. HIERARCHICAL PACKAGES, such as individual licenses, regional bundles, and nationwide bundles, have emerged as a sweet spot. Poorly matched packages can do more harm than good: a bidder who wants ABCD when the packages are ABEF and CDHI has no good strategy.

8.5 Reverse auctions and deferred allocation

New spectrum generally has to be taken from someone else. The FCC's planned double auction (as proposed in 2013) uses a REVERSE AUCTION to buy out TV broadcasters and free spectrum, and a FORWARD AUCTION, likely an SAA with bells and whistles, to resell it to companies that can use it more

valuably. The remaining broadcasters are REPACKED so that the freed spectrum is contiguous: for instance, broadcasters on UHF channels 38 to 51 might be bought out in part and the rest reassigned to channels 38 to 45, freeing channels 46 to 51 (Figure 7).

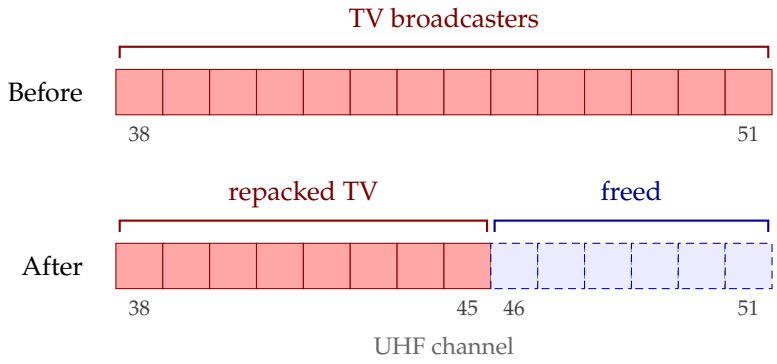


Figure 7: Repacking UHF channels 38 to 51. Before: TV broadcasters use the whole range. After: some broadcasters are bought out and the rest are reassigned to channels 38 to 45, leaving channels 46 to 51 free.

In the proposed single-parameter model, each bidder i (a TV broadcaster, each assumed to be a separate strategic agent) has a private valuation v_i for its license, its minimum acceptable buyout offer. With N the set of bidders, a set $S \subseteq N$ of winners, those bought out, is FEASIBLE if the remaining bidders $N \setminus S$ can be repacked into the target range. The set $S = N$ is feasible and $S = \emptyset$ is not. Checking feasibility is a medium-size NP-hard problem, essentially graph coloring, since stations with overlapping areas cannot get the same or adjacent channels; SAT and integer programming solvers compete to solve it in seconds.

The frontrunner allocation rule is a greedy approximate welfare maximizer, similar to those for Knapsack auctions (Lecture 4), but run in reverse: starting from the feasible set of all bidders, it removes bidders until a minimal feasible set remains. Milgrom and Segal call these DEFERRED ALLOCATION RULES. They are described here in direct-revelation form, though descending implementations with bidder-specific prices are likely, since bidders find them easier to play.

Algorithm 8.1: Deferred allocation rule

Input: Bidders N with bids, and a feasibility test for sets of winners.
Output: The set S of bidders bought out.

```
S ← N                                          // initially feasible
while some i ∈ S has S \ {i} feasible do
    Delete some such i from S                  // i will not be bought out
return S
```

The deletion step is underdetermined. Natural heuristics delete the bidder with the highest bid (least willing to be bought out) or the highest per-capita bid, and the choice will likely be guided by welfare on synthetic data. Suppose the step deletes the bidder with the largest score among those whose removal keeps S feasible, where a bidder's score is increasing in its own bid and independent of the others' bids (ties broken consistently). Then the rule is monotone: bidding lower can only

cause a bidder to win. Indeed, if i wins with some bid, it is never the deleted bidder; lowering its bid lowers only its own score and leaves the feasible deletions unchanged, so every step deletes the same bidder as before and i still wins. By Myerson's Lemma, paying each winner its critical bid, the largest bid with which it would still be bought out, yields a DSIC auction. Deferred allocation rules also have good incentive properties beyond DSIC that forward greedy algorithms lack (Milgrom and Segal).

Source. Tim Roughgarden, Algorithmic Game Theory, Lecture 8: Combinatorial and Wireless Spectrum Auctions.

Lecture 9: Beyond Quasi-Linearity

So far every agent has had quasi-linear utility: it maximizes its valuation $v_i(\omega)$ for the chosen outcome ω minus its payment p_i, where the only restriction on payments is $0 \leq p_i \leq b_i(\omega)$. This lecture studies constrained payments: first budgets, and then the extreme case of no money at all.

9.1 Budget constraints

Budgets matter most when an agent may buy many items. Every keyword auction used in practice, for example, asks a bidder for a bid per click and a daily budget. The simplest model gives agent i with budget B_i the utility

$$u_i(\omega, p_i) = \begin{cases} v_i(\omega) - p_i & \text{if } p_i \leq B_i, \\ -\infty & \text{if } p_i > B_i. \end{cases}$$

Payment constraints join the two other kinds of constraints already in play: incentive constraints (often monotonicity) and allocation constraints (such as giving each good to at most one agent). Surplus maximization, where payments appear neither in the objective nor in the constraints, is special: the VCG mechanism maximizes surplus ex post, as well as if all private data were known in advance. With budgets this is impossible. In a single-item auction where every bidder has a known budget of 1 and a private valuation, the Vickrey auction may charge the winner a second-highest bid larger than 1. Since the Vickrey auction is the unique DSIC surplus-maximizing auction, no DSIC auction that respects budgets maximizes surplus. In fact no such auction approximates the surplus well, so budgets call for new auction formats.

9.2 The market-clearing price auction

There are m identical goods. Bidder i has a private value v_i per good, so k goods are worth $k\,v_i$ to it, and a *public* budget B_i known to the seller in advance. (With private budgets the problem becomes harder, even impossible in some senses.)

Definition 9.1 [Demand]. The demand of bidder i at price p is

$$D_i(p) = \begin{cases} \min\{\lfloor B_i/p \rfloor,\ m\} & \text{if } p < v_i, \\ 0 & \text{if } p > v_i. \end{cases}$$

At $p = v_i$ the bidder is indifferent, and $D_i(v_i)$ may be taken to be any convenient integer in $\{0, 1, \ldots, \lfloor B_i/p \rfloor\}$.

As p rises, demand falls from $D_i(0) = m$ to $D_i(\infty) = 0$. A drop is either from a positive integer to 0 (when p reaches v_i) or by a single unit (when $\lfloor B_i/p \rfloor$ decreases).

The MARKET-CLEARING PRICE p^* is the smallest price with $\sum_i D_i(p^*) = m$; more generally, the smallest p^* with

$$\lim_{p \uparrow p^*} \sum_i D_i(p) \;\geq\; m \;\geq\; \lim_{p \downarrow p^*} \sum_i D_i(p).$$

The auction gives $D_i(p^*)$ goods to each bidder i at price p^* each, choosing $D_i(p^*)$ for bidders with $v_i = p^*$ so that all m goods are allocated. By the definition of demand it respects budgets, but it is not DSIC: like the simultaneous ascending auction, it is vulnerable to demand reduction.

Example: Two goods and two bidders, with $B_1 = +\infty$, $v_1 = 6$, and $B_2 = v_2 = 5$. Under truthful bids, $D_1(p) = 2$ for $p < 6$ and $D_2(p) \geq 1$ for $p < 5$, so total demand is at least 3 until $p = 5$, where $D_1(5) = 2$ and $D_2(5) = 0$. Bidder 1 gets both goods at price 5, for utility $2(6 - 5) = 2$. If bidder 1 instead bids 3, bidder 2's demand drops to 1 as p passes $\frac{5}{2}$ (it can no longer afford two goods), and the auction stops at $p = 3$ with $D_1(3) = 1$. Bidder 1 gets one good at price 3, for utility $3 > 2$.

The allocation rule is monotone, so the example shows that the payment rule is the problem. Myerson's Lemma would supply DSIC payments for this rule, but a more sophisticated allocation rule is better.

9.3 The clinching auction

The CLINCHING AUCTION sells the goods piecemeal at increasing prices. Besides the price p, it tracks the remaining supply s and each bidder's RESIDUAL BUDGET $\hat{B}_i$. The residual demand is $\hat{D}_i(p) = \min\{\lfloor \hat{B}_i/p \rfloor, s\}$ if $p < v_i$ and 0 if $p > v_i$. A bidder *clinches* goods once the others' total demand falls short of the remaining supply.

Algorithm 9.1: Clinching auction for budgeted bidders

Input: Bids b_i (used in place of v_i in $\hat{D}_i$) and public budgets B_i.
Output: An allocation of the m goods and the payments.

$p \leftarrow 0$; $s \leftarrow m$; $\hat{B}_i \leftarrow B_i$ for every i
while $s > 0$ **do**
  Increase p until some bidder i has $k := s - \sum_{j \neq i} \hat{D}_j(p) > 0$
  Give k goods to bidder i at price p each *// clinched*
  $s \leftarrow s - k$; $\hat{B}_i \leftarrow \hat{B}_i - p \cdot k$

Different goods sell at different prices, and prices increase over the course of the auction. Budgets are respected: the auction maintains $\sum_j \hat{D}_j(p) \geq s$, so a bidder clinches $k \leq \hat{D}_i(p)$ goods, which it can afford.

Example: In the previous example with truthful bids, bidder 2's demand drops to 1 as p passes $\frac{5}{2}$, so $s - \hat{D}_2 = 1$ and bidder 1 clinches one good at price $\frac{5}{2}$. Now $s = 1$ and both residual demands equal 1 until p reaches $v_2 = 5$, when bidder 2 drops out and bidder 1 clinches the second good at

price 5. Bidder 1's utility is $(6 - \frac{5}{2}) + (6 - 5) = \frac{9}{2}$, compared with 2 in the market-clearing auction. Bidding 3 instead would end with bidder 2 clinching the second good at price 3, leaving bidder 1 only $\frac{7}{2}$.

In fact no false bid does better here, as the next theorem shows.

Theorem 9.2 [Clinching is DSIC]. *The clinching auction for bidders with public budgets is DSIC.*

Proof. One could check monotonicity and Myerson's payment formula, but it is easier to verify DSIC directly. Fix bidder i and the other bids $\mathbf{b}_{-i}$. Since B_i is public, i cannot affect the term $\lfloor \hat{B}_i/p \rfloor$ of its demand. Its bid only determines when it is kicked out, meaning $\hat{D}_i(p) = 0$ from then on, which happens when p reaches b_i. A good clinched at a price $p < v_i$ adds positive utility, and one clinched at $p > v_i$ adds negative utility.

Run the auction in parallel with bids b_i and v_i. If $b_i < v_i$, the two runs are identical, by induction on the iterations, while the price rises from 0 to b_i; the false bid can only lose goods that the truthful bid would clinch at prices in $[b_i, v_i]$, each with nonnegative utility. If $b_i > v_i$, the false bid can only add goods clinched at prices in $[v_i, b_i]$, each with nonpositive utility. So no false bid beats the truthful one. □

With private budgets run on reported budgets, the clinching auction is no longer DSIC. On its own, the theorem is weak: giving all goods away to random bidders for free is also DSIC and respects budgets. The challenge is to find a benchmark that shows the clinching auction computes a good allocation. Three approaches exist, none fully satisfying.

1. Pareto optimality: no reassignment of goods and payments makes some agent (a bidder, or the seller, whose utility is its revenue) better off without making another worse off. The clinching auction is the unique deterministic DSIC auction that always computes a Pareto-optimal allocation. But Pareto optimality is neither necessary nor sufficient for a desirable auction; Bayesian-optimal mechanisms, for instance, need not be Pareto optimal.
2. Average-case analysis: maximize expected surplus subject to the budget constraints over a distribution of valuations. Progress is slow but steady. With a common budget for all bidders, the clinching auction is provably near-optimal.
3. A modified objective $\sum_i \min\{B_i, v_i x_i\}$ in place of $\sum_i v_i x_i$, with respect to which the clinching auction is provably near-optimal. This objective does not make much sense in some settings.

9.4 Mechanism design without money

In many applications money is infeasible or illegal, which amounts to every agent having budget zero: voting, organ donation, school choice, and labor markets. There is no Vickrey auction, yet some of mechanism design's greatest successes come from this setting.

In the house allocation problem, each of n agents initially owns one house and has a strict total

order over all n houses; an agent need not prefer its own. The goal is to reallocate the houses sensibly.

Algorithm 9.2: Top Trading Cycle Algorithm (TTCA)

Input: Each agent's initial house and preference order.
Output: A reallocation giving each agent exactly one house.
while agents remain **do**
  Each remaining agent points to (the owner of) its favorite remaining house
  // every vertex has out-degree 1*, so the graph has a directed cycle*
  Along each directed cycle (self-loops included), each agent receives the house it points to
  Delete the agents and houses reallocated in this iteration

A directed cycle exists because following outgoing arcs from any vertex must eventually repeat a vertex. Each agent on a cycle gives its house to its predecessor, the agent pointing to it.

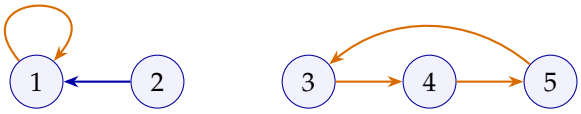


Figure 8: One iteration of the TTCA with two directed cycles: agent 1 points to its own house (a self-loop), and agents $3, 4, 5$ form a cycle. Agent 2 points into the first cycle and remains for the next iteration.

The remaining agents always still own their original houses. So in each iteration an agent points to its own house or to one it likes better, and when deleted it receives the house it points to. Hence no agent ends up worse off than with its initial house.

Theorem 9.3 [TTCA is DSIC**].** *Running the TTCA on reported preferences gives a DSIC mechanism.*

Proof. Let N_j be the agents allocated in iteration j under truthful reports. An agent of N_1 gets its first choice, so it cannot gain. Consider $i \in N_j$. No agent of $N_1 \cup \cdots \cup N_{j-1}$ ever points to i, or i would have been allocated earlier. So i's report cannot change the cycles these agents form, and by induction on the iterations they are removed exactly as before, taking their houses with them. Thus i can never receive a house owned by $N_1 \cup \cdots \cup N_{j-1}$, and with truthful reports the TTCA gives i its favorite house outside this set. □

Keeping everyone's initial house is also DSIC, so optimality needs a separate argument. A coalition S of agents BLOCKS an allocation if the agents of S can reallocate their own initial houses among themselves so that none of them is worse off and at least one is strictly better off. A CORE ALLOCATION is one that no coalition blocks.

Theorem 9.4 [TTCA COMPUTES THE CORE**].** *For every house allocation problem, the TTCA allocation is the unique core allocation.*

Proof. Let N_j be as before. In iteration ℓ the TTCA permutes the houses of N_ℓ among the agents of N_ℓ, and each agent of N_ℓ receives its favorite house outside those of $N_1 \cup \cdots \cup N_{\ell-1}$.
Core. Suppose S blocks via an internal reallocation σ. By induction on ℓ, every agent of $S \cap N_\ell$ receives its TTCA house under σ. Assume this for earlier iterations. Each agent of $S \cap (N_1 \cup \cdots \cup N_{\ell-1})$ then holds its TTCA house, whose owner lies in the same iteration and, as σ is internal, in S. Counting shows that these agents use up exactly the houses of $S \cap (N_1 \cup \cdots \cup N_{\ell-1})$. So an agent of $S \cap N_\ell$ gets a house outside $N_1 \cup \cdots \cup N_{\ell-1}$, which it likes no better than its TTCA house. Being no worse off, with strict preferences, it gets exactly its TTCA house. Then nobody in S is strictly better off, a contradiction.
Uniqueness. Let A be a core allocation different from the TTCA allocation, and let j be the first iteration such that some $a \in N_j$ has a different house under A. Under A the agents of $N_1 \cup \cdots \cup N_{j-1}$ receive their TTCA houses, which are exactly the houses they own, so every agent of N_j gets a house outside them and is weakly worse off than under the TTCA, and a strictly. The coalition N_j blocks A by trading along its TTCA cycles, a contradiction. □

Source. Tim Roughgarden, Algorithmic Game Theory, Lecture 9: Beyond Quasi-Linearity.

Lecture 10: Kidney Exchange and Stable Matching

10.1 Kidney exchange

A healthy person has two kidneys and can live with one, so a patient with kidney failure may have a living donor, such as a family member. Often the pair is INCOMPATIBLE, mainly because of blood and tissue types: an O patient can receive only an O kidney, and an AB donor can give only to an AB patient. Suppose patient P_1 (type A) has donor D_1 (type B), and P_2 (type B) has donor D_2 (type A). Exchanging donors, so that P_1 receives D_2's kidney and P_2 receives D_1's, is a KIDNEY EXCHANGE. National exchanges, where incompatible pairs register to be matched with others, aim to thicken this market and enable as many matches as possible. Payment for organs is illegal in the US (and everywhere except Iran), while exchange is legal, so this is mechanism design without money.

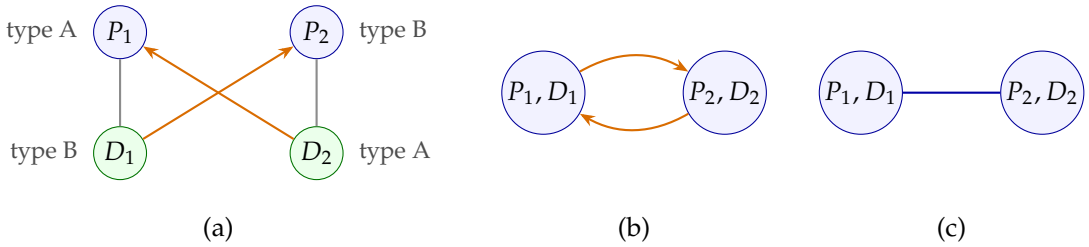


Figure 9: (a) A kidney exchange between two incompatible pairs: each donor gives to the other pair's patient. (b) The same exchange as a TTCA cycle, each pair pointing to the other's donor. (c) The same exchange as an edge of the compatibility graph used for matching.

10.2 First idea: top trading cycles

The first proposal maps the house allocation problem (Lecture 9) to kidney exchange: an agent and its initial house become a patient and its incompatible donor, and each patient ranks donors by the estimated probability of a successful transplant. The TTCA then finds cycles such as the one in Figure 9(b). An extension also handles patients without living donors (agents without houses) and deceased donors (houses without owners) by allocating along suitable chains as well as cycles; chosen correctly, it remains DSIC. The approach has two problems.

- *Long cycles.* In the first iteration the preferred-donor arcs may form a Hamiltonian cycle on all n pairs. Even a 2-cycle requires four surgeries (two extractions, two transplants), and they must be simultaneous: if P_1 and D_2 operate first, D_1 may renege, leaving P_2 as sick as before and without a donor to exchange. Each surgery needs its own operating room and team, so cycles must be short.
- *Preferences.* Patients do not really care which compatible kidney they get, so binary preferences

fit better than total orders.

10.3 Second idea: priority matching

A MATCHING of an undirected graph (V, E) is a set $F \subseteq E$ of edges, no two sharing an endpoint. Let V be the set of incompatible pairs, with an edge between (P_1, D_1) and (P_2, D_2) exactly when P_1 is compatible with D_2 and P_2 with D_1. Restricting to matchings means only pairwise exchanges, hence at most four simultaneous surgeries, and the goal is a MAXIMUM-CARDINALITY MATCHING.

Each vertex i has a true set E_i of incident edges and reports some $F_i \subseteq E_i$; it can implement a misreport by refusing the exchanges in $E_i \setminus F_i$. A vertex cares only about being matched. The goal is a DSIC mechanism, one for which reporting E_i is a dominant strategy, that returns a maximum matching. It must have this form:

1. collect a report F_i from each vertex i;
2. form $E = \{(i, j) : (i, j) \in F_i \cap F_j\}$, keeping an edge only if both endpoints agree;
3. return a maximum matching of $G = (V, E)$.

Tie-breaking in step 3 is essential for DSIC. Different maximum matchings may match the same vertices, which no vertex cares about, or different vertices, as in a star, where the center is always matched but exactly one spoke is.

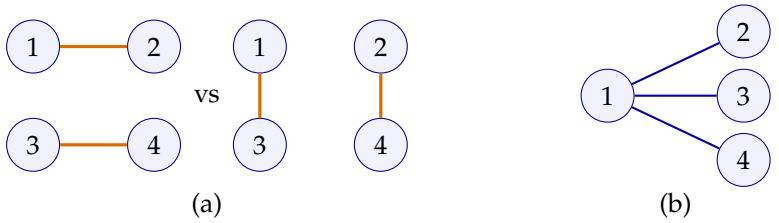


Figure 10: Non-uniqueness of maximum matchings; vertex i is the pair (P_i, D_i). (a) Two maximum matchings that match the same vertices. (b) In a star every maximum matching matches the center 1 and exactly one of $2, 3, 4$.

Order the vertices $1, 2, \ldots, n$ by priority, highest first, and implement step 3 as follows.

Algorithm 10.1: Priority tie-breaking

Input: The graph $G = (V, E)$ with $V = \{1, \ldots, n\}$ in priority order.
Output: A maximum matching of G.

$\mathcal{M}_0 \leftarrow$ the set of maximum matchings of G
for $i \leftarrow 1$ **to** n **do**
  $\mathcal{Z}_i \leftarrow$ the matchings in $\mathcal{M}_{i-1}$ that match i
  if $\mathcal{Z}_i \neq \emptyset$ **then** $\mathcal{M}_i \leftarrow \mathcal{Z}_i$ **else** $\mathcal{M}_i \leftarrow \mathcal{M}_{i-1}$
return any matching in $\mathcal{M}_n$

Vertex i is COMMITTED if $\mathcal{Z}_i \neq \emptyset$; let $S(G)$ be the set of committed vertices. By induction, $\mathcal{M}_j$ is the nonempty set of maximum matchings of G that match every committed vertex among $1, \ldots, j$. Every matching in $\mathcal{M}_n$ matches exactly $S(G)$: one that also matched an uncommitted j would lie in $\mathcal{M}_{j-1}$ and hence in $\mathcal{Z}_j$. So the choice in the last line is irrelevant, and

$$j \notin S(G) \implies \text{no maximum matching of } G \text{ matches } \big(S(G) \cap \{1, \ldots, j-1\}\big) \cup \{j\}. \tag{10.1}$$

Theorem 10.1 [PRIORITY MATCHING IS DSIC]. *For all edge sets $\{E_i\}$ and every priority order, no vertex can go from unmatched to matched by reporting a strict subset $F_i \subsetneq E_i$ instead of E_i.*

Proof. Fix the others' reports. Let G be the graph when i reports E_i, and G' when it reports F_i; then G' is G minus some edges at i. Suppose $i \in S(G')$ but $i \notin S(G)$. If the maximum matching size of G' were smaller than that of G, every maximum matching of G would match i (one avoiding i would lie in G'), so $i \in S(G)$. Hence the two sizes are equal, and every maximum matching of G' is maximum in G.

Take M in the final set for G (matching exactly $S = S(G)$) and M' in the final set for G' (matching exactly $S' = S(G')$). In $M \triangle M'$ the vertex i has degree 1, so it ends an alternating path P that starts with an edge of M'. If P had odd length, $M \triangle P$ would be a larger matching of G; so P has even length and ends at a vertex $w \neq i$ matched by M but not by M'. Then $M \triangle P$ is a maximum matching of G matching $S \setminus \{w\} \cup \{i\}$, and $M' \triangle P$ is a maximum matching of G' matching $S' \setminus \{i\} \cup \{w\}$. (Its new edges are edges of M on P, which avoid i and therefore lie in G'.)

If $i < w$, then $M \triangle P$ matches $(S \cap \{1, \ldots, i-1\}) \cup \{i\}$, contradicting (10.1) for G. If $w < i$, then $M' \triangle P$ matches $(S' \cap \{1, \ldots, w-1\}) \cup \{w\}$ with $w \notin S'$, contradicting (10.1) for G'. □

10.4 Incentives of hospitals

Many pairs are reported to national exchanges by hospitals, which want to match as many of their own patients as possible, not as many patients overall. In the following examples, edges join mutually compatible pairs.

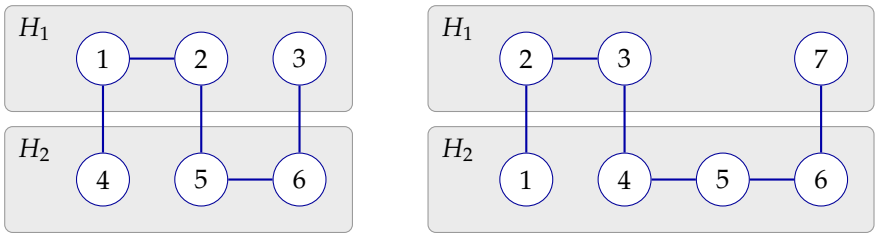


Figure 11: Pairs held by two hospitals H_1 and H_2. (a) Full reporting matches everyone, while internal matches strand 3 and 4. (b) The path $1, 2, \ldots, 7$, where each hospital gains by hiding a pair of its patients.

Example: In Figure 11(a), if H_1 matches $1, 2$ internally and reports only 3, and H_2 matches $5, 6$ internally and reports only 4, the exchange cannot match 3 with 4. If both report everything, the exchange matches $1, 2, 3$ with $4, 5, 6$, and every patient is matched.

Example: In Figure 11(b), with seven vertices every matching leaves some patient unmatched. If H_1 hides 2 and 3, the unique maximum matching of the reported graph is $\{4\text{–}5,\ 6\text{–}7\}$, and H_1 matches $2, 3$ internally, so all of H_1's patients are matched. If instead H_2 hides 5 and 6, the unique maximum matching is $\{1\text{–}2,\ 3\text{–}4\}$, and H_2 matches $5, 6$ internally.

So no DSIC mechanism always computes a maximum matching of the full graph: whichever hospital's patient a maximum matching leaves out, that hospital gains by hiding two patients. The revised goal is an approximately maximum matching in which each hospital gets approximately as many patients matched as in any matching, which is an active research topic.

10.5 Stable matching

Stable matching is the canonical example of mechanism design without money, used to assign medical graduates to hospitals and students to schools. There are two sets U ("men") and V ("women") of n vertices each, and each vertex has a strict total order over the vertices of the other set.

Definition 10.2 [Stable matching]. A stable matching is a perfect bipartite matching with no blocking pair: whenever $u \in U$ and $v \in V$ are not matched to each other, u prefers his mate to v or v prefers her mate to u.

A blocking pair would be tempted to run off together. The definition resembles the core of Lecture 9: an unstable matching is not in the core, and the core allocations are exactly the stable matchings.

Algorithm 10.2: Proposal algorithm (Gale and Shapley)

Input: Preference lists of the men and women.
Output: A stable matching.

```
while some man u ∈ U is unattached do
    u proposes to his favorite woman who has not yet rejected him
    Each woman keeps only her best offer so far and rejects the rest
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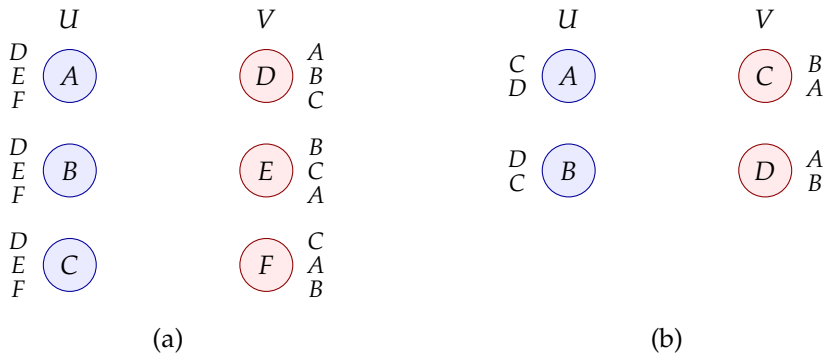


Figure 12: Two instances, each vertex listing its preferences from top to bottom. (a) All men rank D, E, F; the women disagree. (b) Men and women disagree, giving two stable matchings.

Example: In Figure 12(a), suppose C proposes to D, who accepts. Then B proposes to D, who rejects C for B; then A proposes to D, who rejects B for A. Next C proposes to E, then B proposes to E, who rejects C for B; finally C proposes to F, who accepts. The result is A–D, B–E, C–F.

The choice of unattached man is unspecified. Each man goes down his list from the top, the men a woman holds only improve, and every vertex is matched to at most one other at all times.

Theorem 10.3 [Computation of a stable matching]. *The proposal algorithm terminates within n^2 iterations with a stable matching.*

Proof. No man proposes to the same woman twice, so there are at most n^2 proposals. The output is perfect: otherwise some man was rejected by all n women, but a woman rejects only in favor of a better man and stays matched from then on, so all n women, and hence all n men, are matched, a contradiction. For stability, let u and v be unmatched to each other. If u never proposed to v, he is matched to someone he prefers, since he proceeded down his list. If he did, v rejected him, then or later, for a man she preferred, and her partners only improve after that. □

Corollary 10.4 [Existence]. *For all preference lists, at least one stable matching exists.*

This is not obvious a priori: some simple variants of the problem may have no solution. Stable matchings need not be unique. In Figure 12(b), the proposal algorithm gives both men their first choices (A–C, B–D), and giving both women their first choices (A–D, B–C) is also stable. For a man u, let $h(u)$ be his favorite among the women matched to him in some stable matching.

Theorem 10.5 [Male-optimal stable matching]. *The stable matching computed by the proposal algorithm matches every man u to $h(u)$.*

Proof. Fix an execution, and let R be the set of pairs (u, v) such that v rejects u at some point. If u ends matched to v, then $(u, v') \in R$ for every v' that u prefers to v. Since the output is stable, it suffices to show that no stable matching pairs u with v when $(u, v) \in R$.
Induct on the iterations. Suppose v rejects u in favor of u', and the claim holds for all earlier rejections. Every v' that u' prefers to v has already rejected u', so by induction every stable matching pairs u' with v or with someone he likes less. If a stable matching paired u with v, then u' would prefer v to his mate and v would prefer u' to u, a blocking pair. □

So the output is independent of the order of proposals, and a *male-optimal* stable matching exists in which every man simultaneously gets his best case. (A priori, the $h(u)$ need not even be distinct.) The proposal algorithm also matches each woman v to $l(v)$, the worst man matched to her in any stable matching. If preferences are private, the proposal algorithm is DSIC for the men but not for the women.

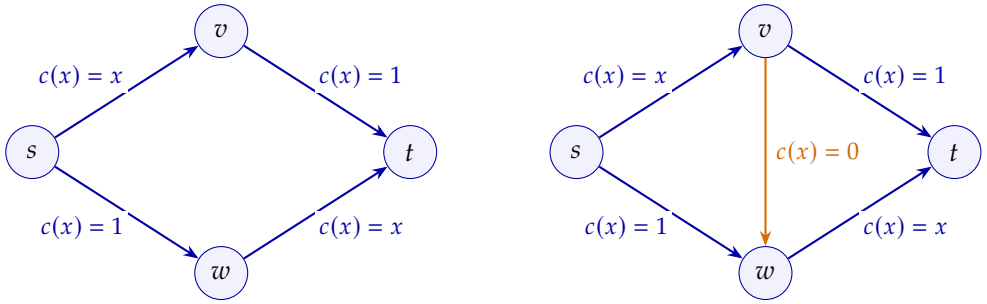
Source. Tim Roughgarden, Algorithmic Game Theory, Lecture 10: Kidney Exchange and Stable Matching.

Lecture 11: Selfish Routing and the Price of Anarchy

The second part of the course asks when games "in the wild," which nobody designed, have near-optimal equilibria. Such games generally have no dominant strategies, so predictions need stronger assumptions about behavior, and there is no reason to expect optimal outcomes. The focus is on full-information games, in which all players' payoffs are common knowledge; in network traffic, for example, everyone prefers shorter routes.

11.1 Examples

In selfish routing, traffic travels from a source s to a sink t, and each edge has a cost function $c(x)$ giving the travel time of all traffic on the edge when an amount x of traffic uses it. The price of anarchy (POA) is the ratio between the travel time in an equilibrium and the minimum possible average travel time.



(a) Initial network (b) Augmented network

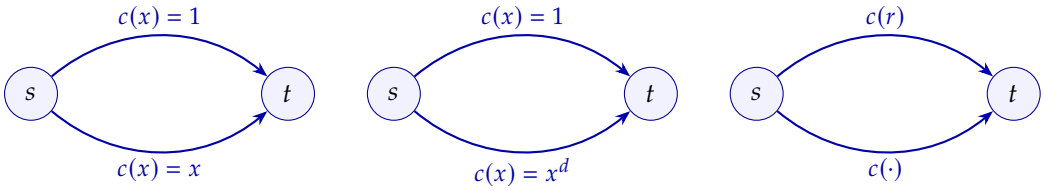
Figure 13: Braess's paradox, with one unit of traffic. Adding the zero-cost edge (v, w) raises the equilibrium travel time from $\frac{3}{2}$ to 2, so the POA of the augmented network is $\frac{4}{3}$.

Braess's paradox (Figure 13). One unit of traffic travels from s to t. In the initial network, by symmetry, half the traffic takes each route, and everyone's travel time is $\frac{3}{2}$. After a zero-cost edge from v to w is added, the route $s \to v \to w \to t$ is a dominant strategy for every driver, and the common travel time becomes 2. The minimum possible travel time is still $\frac{3}{2}$, as there is no profitable way to use the new edge. The POA is $2/\frac{3}{2} = \frac{4}{3}$.

Pigou's example (Figure 14(a)). One unit of traffic chooses between an upper edge with $c(x) = 1$ and a lower edge with $c(x) = x$. The lower edge is a dominant strategy, since even fully congested it is no worse than the upper one. So in equilibrium all traffic uses it, with travel time 1. Splitting the traffic equally gives average travel time $\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4}$, which is optimal, so the POA is again $\frac{4}{3}$.

Nonlinear Pigou's example (Figure 14(b)). Replace the lower cost function by $c(x) = x^d$ with d large.

The lower edge is still dominant, and the equilibrium travel time is still 1. But the optimum is now much better. An equal split has average travel time $\frac{1}{2} + \frac{1}{2} \cdot 2^{-d} \to \frac{1}{2}$, and routing $1 - \epsilon$ on the lower edge and ϵ on the upper one costs $(1-\epsilon)^{d+1} + \epsilon$, which tends to 0 when $\epsilon \to 0$ suitably slowly as $d \to \infty$. So the POA is unbounded as $d \to \infty$.



(a) (b) (c)

Figure 14: (a) Pigou's example. (b) Its nonlinear variant. (c) A Pigou-like network with traffic rate r: an arbitrary cost function $c(\cdot)$ on the lower edge and the constant cost $c(r)$ on the upper edge.

11.2 The main result

The model is a directed graph $G = (V, E)$ with source s, sink t, and r units of traffic (or FLOW) from s to t. Each edge e has a cost function c_e giving the travel time per unit of traffic as a function of the amount of traffic on e; there are no explicit capacities. Cost functions are always *nonnegative, continuous, and nondecreasing*. (Only single-commodity networks are treated; the result and proof extend easily to multiple sources and sinks.)

The examples suggest that highly nonlinear cost functions, not complex networks, cause inefficiency. The result is parameterized by a set C of allowed cost functions, such as the affine functions $\{ax + b : a, b \geq 0\}$.

Theorem 11.1 [TIGHT POA BOUNDS, INFORMAL]. *Among all networks with cost functions in C, the largest POA is achieved in a Pigou-like network.*

Definition 11.2 [PIGOU-LIKE NETWORK]. A PIGOU-LIKE NETWORK has two vertices s, t, two parallel edges from s to t, a traffic rate $r > 0$, a cost function $c(\cdot)$ on the lower edge, and the constant cost function $c(r)$ on the upper edge (Figure 14(c)).

The lower edge is a dominant strategy: even fully congested, it is no worse than the upper edge. So in equilibrium all traffic uses it, at total travel time $r \cdot c(r)$. With x units on the lower edge, the minimum total travel time is

$$\inf_{0 \leq x \leq r} \{x \cdot c(x) + (r - x) \cdot c(r)\}.$$

Letting x range over all $x \geq 0$ does not change this value, since for $x > r$ the expression is at least $r\,c(r)$. Hence the POA of this network is $\sup_{x \geq 0} \{r\,c(r)/(x\,c(x) + (r-x)\,c(r))\}$.

Definition 11.3 [PIGOU BOUND]. For a set C of nonnegative, continuous, nondecreasing cost functions,

the Pigou bound is the worst POA of a Pigou-like network with cost functions in C:

$$\alpha(C) := \sup_{c \in C} \sup_{r \geq 0} \sup_{x \geq 0} \left\{ \frac{r \cdot c(r)}{x \cdot c(x) + (r - x) \cdot c(r)} \right\}. \tag{11.1}$$

The first two suprema choose the free parameters c and r of a Pigou-like network; the third computes its best routing. If C contains the constant functions, then $\alpha(C)$ is a lower bound on the worst-case POA, attained by Pigou-like networks themselves. (It suffices that some $c \in C$ has $c(0) > 0$, since Pigou-like networks can then be simulated by slightly larger networks.) For affine, or even concave, cost functions, $\alpha(C) = \frac{4}{3}$.

Theorem 11.4 [Tight POA bounds, formal]. *For every set C of cost functions and every selfish routing network with cost functions in C, the POA is at most $\alpha(C)$.*

So computing the worst-case POA for a class C reduces to a back-of-the-envelope calculation. With affine cost functions, Pigou's example is the worst case, and the POA is always at most $\frac{4}{3}$. For polynomials with nonnegative coefficients and degree at most d, the worst case is the nonlinear Pigou example: with $r = 1$ and $c(x) = x^d$, the optimal split puts $x = (d+1)^{-1/d}$ on the lower edge, where the derivative of $x^{d+1} + 1 - x$ vanishes, giving cost $1 - x\,d/(d+1)$ and the values in Table 1. The POA is large only for highly nonlinear cost functions; quartic functions, sometimes used to model road traffic, give slightly above 2.

Description	Typical representative	Price of anarchy
Linear	$ax + b$	$4/3$
Quadratic	$ax^2 + bx + c$	$\frac{3\sqrt{3}}{3\sqrt{3} - 2} \approx 1.6$
Cubic	$ax^3 + bx^2 + cx + d$	$\frac{4\sqrt[3]{4}}{4\sqrt[3]{4} - 3} \approx 1.9$
Quartic	$ax^4 + \cdots$	$\frac{5\sqrt[4]{5}}{5\sqrt[4]{5} - 4} \approx 2.2$
Degree $\leq d$	$\sum_{i=0}^{d} a_i x^i$	$\frac{(d+1)\sqrt[d]{d+1}}{(d+1)\sqrt[d]{d+1} - d} \approx \frac{d}{\ln d}$

Table 1: The worst-case POA for cost functions that are polynomials with nonnegative coefficients and degree at most d.

11.3 Flows and equilibria

Let $\mathcal{P}$ be the (nonempty) set of s–t paths of G. A FLOW is a nonnegative vector $\{f_P\}_{P\in\mathcal{P}}$ with $\sum_{P\in\mathcal{P}} f_P = r$. For an edge e, the amount of flow using it is $f_e = \sum_{P\in\mathcal{P}:\, e\in P} f_P$, and the cost of a path is $c_P(f) = \sum_{e\in P} c_e(f_e)$.

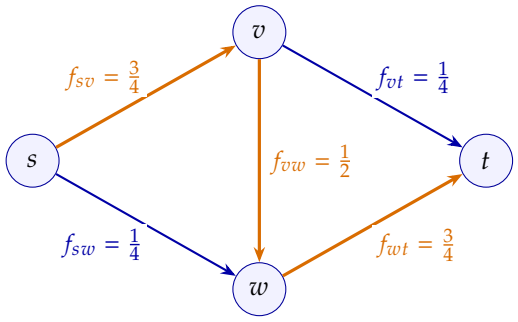


Figure 15: A flow of one unit in the augmented Braess network: half on the zig-zag path $s \to v \to w \to t$ (orange edges) and a quarter on each two-hop path. Edge labels give the edge flows f_e.

Definition 11.5 [EQUILIBRIUM FLOW]. A flow f is an EQUILIBRIUM FLOW if traffic travels only on shortest paths with respect to the travel times it induces: $f_{\hat{P}} > 0$ only if $\hat{P} \in \arg\min_{P\in\mathcal{P}} c_P(f)$.

In the augmented Braess network (Figure 15), this flow gives $c_{svwt} = \frac{3}{4} + 0 + \frac{3}{4} = \frac{3}{2}$ and $c_{svt} = c_{swt} = \frac{7}{4}$. It is not an equilibrium, since the zig-zag path is the only shortest path and some flow avoids it.

Every selfish routing network has at least one equilibrium flow (continuity of the cost functions matters here), and all equilibrium flows have the same cost; both facts are proved later. The objective is the TOTAL TRAVEL TIME, or COST, of a flow, which can be tallied by paths or by edges:

$$C(f) = \sum_{P\in\mathcal{P}} f_P \cdot c_P(f) = \sum_{e\in E} f_e \cdot c_e(f_e),$$

by reversing the order of summation. The POA of a selfish routing network is the cost of an equilibrium flow divided by the cost of an optimal (minimum-cost) flow.

11.4 Proof of the main result

Proof of Theorem 11.4. Let f be an equilibrium flow and f^* an optimal flow.
Step 1: freezing the costs at their equilibrium values $c_e(f_e)$, the flow f is optimal. Every path used by f has the same cost, say L, and $c_P(f) \geq L$ for every path P. Hence

$$\sum_{P\in\mathcal{P}} f_P \cdot c_P(f) = r \cdot L \qquad \text{and} \qquad \sum_{P\in\mathcal{P}} f^*_P \cdot c_P(f) \geq r \cdot L,$$

since both flows send r units. Rewriting both sums over edges and subtracting gives the VARIATIONAL INEQUALITY

$$\sum_{e \in E} (f_e^* - f_e)\, c_e(f_e) \geq 0. \tag{11.2}$$

Step 2: edge by edge, f^* beats f by at most the Pigou bound, up to an error term controlled by (11.2). For each edge e, take $c = c_e$, $r = f_e$, and $x = f_e^*$ in (11.1); this is allowed whether $f_e^* \leq f_e$ or $f_e^* > f_e$. Then

$$\alpha(C) \geq \frac{f_e \cdot c_e(f_e)}{f_e^* \cdot c_e(f_e^*) + (f_e - f_e^*)\, c_e(f_e)},$$

that is,

$$f_e^* \cdot c_e(f_e^*) \geq \frac{1}{\alpha(C)}\, f_e \cdot c_e(f_e) + (f_e^* - f_e)\, c_e(f_e).$$

Summing over $e \in E$ and applying (11.2),

$$C(f^*) \geq \frac{C(f)}{\alpha(C)} + \sum_{e \in E} (f_e^* - f_e)\, c_e(f_e) \geq \frac{C(f)}{\alpha(C)}.$$

So $C(f)/C(f^*) \leq \alpha(C)$. □

Source. Tim Roughgarden, Algorithmic Game Theory, Lecture 11: Selfish Routing and the Price of Anarchy.

Lecture 12: More on Selfish Routing

12.1 Network over-provisioning

Communication networks are often over-provisioned: since capacity is relatively cheap, networks are built with more capacity than needed, both to anticipate growth and because, empirically, networks with spare capacity suffer fewer packet drops and delays. Over-provisioning serves as an alternative to enforcing quality-of-service guarantees directly, for example by admission control. The theory of Lecture 11 corroborates the empirical observation in two ways.

12.2 POA bounds for over-provisioned networks

Suppose every cost function has the form

$$c_e(x) = \begin{cases} \dfrac{1}{u_e - x} & \text{if } x < u_e, \\ +\infty & \text{if } x \geq u_e, \end{cases} \tag{12.1}$$

where u_e is the capacity of edge e. This is the expected delay in an M/M/1 queue, in which jobs arrive as a Poisson process with rate x and have independent exponential service times with mean $1/u_e$. It is the first and simplest model of delay in communication networks. It stays very flat until the load nears the capacity, and then rapidly tends to $+\infty$ (Figure 16(a)).

For $\beta \in (0,1)$, a network with M/M/1 delay functions is β-over-provisioned if $f_e \leq (1-\beta)u_e$ for every edge e, where f is an equilibrium flow; that is, the maximum link utilization at equilibrium is at most $(1-\beta)\cdot 100\%$. When β is not too close to 0, the cost functions on the relevant range behave like low-degree polynomials, which have small POA.

Theorem 12.1 [Over-provisioned networks]. *The POA of every β-over-provisioned network with M/M/1 delay functions is at most*

$$\frac{1}{2}\left(1 + \sqrt{\frac{1}{\beta}}\right), \tag{12.2}$$

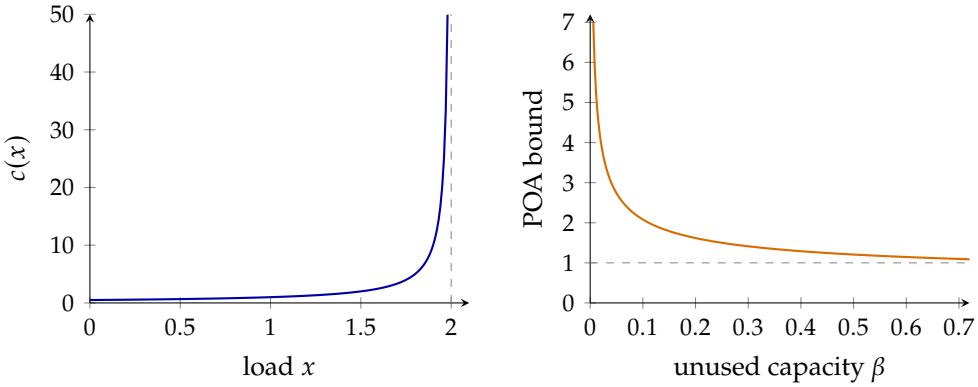
and simple networks show that this bound is tight for every $\beta \in (0,1)$.

Proof. The proof of the main theorem of Lecture 11 uses the Pigou bound only with $c = c_e$, $r = f_e$, and $x = f_e^*$. Here $f_e \leq (1-\beta)u_e$, so it suffices to bound the POA of a Pigou-like network whose lower edge has $c(x) = 1/(u-x)$ and whose rate satisfies $r \leq (1-\beta)u$. Minimize $g(x) = x/(u-x) + (r-x)/(u-r)$: setting $g'(x) = u/(u-x)^2 - 1/(u-r) = 0$ gives $u - x = \sqrt{u(u-r)}$.

With $a = \sqrt{u}$ and $b = \sqrt{u - r}$, this is $x = a(a - b)$, and both terms of g equal $(a - b)/b$, while $r\,c(r) = (a^2 - b^2)/b^2$. So the POA of this network is

$$\frac{(a^2 - b^2)/b^2}{2(a - b)/b} = \frac{a + b}{2b} = \frac{1}{2}\left(1 + \sqrt{\frac{u}{u - r}}\right) \le \frac{1}{2}\left(1 + \sqrt{\frac{1}{\beta}}\right),$$

since $u - r \ge \beta u$. □



(a) (b)

Figure 16: (a) The M/M/1 delay function $c(x) = 1/(u - x)$ for capacity $u = 2$. (b) The worst-case POA bound $\frac{1}{2}(1 + \sqrt{1/\beta})$ as a function of the fraction β of unused capacity.

The bound tends to 1 as $\beta \to 1$ and to $+\infty$ as $\beta \to 0$, where the cost functions act like constants and like very high-degree polynomials, respectively. In between, for $\beta = 0.1$ (maximum utilization 90%) the POA is at most $\frac{1}{2}(1 + \sqrt{10}) \approx 2.1$. A little over-provisioning suffices for near-optimal selfish routing, as Internet Service Providers observe.

12.3 A resource augmentation bound

With arbitrary cost functions the POA is unbounded, by the nonlinear Pigou example (upper edge $c(x) = 1$, lower edge $c(x) = x^d$). A different guarantee compares the equilibrium with an optimal flow forced to route extra traffic. In that example, with one unit the equilibrium costs 1 and the optimum nearly 0. With two units the optimum has nowhere to hide: it routes about $1 - \epsilon$ units on the lower edge and $1 + \epsilon$ on the upper edge, for total cost more than 1.

Theorem 12.2 [Resource augmentation]. *For every selfish routing network and traffic rate r, the cost of an equilibrium flow with rate r is at most the cost of an optimal flow with rate $2r$.*

Equivalently, the equilibrium in a "faster" network, with each $c_e(x)$ replaced by $c_e(x/2)/2$, costs no more than the optimal flow in the original network at the same rate. For M/M/1 delays,

$c_e(x) = 1/(u_e - x)$ becomes $1/(2u_e - x)$, an edge with double the capacity. So a modest technology upgrade improves performance more than dictatorial control.

Proof. Let f be an equilibrium flow at rate r and f^* an optimal flow at rate $2r$. As in Lecture 11, freeze the costs at $c_e(f_e)$. Every path used by f has a common cost L, and $c_P(f) \geq L$ for all paths, so

$$\sum_{P \in \mathcal{P}} f_P \cdot c_P(f) = r \cdot L \qquad \text{and} \qquad \sum_{P \in \mathcal{P}} f^*_P \cdot c_P(f) \geq 2r \cdot L.$$

With respect to the frozen costs, f^* costs at least twice as much as f. It remains to show that freezing overestimates the cost of f^* by at most the cost of f:

$$\underbrace{\sum_{e \in E} f^*_e \cdot c_e(f^*_e)}_{C(f^*)} \geq \underbrace{\sum_{e \in E} f^*_e \cdot c_e(f_e)}_{\geq 2rL} - \underbrace{\sum_{e \in E} f_e \cdot c_e(f_e)}_{=rL},$$

which then gives $C(f^*) \geq rL = C(f)$. It suffices to prove, for every edge,

$$f^*_e \left[c_e(f_e) - c_e(f^*_e) \right] \leq f_e \cdot c_e(f_e). \tag{12.3}$$

If $f^*_e \geq f_e$, the left side is nonpositive since c_e is nondecreasing. If $f^*_e < f_e$, the left side is the area of a rectangle of width f^*_e and height $c_e(f_e) - c_e(f^*_e)$, contained in the rectangle of width f_e and height $c_e(f_e)$ that gives the right side (Figure 17). □

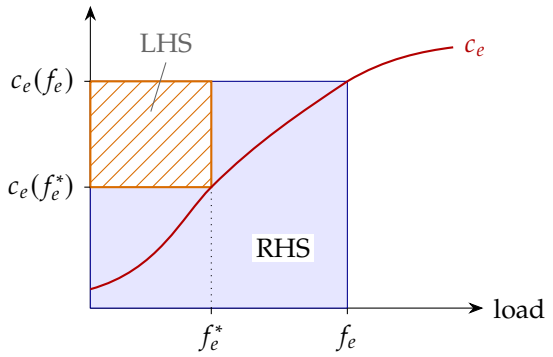
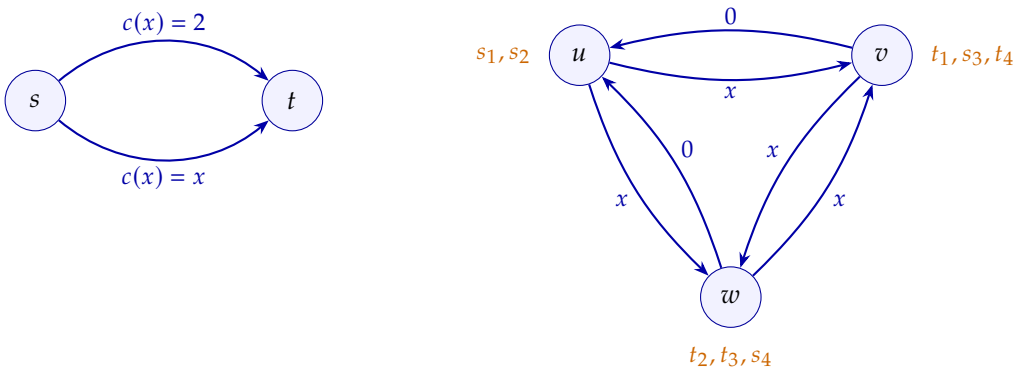


Figure 17: Proof of (12.3) when $f^*_e < f_e$: the hatched rectangle (left side) lies inside the shaded rectangle (right side). The curve is a schematic nondecreasing c_e.

12.4 Atomic selfish routing

So far the model has been NONATOMIC: every player has negligible size, which fits cars on a highway but not an Internet Service Provider controlling much of the traffic. An ATOMIC SELFISH ROUTING NETWORK has k players; player i has a source s_i and a sink t_i (possibly shared with others), routes one unit of traffic on a single s_i–t_i path, and minimizes its cost. Flows, equilibrium flows (no player can decrease its cost by a unilateral deviation), and the cost of a flow are defined as before.



(a) (b)

Figure 18: (a) An atomic Pigou-like network with two players. (b) The bidirected triangle with four players; each player i routes from s_i to t_i, and edge labels are cost functions.

Example: In Figure 18(a), with two players, routing one player on each edge costs $1 + 2 = 3$, which is optimal. It is also an equilibrium: the lower player would pay 2 on the upper edge, and the upper player would raise the lower edge's cost to 2. Both players on the lower edge is a second equilibrium, with cost 4, since either player would still pay 2 by switching.

Different equilibria can thus have different costs, unlike in the nonatomic model. So the PRICE OF ANARCHY of an atomic network is defined with the worst equilibrium:

$$\text{POA} = \frac{\text{cost of worst equilibrium}}{\text{cost of optimal outcome}},$$

which is $\frac{4}{3}$ in this example.

The POA can also exceed its nonatomic counterpart. In the bidirected triangle of Figure 18(b), each player has a one-hop and a two-hop path. When all four use their one-hop paths, each of the four edges with $c(x) = x$ carries one player, the cost is 4, and this is optimal and an equilibrium. When all four use their two-hop paths, the edges (u, w) and (u, v) each carry two players and (w, v) and (v, w) one each, so players 1 and 2 pay 3 and players 3 and 4 pay 2, for a total of 10. This is also an equilibrium: switching to the one-hop path would cost players 1 and 2 exactly 3 and players 3 and 4 exactly 2. So the POA is $10/4 = \frac{5}{2}$, and no example with affine cost functions is worse.

Theorem 12.3 [POA OF ATOMIC SELFISH ROUTING, AFFINE COSTS**].** *In every atomic selfish routing network with affine cost functions, the POA is at most* $\frac{5}{2}$.

Proof. Fix an equilibrium f and an optimal flow f^*, and let f_e and f_e^* be the numbers of players using e in each. Write $c_e(x) = a_e x + b_e$ with $a_e, b_e \geq 0$, and let player i use P_i in f and P_i^* in f^*.
Step 1: let f^* suggest deviations. Since f is an equilibrium, i's cost does not decrease if it switches

to P_i^*, which adds one unit of load to the edges of $P_i^* \setminus P_i$:

$$\sum_{e \in P_i} c_e(f_e) \le \sum_{e \in P_i^* \cap P_i} c_e(f_e) + \sum_{e \in P_i^* \setminus P_i} c_e(f_e + 1) \le \sum_{e \in P_i^*} c_e(f_e + 1),$$

using that cost functions are nondecreasing.

Step 2: sum over players. The term $c_e(f_e + 1)$ appears once for each player whose path P_i^* contains e, that is, f_e^* times, so

$$C(f) = \sum_{i=1}^{k} \sum_{e \in P_i} c_e(f_e) \le \sum_{e \in E} f_e^* \, c_e(f_e + 1) = \sum_{e \in E} \left[a_e f_e^* (f_e + 1) + b_e f_e^* \right]. \tag{12.4}$$

Step 3: disentangle. For all nonnegative integers y, z,

$$y(z + 1) \le \frac{5}{3} y^2 + \frac{1}{3} z^2, \tag{12.5}$$

with equality when $y = z = 1$ and when $y = 1$, $z = 2$. Indeed, for $y = 0$ it is trivial; for $y = 1$ it reads $z^2 - 3z + 2 = (z - 1)(z - 2) \ge 0$, true for integers; and for $y \ge 2$, minimizing $\frac{1}{3} z^2 - yz$ over real z (at $z = \frac{3}{2} y$) shows the slack is at least $\frac{11}{12} y^2 - y > 0$. Applying (12.5) with $y = f_e^*$ and $z = f_e$ in (12.4),

$$C(f) \le \sum_{e \in E} \left[a_e \left(\frac{5}{3} (f_e^*)^2 + \frac{1}{3} f_e^2 \right) + b_e f_e^* \right] \le \frac{5}{3} \sum_{e \in E} f_e^* (a_e f_e^* + b_e) + \frac{1}{3} \sum_{e \in E} a_e f_e^2 \le \frac{5}{3} C(f^*) + \frac{1}{3} C(f).$$

Subtracting $\frac{1}{3} C(f)$ and multiplying by $\frac{3}{2}$ gives $C(f) \le \frac{5}{2} C(f^*)$. □

This is a "canonical" POA proof: apply the equilibrium condition to the deviations suggested by an optimal outcome, sum over players, and disentangle the resulting mixed terms.

Source. Tim Roughgarden, Algorithmic Game Theory, Lecture 12: More on Selfish Routing.

Lecture 13: Potential Games; A Hierarchy of Equilibria

Every pure Nash equilibrium of an atomic selfish routing game with affine cost functions $c_e(x) = a_e x + b_e$ ($a_e, b_e \geq 0$) costs at most $\frac{5}{2}$ times an optimal outcome (Lecture 12). Games such as Rock-Paper-Scissors have no pure Nash equilibrium, so this guarantee is meaningful only if an equilibrium exists. This lecture proves existence for routing games and introduces several equilibrium concepts, asking when each exists, when it is tractable to compute, and why one might prefer one over another.

13.1 Potential games and Rosenthal's theorem

Theorem 13.1 [Rosenthal's theorem]. *Every atomic selfish routing game, with arbitrary real-valued cost functions, has at least one equilibrium flow.*

The proof shows that the game is a potential game: the players are inadvertently and collectively optimizing a single function. This is one of the only general tools for guaranteeing the existence of pure Nash equilibria in a class of games. Define the potential function on flows by

$$\Phi(f) = \sum_{e \in E} \sum_{i=1}^{f_e} c_e(i), \tag{13.1}$$

where f_e is the number of players whose path in f includes e. The inner sum is the area under the staircase of c_e, while the cost objective has the term $f_e \cdot c_e(f_e)$, the bounding box (Figure 19). This similarity between the potential and the objective is useful later.

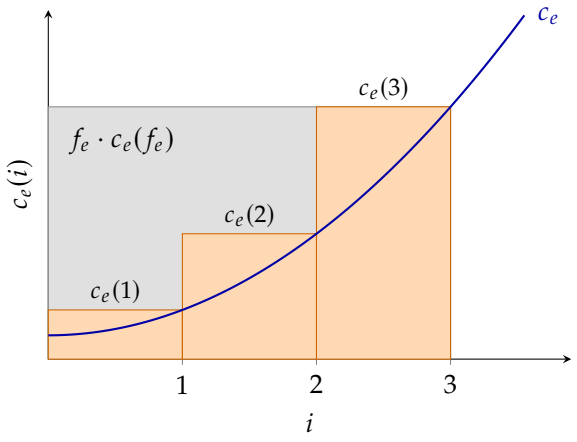


Figure 19: Schematic of one edge with $f_e = 3$. The bars of height $c_e(1), c_e(2), c_e(3)$ make up the potential term $\sum_{i=1}^{f_e} c_e(i)$, which underestimates the box $f_e \cdot c_e(f_e)$ in the cost objective when c_e is nondecreasing.

The defining property is the following. Let player i use the s_i–t_i path P_i in f, let it deviate to another s_i–t_i path $\hat{P}_i$, and let $\hat{f}$ be the resulting flow. Then

$$\Phi(\hat{f}) - \Phi(f) = \sum_{e \in \hat{P}_i} c_e(\hat{f}_e) - \sum_{e \in P_i} c_e(f_e), \tag{13.2}$$

so the change in the potential under a unilateral deviation equals the change in the deviator's cost. One function tracks the effect of deviations by every player.

Proof of Theorem 13.1. First verify (13.2). Edges in both or neither of P_i and $\hat{P}_i$ keep their load and contribute to neither side. An edge $e \in \hat{P}_i \setminus P_i$ has $\hat{f}_e = f_e + 1$, so its inner sum in (13.1) gains the term $c_e(f_e + 1)$; an edge $e \in P_i \setminus \hat{P}_i$ loses its final term $c_e(f_e)$. Both sides of (13.2) therefore equal

$$\sum_{e \in \hat{P}_i \setminus P_i} c_e(f_e + 1) - \sum_{e \in P_i \setminus \hat{P}_i} c_e(f_e).$$

There are only finitely many flows, so some flow f minimizes Φ. No unilateral deviation from f decreases Φ, so by (13.2) no player can decrease its cost by a unilateral deviation, and f is an equilibrium flow. □

13.2 Extensions

The same argument proves several other results.

- The proof never uses that cost functions are nondecreasing, so it holds for arbitrary cost functions, including the decreasing ones of games with positive externalities (Lecture 15).
- The proof never uses the network structure. It applies to CONGESTION GAMES, in which there is an abstract set E of resources, each with a cost function, and each player i has an arbitrary collection $S_i \subseteq 2^E$ of strategies, each a subset of resources.
- For nonatomic selfish routing (Lecture 11), players have negligible size and the inner sum becomes an integral:

  $$\Phi(f) = \sum_{e \in E} \int_0^{f_e} c_e(x)\, dx,$$

  where f_e is the amount of traffic on e. With continuous nondecreasing cost functions, Φ is continuously differentiable and convex, and its first-order optimality conditions are precisely the equilibrium conditions of a nonatomic flow. Since Φ is continuous and the set of flows is compact, Φ has a global minimum, which is an equilibrium flow; this proves existence. Since Φ is convex, its local minima are global minima, so equilibria are essentially unique: when Φ has several global minima, the corresponding equilibrium flows all have the same total cost.

13.3 A hierarchy of equilibrium concepts

Besides games like Rock-Paper-Scissors, atomic selfish routing games in which players have different sizes need not have pure Nash equilibria, even with two players and quadratic cost functions. A meaningful POA analysis of such games needs a larger set of equilibria with guaranteed existence. The rest of the lecture defines three relaxations of pure Nash equilibria, each more permissive and more computationally tractable than the previous one (Figure 20). All three exist in every finite game.

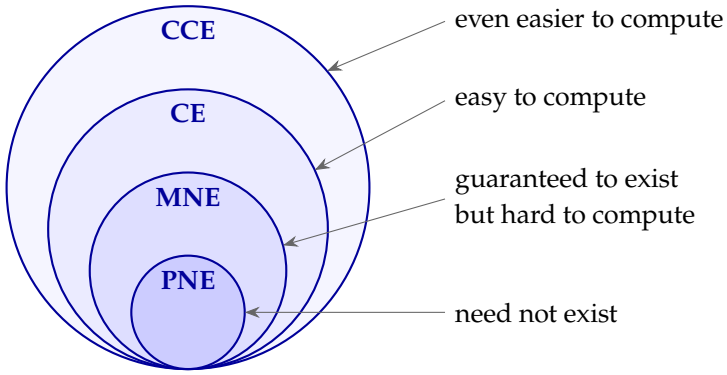


Figure 20: The hierarchy of equilibrium concepts: pure Nash (PNE), mixed Nash (MNE), correlated (CE), and coarse correlated (CCE) equilibria.

A COST-MINIMIZATION GAME has a finite number k of players, a finite strategy set S_i for each player i, and a cost function $C_i(\mathbf{s})$ for each player i, where $\mathbf{s} \in S_1 \times \cdots \times S_k$ is a STRATEGY PROFILE or OUTCOME. For example, in an atomic routing game $C_i(\mathbf{s})$ is i's travel time on its chosen path, given the paths $\mathbf{s}_{-i}$ of the other players. The concepts below are conventionally defined for payoff-maximization games, with all inequalities reversed; the two formulations are equivalent.

Definition 13.2 [PURE NASH EQUILIBRIUM]. A strategy profile $\mathbf{s}$ is a PURE NASH EQUILIBRIUM (PNE) if for every player $i \in \{1, 2, \ldots, k\}$ and every unilateral deviation $s_i' \in S_i$,

$$C_i(\mathbf{s}) \leq C_i(s_i', \mathbf{s}_{-i}). \tag{13.3}$$

PNE are easy to interpret but need not exist. The POA of PNE is left undefined in games without one.

Definition 13.3 [MIXED NASH EQUILIBRIUM]. Distributions $\sigma_1, \ldots, \sigma_k$ over $S_1, \ldots, S_k$ form a MIXED NASH EQUILIBRIUM (MNE) if for every player i and every unilateral deviation $s_i' \in S_i$,

$$\mathbb{E}_{\mathbf{s} \sim \sigma}[C_i(\mathbf{s})] \leq \mathbb{E}_{\mathbf{s} \sim \sigma}[C_i(s_i', \mathbf{s}_{-i})], \tag{13.4}$$

where σ is the product distribution $\sigma_1 \times \cdots \times \sigma_k$.

Allowing mixed-strategy deviations does not change the definition. Every PNE is an MNE in which each player plays deterministically, and Rock-Paper-Scissors shows that a game can have MNE that are not PNE. By Nash's theorem, every cost-minimization game has at least one MNE, so the POA of MNE is well defined in every finite game. However, computing an MNE appears to be computationally intractable, even with two players. If players cannot be expected to reach an equilibrium quickly, performance guarantees for that equilibrium are less meaningful, which motivates still more permissive and tractable concepts.

Definition 13.4 [Correlated equilibrium]. A distribution σ on $S_1 \times \cdots \times S_k$ is a correlated equilibrium (CE) if for every player i, every strategy $s_i \in S_i$, and every deviation $s_i' \in S_i$,

$$\mathbb{E}_{\mathbf{s}\sim\sigma}[C_i(\mathbf{s}) \mid s_i] \le \mathbb{E}_{\mathbf{s}\sim\sigma}[C_i(s_i', \mathbf{s}_{-i}) \mid s_i]. \tag{13.5}$$

The distribution σ need not be a product distribution, so the players' strategies are correlated. The MNE are exactly the CE that are product distributions, so CE exist in every finite game. In the standard interpretation, σ is public and a trusted third party samples $\mathbf{s} \sim \sigma$ and privately suggests s_i to each player i. Player i knows σ and s_i, and hence has a posterior distribution on the others' suggestions $\mathbf{s}_{-i}$. Condition (13.5) says that, assuming the others follow their suggestions, following the suggestion minimizes i's conditional expected cost.

A traffic light is a CE that is not an MNE. In the two-player game below (costs of the row and column player), each player would rather go if the other stops, and both going causes an accident.

	stop	go
stop	$0, 0$	$0, 1$
go	$1, 0$	$-5, -5$

The PNE are (stop, go) and (go, stop). Let σ randomize 50/50 between them. It is not a product distribution, so it is not an MNE. It is a CE: a row player told to go (a green light) knows the column player was told to stop, and if the column player stops, going is a best response; a row player told to stop knows the column player will go, and stopping is then a best response. The column player is symmetric. Unlike MNE, CE are computationally tractable.

Definition 13.5 [Coarse correlated equilibrium]. A distribution σ on $S_1 \times \cdots \times S_k$ is a coarse correlated equilibrium (CCE) if for every player i and every unilateral deviation $s_i' \in S_i$,

$$\mathbb{E}_{\mathbf{s}\sim\sigma}[C_i(\mathbf{s})] \le \mathbb{E}_{\mathbf{s}\sim\sigma}[C_i(s_i', \mathbf{s}_{-i})]. \tag{13.6}$$

A player contemplating a deviation s_i' in (13.6) knows only σ, not its own component s_i of the realization: a CCE protects only against unconditional unilateral deviations, while a CE protects against deviations conditioned on s_i. Every CE is a CCE, since averaging (13.5) over s_i with weights $\mathbb{P}_{\mathbf{s}\sim\sigma}[s_i]$ gives (13.6). Hence CCE exist in every finite game and are computationally tractable, and simple learning dynamics reach them.

13.4 An example with strict inclusions

Consider an atomic selfish routing game with four players, a common source s, a common sink t, and six parallel s–t edges $E = \{0, 1, 2, 3, 4, 5\}$, each with cost function $c(x) = x$. A player's cost is 1 plus the number of other players on its edge.

- *PNE.* The PNE are the $6 \cdot 5 \cdot 4 \cdot 3 = 360$ outcomes in which the players choose distinct edges; every player has cost 1.
- *MNE that is not a PNE.* Each player independently chooses a uniformly random edge. Each of the other three players shares i's edge with probability $\frac{1}{6}$, so every player has expected cost $1 + \frac{3}{6} = \frac{3}{2}$.
- *CE that is not an MNE.* Let σ be uniform over the outcomes in which one edge has two players and two edges have one player each. Given its suggestion s_i, player i shares its edge with probability $\frac{1}{2}$ (it is in the pair for 3 of the 6 possible pairs), so its conditional expected cost is $\frac{3}{2}$. The other three players then occupy the other five edges symmetrically, with 2 or 3 of them off i's edge according as i is or is not in the pair, so a deviation $s_i' \neq s_i$ meets on average $\frac{1}{2} \cdot \frac{2}{5} + \frac{1}{2} \cdot \frac{3}{5} = \frac{1}{2}$ other players and also costs $\frac{3}{2}$. Both sides of (13.5) equal $\frac{3}{2}$. No outcome in the support has four distinct edges, so σ is not a product distribution.
- *CCE that is not a CE.* Restrict σ to the outcomes whose set of chosen edges is $\{0, 2, 4\}$ or $\{1, 3, 5\}$, uniformly. Each player is still in the pair with probability $\frac{1}{2}$, so its expected cost is $\frac{3}{2}$. The distribution is invariant under shifting every edge by 1 modulo 6, so each edge carries on average $\frac{3}{6} = \frac{1}{2}$ of the other three players, and every unconditional deviation also costs $\frac{3}{2}$. Both sides of (13.6) equal $\frac{3}{2}$. It is not a CE: a player suggested edge s_i can deviate to the successive edge $s_i + 1$ (modulo 6), which is never used by the others in this case, and reduce its conditional expected cost to 1.

So all inclusions of Figure 20 can be strict.

The benefit of enlarging the set of equilibria is increased tractability and plausibility. The cost is that fewer desirable properties hold: the POA compares the worst equilibrium to an optimal outcome, so a larger set of equilibria can only have a worse POA. The question is whether some equilibrium concept is large enough to be tractable yet small enough to permit strong worst-case guarantees.

Source. Tim Roughgarden, Algorithmic Game Theory, Lecture 13: Potential Games; A Hierarchy of Equilibria.

Lecture 14: Robust Price-of-Anarchy Bounds in Smooth Games

14.1 Canonical POA proofs

The proof that the POA of every atomic selfish routing game with affine cost functions is at most $\frac{5}{2}$ (Lecture 12) has four steps.

1. Given a PNE $\mathbf{s}$ and an optimal outcome $\mathbf{s}^*$, invoke the PNE hypothesis once per player i, with the hypothetical deviation s_i^*, to get $C_i(\mathbf{s}) \leq C_i(s_i^*, \mathbf{s}_{-i})$. This is the only use of the PNE hypothesis.
2. Sum the k inequalities. The left side is the cost of $\mathbf{s}$; the right side is an entangled function of $\mathbf{s}$ and $\mathbf{s}^*$, with terms of the form $f_e f_e^*$.
3. Relate the entangled term $\sum_{i=1}^k C_i(s_i^*, \mathbf{s}_{-i})$ to the two quantities of interest, here bounding it by $\frac{5}{3}\,\text{cost}(\mathbf{s}^*) + \frac{1}{3}\,\text{cost}(\mathbf{s})$. This is the hardest step, but it is just algebra and does not use that $\mathbf{s}$ is a PNE or that $\mathbf{s}^*$ is optimal.
4. Solve for the POA: subtract $\frac{1}{3}\,\text{cost}(\mathbf{s})$ and multiply by $\frac{3}{2}$ to get a bound of $\frac{5}{2}$.

Many other POA proofs follow this recipe. The point of this lecture is that every such proof automatically gives a robust POA bound, one that applies not only to all PNE but also to all coarse correlated equilibria, the largest set of equilibria in Lecture 13.

14.2 A location game

A location game has the following ingredients.

- A set F of possible locations, for example for a Web cache in a network or a shop in a neighborhood.
- A set of k players. Each player i chooses one location from a set $F_i \subseteq F$ from which to provide a service. All players provide the same service, and a player can serve any number of markets.
- A set M of markets. Each market $j \in M$ has a publicly known value v_j, its maximum willingness to pay for the service.
- For each location $\ell \in F$ and market $j \in M$, a known cost $c_{\ell j}$ of serving j from ℓ.

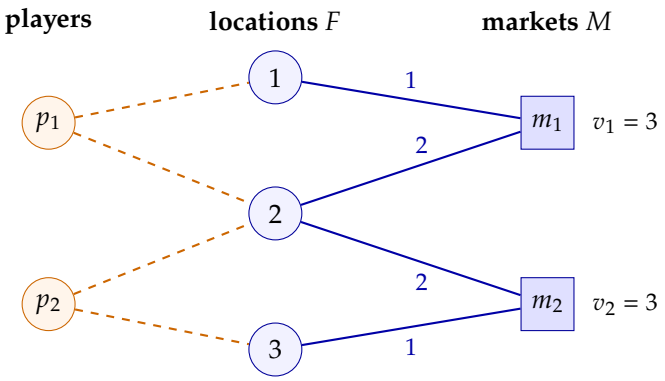


Figure 21: A location game with locations $F = \{1, 2, 3\}$, markets $M = \{1, 2\}$, and two players with $F_1 = \{1, 2\}$ and $F_2 = \{2, 3\}$ (dashed). Solid edges give the service costs $c_{\ell j}$; missing pairs have infinite cost.

In the example of Figure 21, if player 1 chooses location 1 and player 2 chooses location 3, each has a monopoly on its market, charges the full value 3, and has payoff $3 - 1 = 2$. If player 1 switches to location 2, it keeps its monopoly on market 1 with payoff $3 - 2 = 1$. In market 2, player 2 can no longer charge 3: at any price above 2, player 1 would undercut it. So player 2 charges 2 and has payoff $2 - 1 = 1$.

In general, let C be the set of chosen locations in a strategy profile $\mathbf{s}$, and let player i choose $\ell \in C$. Its payoff is $\pi_i(\mathbf{s}) = \sum_{j \in M} \pi_{ij}(\mathbf{s})$, where

$$\pi_{ij}(\mathbf{s}) = \begin{cases} 0 & \text{if } c_{\ell j} \geq v_j \text{ or } \ell \text{ is not the closest location of } C \text{ to } j, \\ d_j^{(2)}(\mathbf{s}) - c_{\ell j} & \text{otherwise,} \end{cases} \tag{14.1}$$

and $d_j^{(2)}(\mathbf{s})$, the highest price i can get away with, is the minimum of v_j and the second-smallest cost between a location of C and j. So each market is served by the provider with the lowest service cost, at the highest competitive price, and $\pi_{ij}(\mathbf{s})$ is i's competitive advantage for market j, capped at v_j minus the service cost. These are payoff-maximization games; their POA is at most 1, and the closer to 1 the better.

The objective is the SOCIAL SURPLUS

$$V(\mathbf{s}) = \sum_{j \in M} \big(v_j - d_j(\mathbf{s})\big), \tag{14.2}$$

where $d_j(\mathbf{s})$ is the minimum of v_j and the smallest cost between a chosen location and j: each market is served by the cheapest chosen location, or not at all if that cost is at least v_j. The surplus depends on $\mathbf{s}$ only through the set of chosen locations, and (14.2) defines $V(T)$ for every set T of locations.

Theorem 14.1 [POA OF LOCATION GAMES**].** *The POA of every location game is at least* $\frac{1}{2}$.

The proof uses only three properties of location games.

(P1) For every $\mathbf{s}$, $\sum_{i=1}^{k} \pi_i(\mathbf{s}) \le V(\mathbf{s})$. Each market j contributes $v_j - d_j(\mathbf{s})$ to the surplus and $d_j^{(2)}(\mathbf{s}) - d_j(\mathbf{s})$ to the payoff of the closest location, and $d_j^{(2)}(\mathbf{s}) \le v_j$.

(P2) For every $\mathbf{s}$, $\pi_i(\mathbf{s}) = V(\mathbf{s}) - V(\mathbf{s}_{-i})$: a player's payoff is exactly the extra surplus created by its location. Market j contributes $d_j(\mathbf{s}_{-i}) - d_j(\mathbf{s})$ to the right side. This is zero unless i's location is the closest to j in $\mathbf{s}$, in which case it is

$$d_j^{(2)}(\mathbf{s}) - d_j(\mathbf{s}). \tag{14.3}$$

This is 0 if $c_{\ell j} \ge v_j$ (both terms equal v_j) and $d_j^{(2)}(\mathbf{s}) - c_{\ell j}$ otherwise. Either way it equals $\pi_{ij}(\mathbf{s})$; summing over j gives (P2).

(P3) V is MONOTONE, meaning $V(T_1) \le V(T_2)$ whenever $T_1 \subseteq T_2$, and SUBMODULAR, meaning

$$V(T_2 \cup \{\ell\}) - V(T_2) \le V(T_1 \cup \{\ell\}) - V(T_1) \tag{14.4}$$

for every location ℓ and all $T_1 \subseteq T_2$. Monotonicity is evident from (14.2). For submodularity, as in (14.3), adding ℓ to T increases the contribution of market j by $\max\{0, d_j(T) - c_{\ell j}\}$, where $d_j(T)$ is the minimum of v_j and the smallest cost from T to j. This is nonzero only if ℓ is closer to j than every location of T, and since $d_j(T)$ only decreases as T grows, so does the increase.

Property (P2) shows that V is a potential function, so every location game is a potential game. The proof below needs only the inequality $\pi_i(\mathbf{s}) \ge V(\mathbf{s}) - V(\mathbf{s}_{-i})$; games with this inequality, (P1), and (P3) are called VALID UTILITY GAMES.

Proof of Theorem 14.1. Follow the four steps. Let $\mathbf{s}$ be a PNE and $\mathbf{s}^*$ a surplus-maximizing outcome. Since $\mathbf{s}$ is a PNE, $\pi_i(\mathbf{s}) \ge \pi_i(s_i^*, \mathbf{s}_{-i})$ for every player i. Summing and using (P1),

$$V(\mathbf{s}) \ge \sum_{i=1}^{k} \pi_i(\mathbf{s}) \ge \sum_{i=1}^{k} \pi_i(s_i^*, \mathbf{s}_{-i}). \tag{14.5}$$

To disentangle the right side, use (P2) and then submodularity, adding the locations $s_1^*, \dots, s_{i-1}^*$ and s_i to the set to which s_i^* is added:

$$\pi_i(s_i^*, \mathbf{s}_{-i}) = V(s_i^*, \mathbf{s}_{-i}) - V(\mathbf{s}_{-i}) \ge V(s_1^*, \dots, s_i^*, \mathbf{s}) - V(s_1^*, \dots, s_{i-1}^*, \mathbf{s}).$$

The sum over i telescopes, and monotonicity gives

$$\sum_{i=1}^{k} \pi_i(s_i^*, \mathbf{s}_{-i}) \ge V(s_1^*, \dots, s_k^*, s_1, \dots, s_k) - V(\mathbf{s}) \ge V(\mathbf{s}^*) - V(\mathbf{s}).$$

With (14.5), $V(\mathbf{s}) \ge V(\mathbf{s}^*) - V(\mathbf{s})$, so $V(\mathbf{s})/V(\mathbf{s}^*) \ge \frac{1}{2}$. □

14.3 Smooth games

The following definition abstracts the third, disentanglement step of these two proofs.

Definition 14.2 [Smooth games].

1. A cost-minimization game is (λ, μ)-smooth if

$$\sum_{i=1}^{k} C_i(s_i^*, \mathbf{s}_{-i}) \leq \lambda \cdot \text{cost}(\mathbf{s}^*) + \mu \cdot \text{cost}(\mathbf{s}) \tag{14.6}$$

   for all strategy profiles $\mathbf{s}, \mathbf{s}^*$, where the objective satisfies $\text{cost}(\mathbf{s}) \leq \sum_{i=1}^{k} C_i(\mathbf{s})$ for every $\mathbf{s}$.

2. A payoff-maximization game is (λ, μ)-smooth if

$$\sum_{i=1}^{k} \pi_i(s_i^*, \mathbf{s}_{-i}) \geq \lambda \cdot V(\mathbf{s}^*) - \mu \cdot V(\mathbf{s}) \tag{14.7}$$

   for all strategy profiles $\mathbf{s}, \mathbf{s}^*$, where the objective satisfies $V(\mathbf{s}) \geq \sum_{i=1}^{k} \pi_i(\mathbf{s})$ for every $\mathbf{s}$.

Every atomic selfish routing game with affine cost functions is a $(\frac{5}{3}, \frac{1}{3})$-smooth cost-minimization game, with $\text{cost}(\mathbf{s}) = \sum_i C_i(\mathbf{s})$ holding with equality, and every location game is a $(1, 1)$-smooth payoff-maximization game, the objective condition being (P1). The disentanglement steps of both proofs were written with $\mathbf{s}$ a PNE and $\mathbf{s}^*$ optimal, but the algebra never used those facts, so it applies to all pairs of strategy profiles.

Note. All consequences below continue to hold if (14.6) holds for some optimal outcome $\mathbf{s}^*$ and all strategy profiles $\mathbf{s}$.

14.4 Robust POA bounds in smooth games

In a (λ, μ)-smooth cost-minimization game with $\mu < 1$, every PNE $\mathbf{s}$ has cost at most $\frac{\lambda}{1-\mu}$ times that of an optimal outcome $\mathbf{s}^*$. Indeed, by the objective assumption, the PNE condition once per player, and smoothness,

$$\text{cost}(\mathbf{s}) \leq \sum_{i=1}^{k} C_i(\mathbf{s}) \leq \sum_{i=1}^{k} C_i(s_i^*, \mathbf{s}_{-i}) \leq \lambda \cdot \text{cost}(\mathbf{s}^*) + \mu \cdot \text{cost}(\mathbf{s}), \tag{14.8}$$

and rearranging gives the bound. Similarly, every PNE of a (λ, μ)-smooth payoff-maximization game has objective value at least $\frac{\lambda}{1+\mu}$ times optimal. These generalize the bounds of $\frac{5}{2}$ and $\frac{1}{2}$ above.

The first sense in which these bounds are robust concerns coarse correlated equilibria (CCE):

distributions σ on $S_1 \times \cdots \times S_k$ such that for every player i and every unilateral deviation $s_i' \in S_i$,

$$\mathbb{E}_{\mathbf{s}\sim\sigma}[C_i(\mathbf{s})] \le \mathbb{E}_{\mathbf{s}\sim\sigma}[C_i(s_i', \mathbf{s}_{-i})]. \tag{14.9}$$

Simple learning algorithms drive the time-averaged history of joint play to the set of CCE, so CCE are highly tractable. In smooth games they are also stringent enough for good worst-case bounds.

Theorem 14.3 [Robust POA bound]. *In every (λ, μ)-smooth cost-minimization game with $\mu < 1$, the POA of coarse correlated equilibria is at most $\frac{\lambda}{1-\mu}$.*

Proof. Let σ be a CCE and $\mathbf{s}^*$ an optimal outcome. Then

$$\begin{aligned}
\mathbb{E}_{\mathbf{s}\sim\sigma}[\text{cost}(\mathbf{s})] &\le \mathbb{E}_{\mathbf{s}\sim\sigma}\left[\sum_{i=1}^{k} C_i(\mathbf{s})\right] = \sum_{i=1}^{k} \mathbb{E}_{\mathbf{s}\sim\sigma}[C_i(\mathbf{s})] \\
&\le \sum_{i=1}^{k} \mathbb{E}_{\mathbf{s}\sim\sigma}[C_i(s_i^*, \mathbf{s}_{-i})] = \mathbb{E}_{\mathbf{s}\sim\sigma}\left[\sum_{i=1}^{k} C_i(s_i^*, \mathbf{s}_{-i})\right] \\
&\le \mathbb{E}_{\mathbf{s}\sim\sigma}[\lambda \cdot \text{cost}(\mathbf{s}^*) + \mu \cdot \text{cost}(\mathbf{s})] = \lambda \cdot \text{cost}(\mathbf{s}^*) + \mu \cdot \mathbb{E}_{\mathbf{s}\sim\sigma}[\text{cost}(\mathbf{s})].
\end{aligned}$$

The first inequality is the objective assumption, the equalities are linearity of expectation, the second inequality is the CCE condition (14.9) once per player with deviation s_i^*, and the last inequality is smoothness. Rearranging completes the proof. □

Likewise, the bound $\frac{\lambda}{1+\mu}$ of smooth payoff-maximization games applies to all CCE. So the bounds of $\frac{5}{2}$ and $\frac{1}{2}$ above hold for all CCE, because their proofs established smoothness.

The second sense of robustness is that the bound degrades gracefully for approximate equilibria. An ϵ-pure Nash equilibrium (ϵ-PNE) of a cost-minimization game is a strategy profile $\mathbf{s}$ in which no player can decrease its cost by more than a $(1+\epsilon)$ factor through a unilateral deviation:

$$C_i(\mathbf{s}) \le (1+\epsilon) \cdot C_i(s_i', \mathbf{s}_{-i}) \qquad \text{for every } i \text{ and } s_i' \in S_i. \tag{14.10}$$

Theorem 14.4 [Approximate equilibria]. *For every (λ, μ)-smooth cost-minimization game, every $\epsilon < \frac{1}{\mu} - 1$, every ϵ-PNE $\mathbf{s}$, and every outcome $\mathbf{s}^*$,*

$$\text{cost}(\mathbf{s}) \le \frac{(1+\epsilon)\lambda}{1 - \mu(1+\epsilon)} \cdot \text{cost}(\mathbf{s}^*).$$

Proof. As in (14.8), but invoking (14.10) once per player, $\text{cost}(\mathbf{s}) \le (1+\epsilon)\sum_{i=1}^{k} C_i(s_i^*, \mathbf{s}_{-i}) \le (1+\epsilon)\big(\lambda \cdot \text{cost}(\mathbf{s}^*) + \mu \cdot \text{cost}(\mathbf{s})\big)$. Rearrange, using $\mu(1+\epsilon) < 1$. □

Similar results hold for smooth payoff-maximization games and for approximate versions of other equilibrium concepts. For example, atomic selfish routing games with affine cost functions are

$(\frac{5}{3}, \frac{1}{3})$-smooth, so for $\epsilon < 2$ every ϵ-PNE has cost at most $\frac{5+5\epsilon}{2-\epsilon}$ times that of an optimal outcome.

Source. Tim Roughgarden, Algorithmic Game Theory, Lecture 14: Robust Price-of-Anarchy Bounds in Smooth Games.

Lecture 15: Best-Case and Strong Nash Equilibria

The externality caused by a player is the difference between its own objective function value and its contribution to the social objective. The models so far have negative externalities: a player in a routing game ignores the extra cost its presence imposes on others using the same edges. Games with positive externalities, such as joining a social network, also benefit the other members. This lecture studies positive externalities in a model of network formation, where different pure Nash equilibria can have wildly different costs. This motivates restricting attention to a subset of "reasonable" equilibria that ideally satisfies better worst-case bounds and has a plausible narrative for why its equilibria deserve study. Two partially successful approaches follow.

15.1 Network cost-sharing games

A network cost-sharing game takes place in a directed or undirected graph $G = (V, E)$ in which each edge e has a fixed cost $\gamma_e \geq 0$, independent of the number of users. There are k players; player i has a source s_i and sink t_i, and its strategies are the s_i–t_i paths. An outcome $\mathbf{P} = (P_1, \ldots, P_k)$ forms the subnetwork $(V, \cup_{i=1}^{k} P_i)$. The $f_e = |\{j : e \in P_j\}|$ players using an edge split its cost equally, so

$$C_i(\mathbf{P}) = \sum_{e \in P_i} \frac{\gamma_e}{f_e}, \qquad \text{cost}(\mathbf{P}) = \sum_{e \in E:\, f_e \geq 1} \gamma_e = \sum_{i=1}^{k} C_i(\mathbf{P}). \tag{15.1}$$

The objective is the total cost of the formed network.

15.2 Two examples

VHS or Betamax. In Figure 22, k players share a source s and a sink t. The optimal outcome sends everyone on the upper edge, at total cost $1 + \epsilon$, and this is a Nash equilibrium. All players on the lower edge is also a Nash equilibrium: each pays $k/k = 1$, while a unilateral deviation to the upper edge costs the full $1 + \epsilon$. So the price of anarchy can be as large as k, the number of players. The POA is distracted by a manifestly unreasonable equilibrium and gives no useful information.

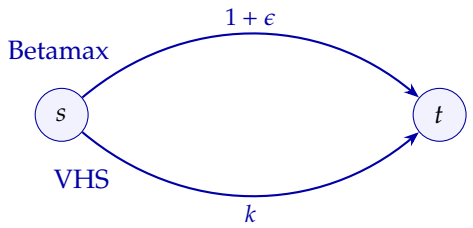


Figure 22: VHS or Betamax. The price of anarchy of a network cost-sharing game can be as large as the number k of players.

Opting out. In Figure 23, players have distinct sources $s_1, \ldots, s_k$ and a common sink t. They can meet at the rendezvous v and continue together to t at joint cost $1 + \epsilon$, or *opt out* and take the direct s_i–t edge, which costs $1/i$. The optimal outcome routes everyone through v, at cost $1 + \epsilon$, but this is not an equilibrium: player k pays $(1 + \epsilon)/k > 1/k$ through v in every outcome, so opting out is a dominant strategy for it. Given that player k opts out, player $k - 1$ pays at least $(1 + \epsilon)/(k - 1)$ through v and prefers to opt out, and so on. This iterated deletion of dominated strategies leaves a single outcome, so everyone opting out is the unique Nash equilibrium. Its cost is the kth harmonic number $\mathcal{H}_k = \sum_{i=1}^{k} \frac{1}{i}$, which lies between $\ln k$ and $\ln k + 1$.

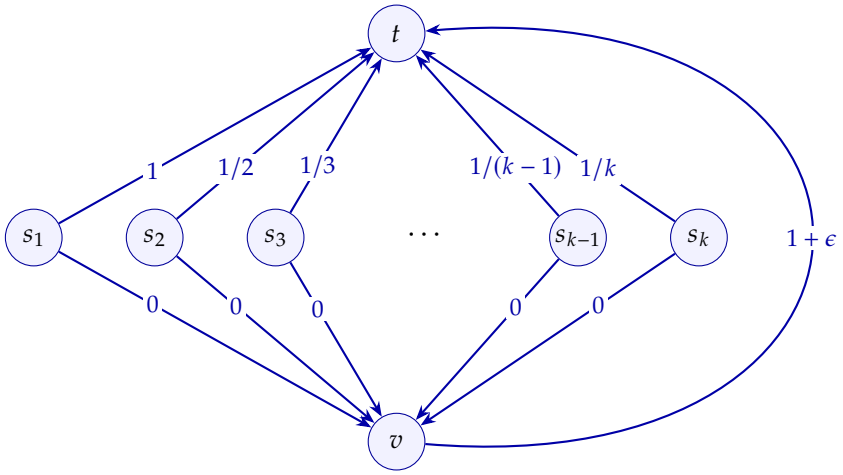


Figure 23: Opting out. The unique Nash equilibrium, in which every player takes its direct edge to t, costs $\mathcal{H}_k$ times as much as an optimal outcome.

The POA here is $\mathcal{H}_k$, much smaller than k, but the example is more compelling: the inefficiency does not come from multiple or unreasonable equilibria, and it captures the efficiency loss from underparticipation in games with positive externalities.

15.3 The price of stability

Definition 15.1 [Price of stability]. The price of stability (POS) is the ratio between the cost of the best Nash equilibrium and the cost of an optimal outcome.

The POS is the optimistic version of the POA. It is a much weaker guarantee, but it is well motivated when a third party can propose an initial outcome, as when setting default parameters of software or a network protocol so that users are not incentivized to change them.

Theorem 15.2 [POS of network cost-sharing games]. *In every network cost-sharing game with k players, there is a pure Nash equilibrium with cost at most $\mathcal{H}_k$ times that of an optimal outcome.*

In particular, every network cost-sharing game has a pure Nash equilibrium, and the opt-out

example shows that $\mathcal{H}_k$ cannot be replaced by anything smaller.

Proof. A network cost-sharing game has the same form as an atomic selfish routing game, with per-player cost γ_e/f_e on an edge with f_e users; the positive externalities make these cost functions decreasing. Rosenthal's potential (Lecture 13) specializes to

$$\Phi(\mathbf{P}) = \sum_{e \in E} \sum_{i=1}^{f_e} \frac{\gamma_e}{i} = \sum_{e \in E} \gamma_e \sum_{i=1}^{f_e} \frac{1}{i}. \tag{15.2}$$

The proof that a minimizer of the potential is a Nash equilibrium works for any cost functions, nondecreasing or not, so a minimizer $\mathbf{P}$ of Φ is a pure Nash equilibrium. In the VHS or Betamax example, the low-cost equilibrium minimizes Φ and the high-cost one does not.
The potential approximates the objective: an edge with $f_e = 0$ contributes nothing to either, and an edge with $1 \le f_e \le k$ satisfies $\gamma_e \le \gamma_e \sum_{i=1}^{f_e} \frac{1}{i} \le \gamma_e \mathcal{H}_k$. Hence

$$\text{cost}(\mathbf{P}) \le \Phi(\mathbf{P}) \le \mathcal{H}_k \cdot \text{cost}(\mathbf{P}) \tag{15.3}$$

for every $\mathbf{P}$. Let $\mathbf{P}^*$ be an outcome of minimum cost. Then

$$\text{cost}(\mathbf{P}) \le \Phi(\mathbf{P}) \le \Phi(\mathbf{P}^*) \le \mathcal{H}_k \cdot \text{cost}(\mathbf{P}^*),$$

using (15.3) twice and the choice of $\mathbf{P}$ as the minimizer of Φ. □

The minimizer of the potential need not be the lowest-cost Nash equilibrium; the proof shows only that it is within $\mathcal{H}_k$ of optimal.

15.4 Strong Nash equilibria and their POA

An alternative approach keeps the worst-case analysis but first identifies a strict subset of the pure Nash equilibria. In games with negative externalities, the POA of PNE is close to 1 and the analysis *zooms out* to ever-larger sets of equilibria. Here worst-case Nash equilibria can be highly suboptimal, so the analysis must *zoom in*. In the VHS or Betamax example, the high-cost equilibrium survives unilateral deviations, but if two players deviated jointly to the upper edge, each would pay only about $\frac{1}{2} < 1$.

Definition 15.3 [Strong Nash equilibrium]. Let $\mathbf{s}$ be an outcome of a cost-minimization game.

(a) Strategies $\mathbf{s}'_A \in \prod_{i \in A} S_i$ are a beneficial deviation for a set A of players if $C_i(\mathbf{s}'_A, \mathbf{s}_{-A}) \le C_i(\mathbf{s})$ for every $i \in A$, with strict inequality for at least one player of A.

(b) The outcome $\mathbf{s}$ is a strong Nash equilibrium if no coalition of players has a beneficial deviation.

Nash equilibria are the special case in which only singleton coalitions may deviate, so every strong

Nash equilibrium is a Nash equilibrium: strong Nash equilibrium is an EQUILIBRIUM REFINEMENT. In VHS or Betamax, the high-cost equilibrium is not strong and the low-cost one is. Since the coalition of all players may deviate, one might expect strong Nash equilibria to be optimal. This holds when all players share the same source and sink, but not in general: the argument showing that everyone opting out is the unique Nash equilibrium also shows that it is a strong Nash equilibrium, with cost $\mathcal{H}_k$ times optimal. No worse example is possible.

Theorem 15.4 [POA OF STRONG NASH EQUILIBRIA**].** *In every network cost-sharing game with k players, every strong Nash equilibrium has cost at most $\mathcal{H}_k$ times that of an optimal outcome.*

This guarantee covers every strong Nash equilibrium rather than one Nash equilibrium, but, unlike Theorem 15.2, it does not assert existence. The two theorems are therefore incomparable.

Proof. Let $\mathbf{P}$ be a strong Nash equilibrium and $\mathbf{P}^*$ an optimal outcome. For a set A of players, write $\mathbf{P}^*_A$ for the outcome in which only the players of A are present, each using its path in $\mathbf{P}^*$. The coalition $A_k = \{1, \ldots, k\}$ does not deviate beneficially to $\mathbf{P}^*$, so some player i has $C_i(\mathbf{P}) \le C_i(\mathbf{P}^*)$; rename the players so that this is player k. Next, the coalition $A_{k-1} = \{1, \ldots, k-1\}$ does not deviate beneficially to $\mathbf{P}^*_{A_{k-1}}$, so some player $i \in A_{k-1}$ has $C_i(\mathbf{P}) \le C_i(\mathbf{P}^*_{A_{k-1}}, P_k)$; rename so that this is player $k-1$, and continue. This gives a renaming in which, for every i and $A_i = \{1, \ldots, i\}$,

$$C_i(\mathbf{P}) \le C_i(\mathbf{P}^*_{A_i}, \mathbf{P}_{-A_i}). \tag{15.4}$$

Summing over the players,

$$\text{cost}(\mathbf{P}) = \sum_{i=1}^{k} C_i(\mathbf{P}) \le \sum_{i=1}^{k} C_i(\mathbf{P}^*_{A_i}, \mathbf{P}_{-A_i}) \le \sum_{i=1}^{k} C_i(\mathbf{P}^*_{A_i}).$$

The second inequality uses positive externalities: deleting the players outside A_i only decreases each f_e, and hence only increases the cost shares of the remaining players.
Now use the potential (15.2). Deleting a player decreases the potential by exactly that player's cost; this is Rosenthal's identity for a deviation to the empty strategy. Writing f_e^i for the number of players of A_i whose path in $\mathbf{P}^*$ contains e,

$$C_i(\mathbf{P}^*_{A_i}) = \sum_{e \in P_i^*} \frac{\gamma_e}{f_e^i} = \Phi(\mathbf{P}^*_{A_i}) - \Phi(\mathbf{P}^*_{A_{i-1}}).$$

The sum telescopes, and by (15.3),

$$\text{cost}(\mathbf{P}) \le \sum_{i=1}^{k} \left[\Phi(\mathbf{P}^*_{A_i}) - \Phi(\mathbf{P}^*_{A_{i-1}})\right] = \Phi(\mathbf{P}^*) \le \mathcal{H}_k \cdot \text{cost}(\mathbf{P}^*). \qquad \square$$

Unlike previous POA analyses of potential games, this proof uses the potential function itself. Both approaches give a bound of $\mathcal{H}_k$ rather than k. Strong Nash equilibria exist in network cost-sharing

games with sufficiently simple network structure, but not in general. In the two-player game of Figure 24, each player has two paths. Checking the four outcomes, the only Nash equilibrium has player 1 on $s_1 \to v \to t_1$ and player 2 on $s_2 \to v \to t_2$, each paying 5. It is not strong: jointly switching to $s_1 \to s_2 \to v \to t_1$ and $s_2 \to v \to t_1 \to t_2$ shares the edges (s_2, v) and (v, t_1), and each player pays $1 + \frac{3}{2} + \frac{3}{2} = \frac{3}{2} + \frac{3}{2} + 1 = 4$.

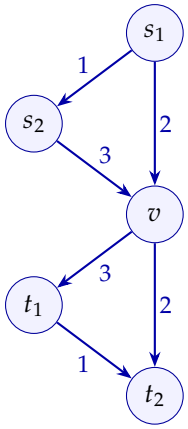


Figure 24: A network cost-sharing game with no strong Nash equilibrium. Player i travels from s_i to t_i; edge labels are the costs γ_e.

Source. Tim Roughgarden, Algorithmic Game Theory, Lecture 15: Best-Case and Strong Nash Equilibria.

Lecture 16: Best-Response Dynamics

This lecture begins the study of three questions. Do strategic players reach an equilibrium of a game, even in principle? If so, does it happen quickly? And by which learning algorithms? Affirmative answers justify equilibrium analysis: properties of equilibria, such as a near-optimal objective value, are not obviously relevant when players fail to find one. Answering requires a behavioral model, or dynamics, for players out of equilibrium. Ideally the same conclusion follows from several simple and natural learning processes, so that it is robust to the details of any particular one.

16.1 Best-response dynamics

Best-response dynamics is a straightforward search for a pure Nash equilibrium (PNE).

Algorithm 16.1: Best-response dynamics

Input: A game and an initial outcome $\mathbf{s}$.
Output: A PNE, if the procedure halts.

while the current outcome $\mathbf{s}$ is not a PNE **do**
  Pick an arbitrary player i and an arbitrary beneficial deviation s_i' for i
  $\mathbf{s} \leftarrow (s_i', \mathbf{s}_{-i})$

The choices of i and s_i' are left unspecified and specialized later. (This version is sometimes called better-response dynamics, with best-response dynamics reserved for the version in which s_i' is chosen to optimize i's payoff given $\mathbf{s}_{-i}$.) Best-response dynamics can halt only at a PNE, so it cycles in every game without one; it can also cycle in games that have a PNE.

It is a perfect fit for potential games. A potential game admits a real-valued function Φ on outcomes such that for every outcome $\mathbf{s}$, every player i, and every unilateral deviation s_i',

$$\Phi(s_i', \mathbf{s}_{-i}) - \Phi(\mathbf{s}) = C_i(s_i', \mathbf{s}_{-i}) - C_i(\mathbf{s}), \tag{16.1}$$

where C_i is player i's cost (or negative payoff). Routing games, location games, and network cost-sharing games are all potential games. A minimizer of Φ is a PNE; best-response dynamics gives a more constructive proof.

Proposition 16.1 [Convergence in potential games]. *In a finite potential game, from an arbitrary initial outcome, best-response dynamics converges to a PNE.*

Proof. In every iteration the deviator's cost strictly decreases, so by (16.1) the potential strictly decreases, and no outcome repeats. Since the game is finite, the dynamics halts, necessarily at a PNE. □

How fast does this happen? Three notions of fast convergence follow, from strongest to weakest. The strongest is convergence to a PNE in a polynomial number of iterations, polynomial in the number of players and the total number of strategies. (The number of outcomes is exponential in the number of players, so the trivial bound by the number of outcomes is uninteresting.) This holds when Φ takes only polynomially many distinct values, for example when it is integer-valued and polynomially bounded. In general, however, Φ can take exponentially many values and best-response dynamics can decrease it very slowly, requiring exponentially many iterations.

16.2 Fast convergence to ϵ-PNE in symmetric routing games

The second notion settles for an approximate equilibrium.

Definition 16.2 [ϵ-pure Nash equilibrium]. For $\epsilon \in [0, 1]$, an outcome $\mathbf{s}$ of a cost-minimization game is an *ϵ-pure Nash equilibrium* (ϵ-PNE) if for every player i and deviation $s_i' \in S_i$,

$$C_i(s_i', \mathbf{s}_{-i}) \geq (1 - \epsilon) \cdot C_i(\mathbf{s}). \tag{16.2}$$

This is the definition of Lecture 14 reparametrized: an ϵ-PNE here is an $\frac{\epsilon}{1-\epsilon}$-PNE there. An *$\epsilon$-move* for player i is a deviation s_i' with $C_i(s_i', \mathbf{s}_{-i}) < (1 - \epsilon)C_i(\mathbf{s})$, and *$\epsilon$-best-response dynamics* permits only such significant improvements, which is the key to converging much faster.

Algorithm 16.2: ϵ-best-response dynamics

Input: A game, an initial outcome $\mathbf{s}$, and $\epsilon \in [0, 1]$.
Output: An ϵ-PNE.

```
while the current outcome s is not an ε-PNE do
    Pick an arbitrary player i that has an ε-move, and an arbitrary ε-move s_i' for i
    s ← (s_i', s_{-i})
```

It can halt only at an ϵ-PNE and, by the argument of Proposition 16.1, converges in every finite potential game.

Theorem 16.3 [Convergence of ϵ-best-response dynamics]. *Consider an atomic selfish routing game in which:*

1. *all players have a common source vertex and a common sink vertex;*
2. *cost functions satisfy the α-bounded jump condition: $c_e(x + 1) \in [c_e(x), \alpha \cdot c_e(x)]$ for every edge e and positive integer x;*
3. *the MaxGain variant of ϵ-best-response dynamics is used: in every iteration, among players with an ϵ-move, the player who can obtain the largest absolute cost decrease moves to its minimum-cost deviation.*

Then an ϵ-PNE is reached in $O\left(\frac{k\alpha}{\epsilon} \log \frac{\Phi(\mathbf{s}^0)}{\Phi_{\min}}\right)$ iterations, where $\mathbf{s}^0$ is the initial outcome and $\Phi_{\min} > 0$ is the minimum potential value.

The first two hypotheses are necessary: if either is dropped, all variants of best-response dynamics take an exponential number of iterations in the worst case. In the third, it matters that ϵ-best-response dynamics is used rather than standard best-response dynamics, but the result holds for all natural variants of ϵ-best-response dynamics; essentially, every player needs the opportunity to move sufficiently often.

The plan is to strengthen the proof of Proposition 16.1 quantitatively, showing that every iteration decreases the potential by a lot. The first lemma provides a player with high cost, whose move would decrease the potential significantly. The second, which uses the hypotheses, shows that the player actually chosen to move has a cost decrease within an α factor of that.

Lemma 16.4 [A HIGH-COST PLAYER]. *In every outcome* $\mathbf{s}$*, some player* i *has* $C_i(\mathbf{s}) \geq \Phi(\mathbf{s})/k$.

Proof. The objective of a selfish routing game is $\text{cost}(\mathbf{s}) = \sum_{e \in E} f_e \cdot c_e(f_e)$, where f_e is the number of players using e, and the potential is $\Phi(\mathbf{s}) = \sum_{e \in E} \sum_{i=1}^{f_e} c_e(i)$. Since cost functions are nondecreasing, $\Phi(\mathbf{s}) \leq \text{cost}(\mathbf{s})$. Since $\text{cost}(\mathbf{s}) = \sum_{i=1}^{k} C_i(\mathbf{s})$, some player has at least the average cost:

$$\max_{i=1}^{k} C_i(\mathbf{s}) \geq \frac{\text{cost}(\mathbf{s})}{k} \geq \frac{\Phi(\mathbf{s})}{k}.$$

□

Lemma 16.5 [THE MOVER GAINS ENOUGH]. *Suppose MaxGain* ϵ*-best-response dynamics chooses player* i *in the outcome* $\mathbf{s}$*, and* i *takes the* ϵ*-move* s_i'*. Then for every other player* j*,*

$$C_i(\mathbf{s}) - C_i(s_i', \mathbf{s}_{-i}) \geq \frac{\epsilon}{\alpha} C_j(\mathbf{s}). \tag{16.3}$$

Since an ϵ-move for j decreases its cost by at least $\epsilon C_j(\mathbf{s})$, the lemma says that, up to a factor of α, the decrease enjoyed by i is at least that of any other player taking an ϵ-move, whether or not one is available.

Proof. Fix j. If j has an ϵ-move, it would decrease j's cost by at least $\epsilon C_j(\mathbf{s})$, and MaxGain chose i over j, so (16.3) holds even without the α (note $\alpha \geq 1$).
Otherwise j has no ϵ-move. All players have the same strategy set, so s_i' is available to j, and

$$C_i(s_i', \mathbf{s}_{-i}) < (1-\epsilon) C_i(\mathbf{s}) \quad \text{while} \quad C_j(s_i', \mathbf{s}_{-j}) \geq (1-\epsilon) C_j(\mathbf{s}). \tag{16.4}$$

The outcomes $(s_i', \mathbf{s}_{-i})$ and $(s_i', \mathbf{s}_{-j})$ share at least $k-1$ strategies: s_i', played by i in the first and by j in the second, and the $k-2$ strategies of the players other than i and j. So the load on every edge differs by at most one between them, and by the α-bounded jump condition every edge cost differs by at most a factor of α. Comparing the costs of the player using s_i' in each,

$$C_j(s_i', \mathbf{s}_{-j}) \leq \alpha \cdot C_i(s_i', \mathbf{s}_{-i}).$$

Chaining this with (16.4), $(1-\epsilon)C_j(\mathbf{s}) \le \alpha(1-\epsilon)C_i(\mathbf{s})$, so $C_j(\mathbf{s}) \le \alpha C_i(\mathbf{s})$. Since s_i' is an ϵ-move,

$$C_i(\mathbf{s}) - C_i(s_i', \mathbf{s}_{-i}) \ge \epsilon\, C_i(\mathbf{s}) \ge \frac{\epsilon}{\alpha}\, C_j(\mathbf{s}). \qquad \square$$

Proof of Theorem 16.3. In an iteration in which player i makes the ϵ-move s_i',

$$\Phi(\mathbf{s}) - \Phi(s_i', \mathbf{s}_{-i}) = C_i(\mathbf{s}) - C_i(s_i', \mathbf{s}_{-i}) \ge \frac{\epsilon}{\alpha} \max_{j=1}^{k} C_j(\mathbf{s}) \ge \frac{\epsilon}{\alpha k}\, \Phi(\mathbf{s}),$$

by the potential property, Lemma 16.5 (which also holds for $j = i$, as $\alpha \ge 1$), and Lemma 16.4. So every iteration multiplies the potential by at most $1 - \frac{\epsilon}{\alpha k}$. Since $(1-x)^{1/x} \le (e^{-x})^{1/x} = \frac{1}{e}$ for $x \in (0,1)$, every $\frac{k\alpha}{\epsilon}$ iterations decrease the potential by a constant factor. The potential starts at $\Phi(\mathbf{s}^0)$ and never drops below $\Phi_{\min}$, so the dynamics converges within $O\big(\frac{k\alpha}{\epsilon} \log \frac{\Phi(\mathbf{s}^0)}{\Phi_{\min}}\big)$ iterations. $\square$

The theorem justifies equilibrium analysis in atomic selfish routing games with a shared source and sink, but it cannot be extended much further: with multiple sources and sinks, ϵ-best-response dynamics can require exponentially many iterations, however the deviating player and deviation are chosen in each iteration.

16.3 Fast convergence to low-cost outcomes in smooth potential games

The third notion asks only to quickly reach outcomes whose objective value is as good as if the players had converged to an approximate equilibrium. This does not imply convergence to an approximate equilibrium, but when the main reason for equilibrium analysis is a POA bound, it is a costless surrogate. It applies to all smooth potential games (Lecture 14), including routing games with arbitrary sources, sinks, and cost functions, and location games. Recall that (λ, μ)-smoothness means $\sum_{i=1}^{k} C_i(s_i^*, \mathbf{s}_{-i}) \le \lambda \cdot \text{cost}(\mathbf{s}^*) + \mu \cdot \text{cost}(\mathbf{s})$ for every pair of outcomes $\mathbf{s}, \mathbf{s}^*$.

The dynamics is MaxGain best-response dynamics, without the restriction to ϵ-moves: each iteration chooses the player i and deviation s_i' maximizing $C_i(\mathbf{s}) - C_i(s_i', \mathbf{s}_{-i})$.

Theorem 16.6 [Fast convergence to low-cost outcomes]. *Consider a (λ, μ)-smooth cost-minimization game with $\mu < 1$ and a positive potential function Φ satisfying $\Phi(\mathbf{s}) \le \text{cost}(\mathbf{s})$ for every outcome $\mathbf{s}$. Let $\mathbf{s}^0, \ldots, \mathbf{s}^T$ be generated by MaxGain best-response dynamics, let $\mathbf{s}^*$ be a minimum-cost outcome, and let $\eta \in (0,1)$. Then all but*

$$O\Big(\frac{k}{\eta(1-\mu)} \log \frac{\Phi(\mathbf{s}^0)}{\Phi_{\min}}\Big) \tag{16.5}$$

outcomes $\mathbf{s}^t$ satisfy

$$\text{cost}(\mathbf{s}^t) \le \frac{\lambda}{(1-\mu)(1-\eta)} \cdot \text{cost}(\mathbf{s}^*),$$

where $\Phi_{\min}$ is the minimum potential value of an outcome and k is the number of players.

For small η the guarantee is $\frac{\lambda}{1-\mu}(1 + O(\eta))$, the robust POA bound, so all but a few outcomes in the sequence are essentially as good as a PNE. These bad outcomes need not be consecutive: an iteration of best-response dynamics can increase the overall cost, so a good outcome can be followed by a bad one.

Proof. Call $\mathbf{s}^t$ *bad* if it violates the guarantee. The plan is to show that the potential decreases significantly at every bad state. For an outcome $\mathbf{s}^t$, let $\delta_i(\mathbf{s}^t) = C_i(\mathbf{s}^t) - C_i(s_i^*, \mathbf{s}_{-i}^t)$ be the cost decrease i would get by switching to s_i^*, and let $\Delta(\mathbf{s}^t) = \sum_{i=1}^k \delta_i(\mathbf{s}^t)$. This is nonpositive at a PNE but can in general have either sign. By smoothness,

$$\text{cost}(\mathbf{s}^t) \leq \sum_{i=1}^k C_i(\mathbf{s}^t) = \sum_{i=1}^k \left[C_i(s_i^*, \mathbf{s}_{-i}^t) + \delta_i(\mathbf{s}^t)\right] \leq \lambda \cdot \text{cost}(\mathbf{s}^*) + \mu \cdot \text{cost}(\mathbf{s}^t) + \Delta(\mathbf{s}^t),$$

and hence

$$\text{cost}(\mathbf{s}^t) \leq \frac{\lambda}{1-\mu} \cdot \text{cost}(\mathbf{s}^*) + \frac{1}{1-\mu}\Delta(\mathbf{s}^t). \tag{16.6}$$

The cost of an outcome is large only when the players have a lot to gain from unilateral deviations. At a bad state, $\frac{\lambda}{1-\mu}\text{cost}(\mathbf{s}^*) < (1-\eta)\,\text{cost}(\mathbf{s}^t)$, so by (16.6) and since Φ underestimates the cost,

$$\Delta(\mathbf{s}^t) \geq \eta(1-\mu)\,\text{cost}(\mathbf{s}^t) \geq \eta(1-\mu)\Phi(\mathbf{s}^t).$$

A best response by player i decreases its cost by at least $\delta_i(\mathbf{s}^t)$, and some player has $\delta_i(\mathbf{s}^t) \geq \Delta(\mathbf{s}^t)/k$. So at a bad state the player chosen by MaxGain decreases its cost, and hence the potential, by at least $\frac{\eta(1-\mu)}{k}\Phi(\mathbf{s}^t)$:

$$\Phi(\mathbf{s}^{t+1}) \leq \left(1 - \frac{\eta(1-\mu)}{k}\right)\Phi(\mathbf{s}^t).$$

The potential never increases, so after every $\frac{k}{\eta(1-\mu)}$ bad states it has decreased by a constant factor. It starts at $\Phi(\mathbf{s}^0)$ and never drops below $\Phi_{\min}$, which bounds the number of bad states by (16.5). □

Source. Tim Roughgarden, Algorithmic Game Theory, Lecture 16: Best-Response Dynamics.

Lecture 17: No-Regret Dynamics

Best-response dynamics are most relevant for potential games. A second fundamental class of learning dynamics, no-regret dynamics, converges rapidly in *arbitrary* games to an approximate coarse correlated equilibrium (CCE), though not in general to a Nash equilibrium.

17.1 The regret-minimization problem

The regret-minimization problem concerns a single decision-maker playing against an adversary. There is a set A of $n \geq 2$ actions, and at each time $t = 1, 2, \ldots, T$:

1. the decision-maker picks a mixed strategy p^t, a probability distribution over A;
2. an adversary picks a cost vector $c^t : A \to [0, 1]$;
3. an action a^t is drawn from p^t, the decision-maker incurs cost $c^t(a^t)$, and it learns the entire cost vector c^t, not just the realized cost.

The important assumption is that costs are bounded. In a multi-player game, A will be one player's strategy set and c^t will be induced by the strategies of the other players.

Example: The best action *sequence* in hindsight, with cost $\sum_{t=1}^{T} \min_{a \in A} c^t(a)$, is too strong a benchmark. With $n = 2$, let the adversary set $c^t = (1, 0)$ if the algorithm plays the first action with probability at least $\frac{1}{2}$, and $c^t = (0, 1)$ otherwise. The algorithm's expected cost is at least $T/2$, while the best action sequence has cost 0.

So the benchmark is instead the best *fixed* action in hindsight, $\min_{a \in A} \sum_{t=1}^{T} c^t(a)$.

Definition 17.1 [Regret]. The (time-averaged) regret of the action sequence $a^1, \ldots, a^T$ with respect to the action a is

$$\frac{1}{T}\left[\sum_{t=1}^{T} c^t(a^t) - \sum_{t=1}^{T} c^t(a)\right]. \tag{17.1}$$

Definition 17.2 [No-regret algorithm]. Let $\mathcal{A}$ be an online decision-making algorithm.

(a) An adversary for $\mathcal{A}$ is a function that takes as input the day t, the mixed strategies $p^1, \ldots, p^t$ produced by $\mathcal{A}$ on the first t days, and the realized actions $a^1, \ldots, a^{t-1}$ of the first $t-1$ days, and outputs a cost vector $c^t : A \to [0, 1]$.

(b) $\mathcal{A}$ has no (external) regret if for every adversary for it, the expected regret (17.1) with respect to every action $a \in A$ is $o(1)$ as $T \to \infty$.

Such an adversary is adaptive; an oblivious adversary is the special case in which c^t depends only

on t (and on $\mathcal{A}$). Designing a no-regret algorithm is also called COMBINING EXPERT ADVICE: thinking of each action as an expert, a no-regret algorithm does asymptotically as well as the best expert. The number n of actions is fixed as $T \to \infty$, so the regret may depend on n but must tend to 0 as the horizon grows.

17.2 Lower bounds

Proposition 17.3 [RANDOMIZATION IS NECESSARY]. *No deterministic algorithm has no regret.*

Proof. A deterministic algorithm commits to a single action a^t at each step, and the adversary can compute it. Set $c^t(a^t) = 1$ and $c^t(a) = 0$ for every other action. The algorithm pays T. Each day the costs sum to 1 over the actions, so the total cost $\sum_a \sum_t c^t(a) = T$ and the best fixed action pays at most T/n. The regret with respect to that action is at least $1 - 1/n \geq \frac{1}{2}$, a constant. □

Proposition 17.4 [$\Omega(\sqrt{(\ln n)/T})$ LOWER BOUND]. *Even with $n = 2$ actions, for every algorithm there is an oblivious adversary for which the expected regret is $\Omega(1/\sqrt{T})$. With n actions, the expected regret cannot vanish faster than $\Theta(\sqrt{(\ln n)/T})$.*

Proof for $n = 2$. Let the adversary choose $(1, 0)$ or $(0, 1)$ independently and uniformly at random each day. Whatever the algorithm does, its cumulative expected cost is exactly $T/2$. The two actions' cumulative costs are H and $T - H$, where H is the number of heads in T fair coin flips; since H has standard deviation $\Theta(\sqrt{T})$, with constant probability one action has cumulative cost only $T/2 - \Theta(\sqrt{T})$. So over this distribution on 2^T oblivious adversaries, every algorithm has expected regret $\Omega(1/\sqrt{T})$, and hence some fixed adversary in the support forces expected regret $\Omega(1/\sqrt{T})$. A similar argument handles n actions. □

17.3 The multiplicative weights algorithm

Theorem 17.5 [NO-REGRET ALGORITHMS EXIST]. *There exist simple no-regret algorithms with expected regret $O(\sqrt{(\ln n)/T})$ with respect to every fixed action.*

Corollary 17.6. *There is an online decision-making algorithm that, for every $\epsilon > 0$, has expected regret at most ϵ with respect to every fixed action after $O((\ln n)/\epsilon^2)$ iterations.*

By Proposition 17.4, the bound of Theorem 17.5 is optimal up to the constant factor. The MULTIPLICATIVE WEIGHTS (MW) algorithm, also called Randomized Weighted Majority or Hedge, achieves it. It follows two principles: since play must be randomized, the probability of an action should decrease with its cumulative cost; and, for optimal regret, bad actions should be punished aggressively, so that when a previously good action turns sour its probability decreases at an

exponential rate. Each action keeps a weight, intuitively its credibility, which can only decrease.

Algorithm 17.1: Multiplicative weights (MW)

Input: Actions A with $|A| = n$, a parameter $\epsilon \in (0, \frac{1}{2}]$.

$w^1(a) \leftarrow 1$ for every $a \in A$

for $t \leftarrow 1$ **to** T **do**

  Play an action drawn from $p^t := w^t/\Gamma^t$, where $\Gamma^t = \sum_{a \in A} w^t(a)$

  Receive the cost vector c^t

  $w^{t+1}(a) \leftarrow w^t(a) \cdot (1-\epsilon)^{c^t(a)}$ for every $a \in A$

If all costs are 0 or 1, a weight either stays the same or is multiplied by $1-\epsilon$. As $\epsilon \to 0$, p^t tends to the uniform distribution (exploration); as $\epsilon \to 1$, p^t tends to put all its mass on the action with the smallest cumulative cost so far (exploitation). Other updates, such as $w^{t+1}(a) = w^t(a)(1-\epsilon c^t(a))$, also work.

17.4 Analysis of multiplicative weights

It suffices to consider oblivious adversaries. The distribution p^t is a deterministic function of $c^1, \dots, c^{t-1}$, independent of the realized actions, so a worst-case adversary gains nothing by conditioning on them, and the distributions $p^1, \dots, p^t$ are determined by its own earlier cost vectors. (Formally, backward induction on the worst adaptive adversary yields an oblivious one.) Fix $c^1, \dots, c^T$.

Proof of Theorem 17.5. The plan is to relate both the algorithm's expected cost and the cost of the best fixed action to the total weight Γ^{T+1} after the final update; the weights, and hence Γ^t, only decrease.

Step 1: a good fixed action keeps the final weight large. Let a^* be a best fixed action and $OPT := \sum_{t=1}^{T} c^t(a^*)$. Then

$$\Gamma^{T+1} \;\geq\; w^{T+1}(a^*) = \underbrace{w^1(a^*)}_{=1} \prod_{t=1}^{T} (1-\epsilon)^{c^t(a^*)} = (1-\epsilon)^{OPT}.$$

Step 2: Γ^t decreases exponentially with the algorithm's cost. The algorithm's expected cost at time t is

$$v^t := \sum_{a \in A} p^t(a)\, c^t(a) = \sum_{a \in A} \frac{w^t(a)}{\Gamma^t}\, c^t(a).$$

Since $(1-\epsilon)^x \leq 1 - \epsilon x$ for $\epsilon \in [0, \frac{1}{2}]$ and $x \in [0,1]$ (the left side is convex in x and agrees with the right side at $x = 0$ and $x = 1$),

$$\Gamma^{t+1} = \sum_{a \in A} w^t(a)(1-\epsilon)^{c^t(a)} \leq \sum_{a \in A} w^t(a)\big(1 - \epsilon c^t(a)\big) = \Gamma^t(1 - \epsilon v^t).$$

Step 3: combine. With $\Gamma^1 = n$,

$$(1-\epsilon)^{OPT} \le \Gamma^{T+1} \le n \prod_{t=1}^{T}(1-\epsilon v^t), \qquad \text{so} \qquad OPT \cdot \ln(1-\epsilon) \le \ln n + \sum_{t=1}^{T} \ln(1-\epsilon v^t).$$

From the Taylor expansion $\ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \cdots$, we have $\ln(1-\epsilon v^t) \le -\epsilon v^t$ (drop all terms but the first) and $\ln(1-\epsilon) \ge -\epsilon - \epsilon^2$ for $\epsilon \le \frac{1}{2}$ (keep the first two terms and double the second). Hence $OPT(-\epsilon - \epsilon^2) \le \ln n - \epsilon \sum_t v^t$, and since $OPT \le T$,

$$\sum_{t=1}^{T} v^t \le OPT(1+\epsilon) + \frac{\ln n}{\epsilon} \le OPT + \epsilon T + \frac{\ln n}{\epsilon}. \tag{17.2}$$

Choose $\epsilon = \sqrt{(\ln n)/T}$ to equalize the two error terms (this is at most $\frac{1}{2}$ once $T \ge 4 \ln n$). The expected cumulative cost of MW is then at most $OPT + 2\sqrt{T \ln n}$, and dividing by T gives expected regret at most $2\sqrt{(\ln n)/T}$ with respect to every fixed action. □

In particular, $T \ge 4(\ln n)/\epsilon^2$ iterations give expected regret at most ϵ, proving Corollary 17.6. If T is not known in advance, use $\epsilon = \sqrt{(\ln n)/\hat{T}}$ on day t, where $\hat{T}$ is the smallest power of 2 larger than t; the guarantee of Theorem 17.5 carries over.

17.5 No-regret dynamics and coarse correlated equilibria

Consider a cost-minimization game with players $1, \dots, k$, strategy sets S_i, and cost functions C_i. In each step $t = 1, 2, \dots, T$ of NO-REGRET DYNAMICS:

1. each player i simultaneously and independently chooses a mixed strategy p_i^t using a no-regret algorithm of its choice;
2. each player i receives the cost vector c_i^t with $c_i^t(s_i) = \mathbb{E}_{\mathbf{s}_{-i} \sim \sigma_{-i}}[C_i(s_i, \mathbf{s}_{-i})]$, the expected cost of s_i when every other player j plays p_j^t; here $\sigma_{-i} = \prod_{j \ne i} p_j^t$.

These dynamics are well defined by Theorem 17.5, and are efficient: with MW, each player updates one weight per strategy, and $O((\ln n)/\epsilon^2)$ iterations bring every player's expected regret to at most ϵ for every strategy, where n is the largest strategy set. The results below also extend to players moving sequentially.

Proposition 17.7 [NO-REGRET DYNAMICS CONVERGE TO CCE]. *Suppose after T iterations of no-regret dynamics every player of a cost-minimization game has regret at most ϵ for each of its strategies. Let $\sigma^t = \prod_{i=1}^{k} p_i^t$ be the outcome distribution at time t and $\sigma = \frac{1}{T}\sum_{t=1}^{T} \sigma^t$ the time-averaged history. Then σ is an ϵ-APPROXIMATE COARSE CORRELATED EQUILIBRIUM:*

$$\mathbb{E}_{\mathbf{s} \sim \sigma}[C_i(\mathbf{s})] \le \mathbb{E}_{\mathbf{s} \sim \sigma}[C_i(s_i', \mathbf{s}_{-i})] + \epsilon$$

for every player i and unilateral deviation s_i'.

Proof. By definition of σ,

$$\mathbb{E}_{\mathbf{s}\sim\sigma}[C_i(\mathbf{s})] = \frac{1}{T}\sum_{t=1}^{T}\mathbb{E}_{\mathbf{s}\sim\sigma^t}[C_i(\mathbf{s})] \quad \text{and} \quad \mathbb{E}_{\mathbf{s}\sim\sigma}[C_i(s_i', \mathbf{s}_{-i})] = \frac{1}{T}\sum_{t=1}^{T}\mathbb{E}_{\mathbf{s}\sim\sigma^t}[C_i(s_i', \mathbf{s}_{-i})].$$

The right-hand sides are player i's time-averaged expected cost when following its no-regret algorithm and when playing s_i' every day. Since its regret is at most ϵ, the first exceeds the second by at most ϵ. □

It matters here that the algorithms have no regret against *adaptive* adversaries. Player i's mixed strategy at time t affects the cost vectors the other players receive, hence their later strategies, hence the cost vectors player i receives later: other learning players act as an adaptive adversary.

Lecture 14 showed that POA bounds for (λ, μ)-smooth games, where $\sum_{i=1}^{k} C_i(s_i^*, \mathbf{s}_{-i}) \le \lambda \cdot cost(\mathbf{s}^*) + \mu \cdot cost(\mathbf{s})$ for all profiles $\mathbf{s}, \mathbf{s}^*$ and $cost(\mathbf{s}) \le \sum_i C_i(\mathbf{s})$, extend to CCE. With approximate equilibria they remain approximately correct.

Corollary 17.8 [POA OF NO-REGRET DYNAMICS**].** *In a (λ, μ)-smooth game with $\mu < 1$, suppose after T iterations of no-regret dynamics player i has expected regret at most R_i for each of its strategies. Then the time-averaged expected objective value satisfies*

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}_{\mathbf{s}\sim\sigma^t}[cost(\mathbf{s})] \le \frac{\lambda}{1-\mu}\, cost(\mathbf{s}^*) + \frac{\sum_{i=1}^{k} R_i}{1-\mu}.$$

As $T \to \infty$, $\sum_i R_i \to 0$ and the guarantee converges to the POA bound $\lambda/(1-\mu)$.

Proof. Write $\mathbf{s}^t \sim \sigma^t$. Using $cost \le \sum_i C_i$, then the regret bound of each player against the fixed strategy s_i^*, then smoothness,

$$\begin{aligned}\sum_{t=1}^{T}\mathbb{E}[cost(\mathbf{s}^t)] &\le \sum_{i=1}^{k}\sum_{t=1}^{T}\mathbb{E}[C_i(\mathbf{s}^t)] \le \sum_{t=1}^{T}\mathbb{E}\Big[\sum_{i=1}^{k} C_i(s_i^*, \mathbf{s}_{-i}^t)\Big] + T\sum_{i=1}^{k} R_i \\ &\le \sum_{t=1}^{T}\big(\lambda\, cost(\mathbf{s}^*) + \mu\,\mathbb{E}[cost(\mathbf{s}^t)]\big) + T\sum_{i=1}^{k} R_i.\end{aligned}$$

Divide by T and move the μ term to the left. □

Source. Tim Roughgarden, Algorithmic Game Theory, Lecture 17: No-Regret Dynamics.

Lecture 18: From External Regret to Swap Regret and the Minimax Theorem

Lecture 17 showed that coarse correlated equilibria (CCE), the biggest set in the hierarchy PNE $\subseteq$ MNE $\subseteq$ CE $\subseteq$ CCE, are tractable: simple learning procedures converge quickly to them. This lecture shows that the next-biggest set, the correlated equilibria (CE), is tractable in the same sense, and that mixed Nash equilibria of two-player zero-sum games are tractable too.

These equilibria, and MNE of two-player zero-sum games, can also be computed exactly in polynomial time by linear programming, but unlike learning procedures, linear programming is no model of how agents learn.

18.1 Swap regret and correlated equilibria

Recall from Lecture 13 that a correlated equilibrium of a cost-minimization game is a distribution σ over outcomes such that, for every player i with strategy set S_i and every switching function $\delta : S_i \to S_i$,

$$\mathbb{E}_{\mathbf{s}\sim\sigma}[C_i(\mathbf{s})] \leq \mathbb{E}_{\mathbf{s}\sim\sigma}[C_i(\delta(s_i), \mathbf{s}_{-i})].$$

For example, in the traffic intersection game of Lecture 13, mixing 50/50 between the two pure Nash equilibria gives a correlated equilibrium that is not a Nash equilibrium.

In the online decision-making setting of Lecture 17, each day $t = 1, \ldots, T$ a decision-maker commits to a distribution p^t over its n actions A, an adversary chooses $c^t : A \to [0, 1]$, and an action $a^t \sim p^t$ costs $c^t(a^t)$. External regret compares with every fixed action; the analog of correlated equilibria compares with every switching function.

Definition 18.1 [No swap regret]. An online decision-making algorithm has no swap regret if for every adversary for it, the expected swap regret

$$\frac{1}{T}\left[\sum_{t=1}^{T} c^t(a^t) - \sum_{t=1}^{T} c^t(\delta(a^t))\right] \tag{18.1}$$

with respect to every switching function $\delta : A \to A$ is $o(1)$ as $T \to \infty$.

Fixed actions are constant switching functions, so an algorithm with no swap regret also has no external regret. In no-swap-regret dynamics, each player i independently chooses p_i^t with a no-swap-regret algorithm, and cost vectors are defined as in no-regret dynamics: $c_i^t(s_i)$ is the expected cost of $s_i \in S_i$ when every other player j plays p_j^t.

Proposition 18.2 [No-swap-regret dynamics converge to CE]. *Suppose after T iterations of no-swap-regret dynamics every player of a cost-minimization game has swap regret at most ϵ for each of its switching functions. Let $\sigma^t = \prod_{i=1}^{k} p_i^t$ be the outcome distribution at time t and $\sigma = \frac{1}{T}\sum_{t=1}^{T} \sigma^t$. Then σ is an ϵ-approximate correlated equilibrium:*

$$\mathbb{E}_{\mathbf{s}\sim\sigma}[C_i(\mathbf{s})] \le \mathbb{E}_{\mathbf{s}\sim\sigma}[C_i(\delta(s_i), \mathbf{s}_{-i})] + \epsilon$$

for every player i and switching function $\delta : S_i \to S_i$.

Proof. As for no-regret dynamics in Lecture 17, with $\delta(s_i)$ in place of a fixed deviation s_i': both sides are time averages over t of expectations under σ^t, namely player i's expected cost when following its algorithm and when applying δ to its choices, and these differ by at most its swap regret ϵ. □

18.2 A black-box reduction from swap regret to external regret

Theorem 18.3 [Blum–Mansour]. *If there is a no-external-regret algorithm, then there is a no-swap-regret algorithm.*

The reduction preserves computational efficiency: plugging in multiplicative weights gives a polynomial-time no-swap-regret algorithm. So correlated equilibria are tractable in the same strong sense as CCE.

Let $M_1, \dots, M_n$ be n no-external-regret algorithms, for example n copies of multiplicative weights, one per action. Roughly, M_j is responsible for protecting against profitable deviations from action j to other actions. The master algorithm M is as follows; see Figure 25.

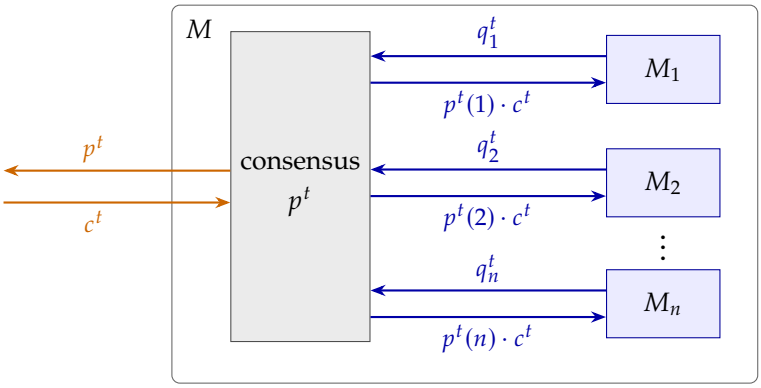


Figure 25: The black-box reduction from swap regret to external regret. Each M_j recommends q_j^t to the master algorithm and receives the true cost vector scaled by $p^t(j)$.

Algorithm 18.1: Master algorithm M

Input: No-external-regret algorithms $M_1, \dots, M_n$ over the actions $A = \{1, \dots, n\}$.

for $t \leftarrow 1$ **to** T **do**
- Receive distributions $q_1^t, \dots, q_n^t$ over A from $M_1, \dots, M_n$
- Compute and output a consensus distribution p^t (chosen below)
- Receive the cost vector c^t from the adversary
- Give each algorithm M_j the scaled cost vector $p^t(j) \cdot c^t$

Proof of Theorem 18.3. The time-averaged expected cost of M, and its cost under a switching function $\delta : A \to A$, are

$$\frac{1}{T}\sum_{t=1}^{T}\sum_{i=1}^{n} p^t(i)\, c^t(i) \qquad \text{and} \qquad \frac{1}{T}\sum_{t=1}^{T}\sum_{i=1}^{n} p^t(i)\, c^t(\delta(i)). \tag{18.2}$$

The goal is to show that the first is at most the second plus a term that tends to 0 as $T \to \infty$, for every δ.

From M_j's perspective, actions are chosen according to $q_j^1, \dots, q_j^T$ and the cost vectors are $p^1(j)c^1, \dots, p^T(j)c^T$. Since M_j has no external regret, its perceived cost is at most that of every fixed action $k \in A$, up to a regret term $R_j \to 0$:

$$\frac{1}{T}\sum_{t=1}^{T}\sum_{i=1}^{n} q_j^t(i)\big(p^t(j)\, c^t(i)\big) \le \frac{1}{T}\sum_{t=1}^{T} p^t(j)\, c^t(k) + R_j. \tag{18.3}$$

Fix δ, take $k = \delta(j)$ in (18.3), and sum over $j = 1, \dots, n$:

$$\frac{1}{T}\sum_{t=1}^{T}\sum_{i=1}^{n}\sum_{j=1}^{n} q_j^t(i)\, p^t(j)\, c^t(i) \le \frac{1}{T}\sum_{t=1}^{T}\sum_{j=1}^{n} p^t(j)\, c^t(\delta(j)) + \sum_{j=1}^{n} R_j. \tag{18.4}$$

The right-hand side is exactly the cost under δ in (18.2), up to $\sum_j R_j$, which tends to 0 because n is fixed as $T \to \infty$; this is why c^t was split as it was. It remains to choose each p^t so that the left-hand side of (18.4) equals the cost of M. Comparing coefficients of $c^t(i)$, it suffices that

$$p^t(i) = \sum_{j=1}^{n} q_j^t(i)\, p^t(j) \qquad \text{for every } i \in A. \tag{18.5}$$

These are the equations of a stationary distribution: form the Markov chain on states A in which the transition probability from j to i is $q_j^t(i)$, so that q_j^t gives the transitions out of state j. A distribution p^t satisfies (18.5) if and only if it is stationary for this chain. At least one stationary distribution exists, and one can be computed in polynomial time by an eigenvector computation. □

The consensus distribution also arises from a natural process. Ask some algorithm M_{j_1} for a recommendation $j_2 \sim q_{j_1}^t$, then ask M_{j_2} for a recommendation $j_3 \sim q_{j_2}^t$, and so on. This random

walk converges to a stationary distribution p^t when the chain is ergodic.

18.3 The minimax theorem for two-player zero-sum games

While mixed Nash equilibria always exist, computing one is intractable in general (Lecture 20). Two-player zero-sum games are a happier special case. In a TWO-PLAYER ZERO-SUM GAME, each player's payoff is the negative of the other's, so the game is a single matrix A: rows and columns are the two players' strategies, and a_{ij} is the row player's payoff and the column player's loss in outcome (i, j). The row player prefers bigger numbers and the column player smaller ones. For Rock-Paper-Scissors, with rows and columns ordered Rock, Paper, Scissors,

$$A = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}.$$

Pure Nash equilibria generally do not exist, so the focus is on mixed ones. Let $\mathbf{x}$ and $\mathbf{y}$ be mixed strategies over rows and columns. With independent randomization, the row player's expected payoff is

$$\sum_{i,j} \mathbb{P}_{\mathbf{x}}[i]\, \mathbb{P}_{\mathbf{y}}[j]\, a_{ij} = \mathbf{x}^T A \mathbf{y},$$

and a MIXED NASH EQUILIBRIUM is a pair $(\hat{\mathbf{x}}, \hat{\mathbf{y}})$ with

$$\hat{\mathbf{x}}^T A \hat{\mathbf{y}} \geq \mathbf{x}^T A \hat{\mathbf{y}} \text{ for all row strategies } \mathbf{x} \qquad \text{and} \qquad \hat{\mathbf{x}}^T A \hat{\mathbf{y}} \leq \hat{\mathbf{x}}^T A \mathbf{y} \text{ for all column strategies } \mathbf{y}.$$

Intuitively, committing to a mixed strategy first can only be a disadvantage, since the second player adapts to it. The minimax theorem says that it does not matter.

Theorem 18.4 [MINIMAX THEOREM]. *For every two-player zero-sum game A,*

$$\max_{\mathbf{x}} \left(\min_{\mathbf{y}} \mathbf{x}^T A \mathbf{y} \right) = \min_{\mathbf{y}} \left(\max_{\mathbf{x}} \mathbf{x}^T A \mathbf{y} \right). \tag{18.6}$$

On the left the row player moves first and the column player responds optimally; on the right the roles are reversed. Beyond fixed-point theorems or linear programming duality, the existence of a no-external-regret algorithm suffices to prove it.

Proof. The left side of (18.6) is at most the right side: if $\hat{\mathbf{x}}$ is optimal for the row player moving first, it can still play $\hat{\mathbf{x}}$ when moving second.
For the reverse inequality, let both players use no-regret algorithms for T steps, long enough that each has expected regret at most ϵ with respect to every fixed strategy. Costs and regret adapt directly to payoffs, and multiplicative weights still has regret $O(\sqrt{(\ln n)/T})$, so $T = \Theta((\ln n)/\epsilon^2)$

suffices. Let $\mathbf{p}^1, \ldots, \mathbf{p}^T$ and $\mathbf{q}^1, \ldots, \mathbf{q}^T$ be the row and column players' mixed strategies; the inputs to their algorithms at time t are the payoff vectors $A\mathbf{q}^t$ and $(\mathbf{p}^t)^T A$. Set

$$\hat{\mathbf{x}} = \frac{1}{T}\sum_{t=1}^{T} \mathbf{p}^t, \qquad \hat{\mathbf{y}} = \frac{1}{T}\sum_{t=1}^{T} \mathbf{q}^t, \qquad v = \frac{1}{T}\sum_{t=1}^{T} (\mathbf{p}^t)^T A \mathbf{q}^t.$$

Since the row player's regret is at most ϵ with respect to every row i, with pure strategy e_i,

$$e_i^T A\hat{\mathbf{y}} = \frac{1}{T}\sum_{t=1}^{T} e_i^T A\mathbf{q}^t \le \frac{1}{T}\sum_{t=1}^{T} (\mathbf{p}^t)^T A\mathbf{q}^t + \epsilon = v + \epsilon. \tag{18.7}$$

Every row strategy $\mathbf{x}$ is a distribution over the e_i, so by linearity $\mathbf{x}^T A\hat{\mathbf{y}} \le v + \epsilon$ for every $\mathbf{x}$. Symmetrically, the column player's regret bound gives $\hat{\mathbf{x}}^T A\mathbf{y} \ge v - \epsilon$ for every $\mathbf{y}$. Hence

$$\max_{\mathbf{x}} \min_{\mathbf{y}} \mathbf{x}^T A\mathbf{y} \ge \min_{\mathbf{y}} \hat{\mathbf{x}}^T A\mathbf{y} \ge v - \epsilon \ge \max_{\mathbf{x}} \mathbf{x}^T A\hat{\mathbf{y}} - 2\epsilon \ge \min_{\mathbf{y}} \max_{\mathbf{x}} \mathbf{x}^T A\mathbf{y} - 2\epsilon.$$

Letting $\epsilon \to 0$ (and $T \to \infty$) completes the proof. □

Corollary 18.5 [Consequences of the minimax theorem]. *In every two-player zero-sum game:*

(a) a mixed Nash equilibrium exists;

(b) the Nash equilibria are exactly the minimax pairs*: pairs $(\hat{\mathbf{x}}, \hat{\mathbf{y}})$ in which $\hat{\mathbf{x}}$ optimizes the left side and $\hat{\mathbf{y}}$ the right side of (18.6);*

(c) (mix and match) if $(\mathbf{x}^1, \mathbf{y}^1)$ and $(\mathbf{x}^2, \mathbf{y}^2)$ are Nash equilibria, then so are $(\mathbf{x}^1, \mathbf{y}^2)$ and $(\mathbf{x}^2, \mathbf{y}^1)$.

Proof. Let v^* be the common value in (18.6). If $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ optimize the two sides, then $v^* = \min_{\mathbf{y}} \hat{\mathbf{x}}^T A\mathbf{y} \le \hat{\mathbf{x}}^T A\hat{\mathbf{y}} \le \max_{\mathbf{x}} \mathbf{x}^T A\hat{\mathbf{y}} = v^*$, so equality holds throughout and neither player can improve: $(\hat{\mathbf{x}}, \hat{\mathbf{y}})$ is an equilibrium. Optimizers exist by compactness, which gives (a). Conversely, at an equilibrium $\min_{\mathbf{y}} \hat{\mathbf{x}}^T A\mathbf{y} = \hat{\mathbf{x}}^T A\hat{\mathbf{y}} = \max_{\mathbf{x}} \mathbf{x}^T A\hat{\mathbf{y}}$; the first quantity is at most the left side of (18.6) and the last is at least the right side, so by the theorem all are equal to v^* and both strategies are optimal. This proves (b), and (c) follows because any optimizer of the left side paired with any optimizer of the right side is a minimax pair. □

In the proof of the theorem, $(\hat{\mathbf{x}}, \hat{\mathbf{y}})$ is already a 2ϵ-approximate equilibrium: $\hat{\mathbf{x}}^T A\hat{\mathbf{y}}$ lies in $[v - \epsilon, v + \epsilon]$, so neither player gains more than 2ϵ by deviating. Existence of Nash equilibria here thus needs no fixed-point theorem.

Source. Tim Roughgarden, Algorithmic Game Theory, Lecture 18: From External Regret to Swap Regret and the Minimax Theorem.

Lecture 19: Pure Nash Equilibria and PLS-Completeness

19.1 The big picture

Several equilibrium concepts have polynomial-time algorithms and quickly converging learning dynamics (Figure 26):

- in general games, no-(external-)regret dynamics converge quickly to an approximate coarse correlated equilibrium (Lecture 17);
- in general games, no-swap-regret dynamics converge quickly to an approximate correlated equilibrium (Lecture 18);
- in two-player zero-sum games, no-regret dynamics converge quickly to an approximate mixed Nash equilibrium (Lecture 18);
- in symmetric atomic routing games, where all players share the same source and sink, ϵ-best-response dynamics converge quickly to an approximate pure Nash equilibrium (Lecture 16).

Moreover, linear programming computes an exact CCE or CE of a general game and an exact MNE of a two-player zero-sum game, and minimum-cost flow computes an exact PNE of a symmetric atomic selfish routing game.

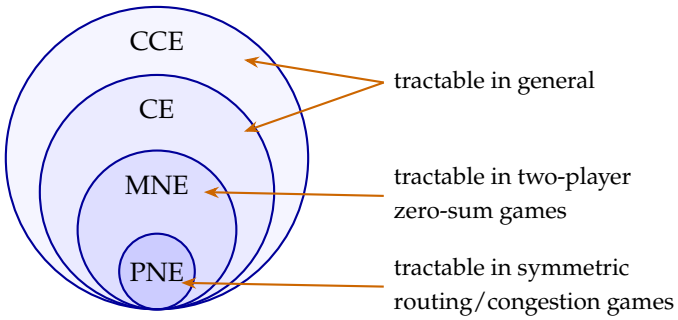


Figure 26: The hierarchy of equilibrium concepts and where each is known to be tractable.

No such results are known for MNE of non-zero-sum two-player games or for PNE of non-symmetric atomic routing games. This lecture and the next develop analogs of NP-completeness for equilibrium computation, starting with the technically simpler case of PNE in routing and congestion games. Computing a PNE of a congestion game is equivalent to computing a local minimum of the Rosenthal potential function (Lecture 13), so the complexity theory of local search applies.

19.2 Local search and maximum cut

In the MAXIMUM CUT problem, the input is an undirected graph $G = (V, E)$ with a weight $w_e \geq 0$ on each edge. Feasible solutions are cuts $(S, \bar{S})$, partitions of V into two non-empty sets, and the objective is to maximize the total weight of the cut edges, those with one endpoint on each side. Unlike minimum cut, maximum cut is NP-hard.

Algorithm 19.1: Local search for maximum cut

Input: Graph $G = (V, E)$ with edge weights $w_e \geq 0$.
Output: A locally optimal cut.

```
Start with an arbitrary cut (S, S̄)
while some local move is improving do
    Make an improving local move
```

A LOCAL MOVE moves one vertex v to the other side, keeping both sides non-empty. Moving v from S to $\bar{S}$ increases the objective by

$$\underbrace{\sum_{u \in S} w_{uv}}_{\text{newly cut}} - \underbrace{\sum_{u \in \bar{S}} w_{uv}}_{\text{newly uncut}},$$

and the move is IMPROVING if this is positive. Local search terminates at a LOCAL OPTIMUM, a cut with no improving move, which need not be a global optimum (Figure 27).

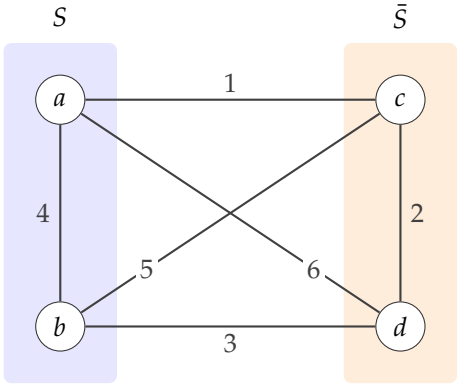


Figure 27: A maximum cut instance. The cut $S = \{a, b\}$, $\bar{S} = \{c, d\}$ has weight $1 + 3 + 5 + 6 = 15$, and moving any single vertex decreases it, so it is a local optimum. The cut $\{a, c\}$, $\{b, d\}$ has weight $4 + 2 + 5 + 6 = 17$.

Local optima are generally easier to find than global ones. With unit weights, computing a global maximum is still NP-hard, but the objective takes values in $\{0, 1, \ldots, |E|\}$, so local search stops within $|E|$ iterations. With general nonnegative weights, no polynomial-time algorithm (local search or otherwise) is known for computing a local optimum. Proving that none exists would separate P from NP, and NP-hardness turns out to be too strong a goal as well (Lecture 20). Instead, the theory below is an analog of NP-completeness tailored to local search.

19.3 PLS: abstract local search problems

The goal is to show that computing a local optimum of maximum cut is as hard as any other local search problem. Just as an efficient verifier is the minimal ingredient for brute-force search over NP witnesses, an ABSTRACT LOCAL SEARCH PROBLEM is specified by three polynomial-time algorithms:

1. one that takes an instance and outputs an arbitrary feasible solution (for maximum cut, an arbitrary cut);
2. one that takes an instance and a feasible solution and returns its objective function value (the total weight of the cut edges);
3. one that takes an instance and a feasible solution and either reports "locally optimal" or produces a better solution (it checks all $|V|$ local moves and executes an improving one if there is one).

An invalid output of the third algorithm is read as "locally optimal" unless it is a feasible solution, produced within a specified polynomial time and confirmed better by the second algorithm.

Every such problem admits a local search algorithm: obtain a start from the first algorithm and apply the third until a local optimum is reached. The objective strictly improves at each step, and feasible solutions have polynomial description length, so there are at most exponentially many and the procedure terminates. It need not run in polynomial time. The goal is to compute a local optimum, by local search or by any other correct algorithm. The complexity class PLS is the set of all abstract local search problems; most local search problems one meets can be cast as PLS problems.

Definition 19.1 [PLS REDUCTION AND PLS-COMPLETENESS**].** A REDUCTION from $L_1 \in$ PLS to $L_2 \in$ PLS is a pair of polynomial-time algorithms:

1. A maps every instance x of L_1 to an instance $A(x)$ of L_2;
2. B maps every local optimum of $A(x)$ to a local optimum of x.

A problem $L \in$ PLS is PLS-COMPLETE if every problem of PLS reduces to it.

If L_2 is solvable in polynomial time, so is L_1, by composing with A and B. Hence a PLS-complete problem has no polynomial-time algorithm unless PLS $\subseteq$ P, which most researchers believe false, though less confidently than P $\neq$ NP. A PLS-complete problem is a single local search problem that simultaneously encodes every local search problem.

Theorem 19.2 [MAXIMUM CUT IS PLS-COMPLETE**].** *Computing a local maximum of a maximum cut instance with general nonnegative weights is a PLS-complete problem.*

The analog of Cook's theorem is that a problem about Boolean circuits, called CIRCUITFLIP, is PLS-complete; reductions then give further complete problems, including maximum cut. Even if

PLS $\subseteq$ P, local search itself needs exponential worst-case time on all known PLS-complete problems: there is an unconditional exponential lower bound for local search on CircuitFlip, and all reductions used to establish PLS-completeness preserve it. In particular, local search can require exponentially many iterations on maximum cut instances.

19.4 Congestion games

A CONGESTION GAME generalizes atomic selfish routing by making strategies abstract subsets of a ground set rather than paths. It has a set E of RESOURCES, an explicitly described strategy set $\mathcal{S}_i \subseteq 2^E$ for each player $i = 1, \ldots, k$, and a cost $c_e(i)$ for each resource e and load $i \in \{1, \ldots, k\}$. The results proved for atomic selfish routing games (the POA bound of $\frac{5}{2}$ for affine costs, existence of PNE for arbitrary costs, and fast convergence of ϵ-best-response dynamics in symmetric games with α-bounded jump costs) hold for congestion games with the same proofs.

Computing a PNE of a congestion game is a PLS problem, by the correspondence between best-response dynamics and local search on the Rosenthal potential. The first algorithm outputs an arbitrary profile, for example every player's first strategy. The second returns the potential

$$\Phi(\mathbf{s}) = \sum_{e \in E} \sum_{i=1}^{n_e(\mathbf{s})} c_e(i), \tag{19.1}$$

where $n_e(\mathbf{s})$ is the number of players whose strategy in $\mathbf{s}$ includes e. The third checks whether $\mathbf{s}$ is a PNE; if not, it executes one iteration of best-response dynamics, which lowers the potential. All of this runs in time polynomial in the description of the game.

Theorem 19.3 [FABRIKANT–PAPADIMITRIOU–TALWAR]. *Computing a PNE of a congestion game is PLS-complete.*

Proof. Reduce from maximum cut. Given $G = (V, E)$ with weights $\{w_e\}_{e \in E}$, algorithm A constructs the following congestion game.

1. The players are the vertices V.
2. Each edge $e \in E$ gives two resources, r_e and $\bar{r}_e$.
3. Player v has two strategies, $\{r_e\}_{e \in \delta(v)}$ and $\{\bar{r}_e\}_{e \in \delta(v)}$, where $\delta(v)$ is the set of edges incident to v.
4. A resource r_e or $\bar{r}_e$ costs 0 if one player uses it and w_e if two players use it.

A player's choice is a side of the cut. For $e = (u, v)$, only u and v have strategies containing r_e or $\bar{r}_e$, so the combined load on the twin resources is always exactly two: both on one resource (same side) or one on each (different sides). This is why costs are needed only for loads one and two.

Strategy profiles correspond bijectively to cuts of G (allowing a side to be empty): $(S, \bar{S})$ corresponds to the profile in which $v \in S$ uses the r_e's and $v \in \bar{S}$ uses the $\bar{r}_e$'s. An edge cut by $(S, \bar{S})$ puts load 1 on each twin and contributes 0 to Φ; an uncut edge puts load 2 on one twin and 0 on the other and contributes w_e. Hence

$$\Phi(\mathbf{s}) = \sum_{e \in E} w_e - w(S, \bar{S}),$$

the total weight of uncut edges. Likewise, player v's cost is the weight of the uncut edges at v, so switching sides changes its cost by exactly minus the change in cut weight from moving v. Thus a profile is a PNE if and only if no single vertex move improves the cut, that is, PNE are exactly the local maxima of G (local minima of Φ). Mapping a PNE back to a local optimum of G is trivial, which gives B. □

The reduction is a bijection between local search for maximum cut and best-response dynamics in congestion games, so the latter can also need exponentially many iterations: PLS-completeness also yields concrete lower bounds for specific dynamics, which are hard to prove from scratch.

Theorem 19.4 [Symmetric congestion games]. *Computing a PNE of a symmetric congestion game is PLS-complete.*

This does not contradict Lecture 16, whose convergence result for ϵ-best-response dynamics extends to symmetric congestion games: there, approximate PNE are tractable while exact PNE are intractable, and exact best-response dynamics can need exponentially many iterations. (The reduction below violates the α-bounded jump assumption, which can be fixed.) Nor can the minimum-cost flow algorithm for symmetric atomic selfish routing games extend to symmetric congestion games, unless PLS $\subseteq$ P.

Proof of Theorem 19.4. Reduce from the asymmetric case (Theorem 19.3). Given a congestion game with k players and strategy sets $\mathcal{S}_1, \ldots, \mathcal{S}_k$, build a symmetrized version with the same players. Add k new resources $r_1, \ldots, r_k$, each costing 0 when used by one player and a huge amount M when used by two or more, with M larger than the total cost any player can incur from the original resources. Supplement each strategy of $\mathcal{S}_i$ with r_i, and let every player choose any of these augmented strategies.

At a PNE, each player adopts the identity of exactly one original player: if two players shared some r_i, then some r_j would be unused (there are k players and k new resources), and either of them would gain by switching to a strategy containing r_j. So the players use $r_1, \ldots, r_k$ once each, and a PNE of the symmetric game maps directly to a PNE of the asymmetric game: a profitable deviation for original player i would also be profitable for the player holding r_i. □

Source. Tim Roughgarden, Algorithmic Game Theory, Lecture 19: Pure Nash Equilibria and PLS-Completeness.

Lecture 20: Mixed Nash Equilibria and PPAD-Completeness

Coarse correlated and correlated equilibria are tractable in arbitrary games, while positive results for mixed Nash equilibria (MNE) so far cover only two-player zero-sum games (Lecture 18). This lecture develops the theory suggesting that significantly more general positive results are not possible.

20.1 Computing an MNE of a bimatrix game

The input of a BIMATRIX GAME is two $m \times n$ payoff matrices, A for the row player and B for the column player; zero-sum games have $B = -A$. The goal is to compute mixed strategies $\hat{\mathbf{x}}, \hat{\mathbf{y}}$ such that

$$\hat{\mathbf{x}}^T A \hat{\mathbf{y}} \geq \mathbf{x}^T A \hat{\mathbf{y}} \quad \text{for all row strategies } \mathbf{x}, \qquad \hat{\mathbf{x}}^T B \hat{\mathbf{y}} \geq \hat{\mathbf{x}}^T B \mathbf{y} \quad \text{for all column strategies } \mathbf{y}. \tag{20.1}$$

No polynomial-time algorithm for this problem is known. The aim is to show that it is complete for a suitable complexity class, and the difficulty is identifying which class.

20.2 NP search problems (FNP)

NP problems are decision problems, but the output of an equilibrium computation should be an equilibrium. The class FNP (functional NP) consists of SEARCH PROBLEMS: an algorithm receives an instance of an NP problem, such as a SAT formula or a graph, and must output a witness, such as a satisfying assignment or a Hamiltonian cycle, if one exists, and "no" otherwise. A reduction between search problems is a pair of polynomial-time algorithms: A maps instances x of one problem to instances $A(x)$ of the other, and B maps solutions of $A(x)$ to solutions of x (and "no" to "no"). As with NP, the functional version of SAT is FNP-complete: the proof of Cook's theorem gives a correspondence between the witnesses of a verifier and the satisfying assignments of the constructed formula.

PLS is a subset of FNP. Its witnesses are local optima, and its third algorithm is an efficient verifier that does more than an NP verifier requires: when a solution is not locally optimal, it produces a better one.

Computing an MNE of a bimatrix game is also in FNP: given $\hat{\mathbf{x}}, \hat{\mathbf{y}}$ one can check (20.1) efficiently, although it quantifies over infinitely many mixed strategies, because it suffices to check pure-strategy deviations. Equivalently, $\hat{\mathbf{x}}$ is a best response to $\hat{\mathbf{y}}$ if and only if it randomizes only over rows with maximum expected payoff against $\hat{\mathbf{y}}$. Moreover, some MNE has description length polynomial in the input: given the supports of an MNE, suitable mixing probabilities can be recovered by solving a linear system.

20.3 Guaranteed witnesses (TFNP)

Theorem 20.1 [Megiddo–Papadimitriou]. *If computing an MNE of a bimatrix game is FNP-complete, then* NP = coNP.

NP = coNP is believed as unlikely as P = NP: it would give short, efficiently verifiable proofs that a formula is unsatisfiable, for the coNP-complete UNSAT problem.

Proof. Suppose functional SAT reduces to computing an MNE of a bimatrix game. Then there are polynomial-time algorithms A, mapping every SAT formula φ to a bimatrix game $A(\varphi)$, and B, mapping every MNE $(\hat{\mathbf{x}}, \hat{\mathbf{y}})$ of $A(\varphi)$ to a satisfying assignment of φ if one exists and to "no" otherwise.
Let φ be an unsatisfiable formula. By Nash's theorem, $A(\varphi)$ has an MNE $(\hat{\mathbf{x}}, \hat{\mathbf{y}})$, and one of polynomial description length. This MNE is a short, efficiently verifiable proof that φ is unsatisfiable: compute $A(\varphi)$ and verify that $(\hat{\mathbf{x}}, \hat{\mathbf{y}})$ is an MNE of it, then verify that $B(\hat{\mathbf{x}}, \hat{\mathbf{y}})$ is "no". If both checks pass, the correctness of A and B implies that φ is unsatisfiable. Hence UNSAT is in NP and NP = coNP. □

The proof exploits a mismatch: an instance of FSAT may have no witness, while every instance of computing an MNE has one, by Nash's theorem. The class TFNP (total functional NP) consists of the FNP problems in which every instance has at least one witness. The same proof shows that if any TFNP problem is FNP-complete, then NP = coNP. Every PLS instance has a witness, since local search must stop at a local optimum, so PLS ⊆ TFNP (Figure 28). Hence no PLS problem, such as computing a PNE of a congestion game, is FNP-complete unless NP = coNP, which justifies PLS-completeness as the weaker notion used in Lecture 19.

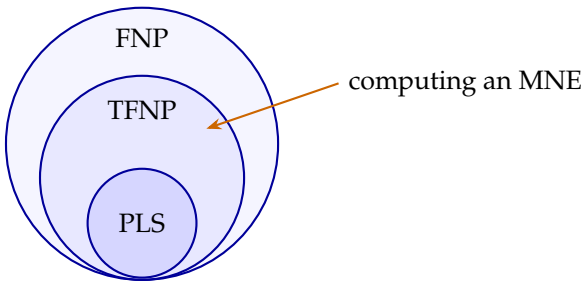


Figure 28: Computing an MNE of a bimatrix game lies in TFNP, which also contains PLS.

20.4 Syntactic and semantic complexity classes

Since computing an MNE is in TFNP, the sensible refined goal would be TFNP-completeness, but TFNP does not seem to have complete problems. Classes with complete problems, such as NP, P, and PSPACE, are syntactic: membership is characterized by acceptance by a concrete computational model, such as polynomial-time or polynomial-space Turing machines, which gives a generic reason

for membership. TFNP is a SEMANTIC class with no obvious generic reason. Computing an MNE is in TFNP because of Nash's theorem, essentially for topological reasons; factoring a positive integer is in TFNP for number-theoretic reasons. No one knows whether these are instances of one generic argument.

20.5 PPAD: a syntactic subclass of TFNP

Papadimitriou proposed identifying subclasses of TFNP that contain interesting problems not known to be in P (like computing an MNE) and that have complete problems, roughly equivalently a syntactic definition giving a generic reason for membership. PLS is one example: local search, defined by its three algorithms, is guaranteed to terminate at a local optimum. The right class for MNE of bimatrix games is PPAD ("polynomial parity argument, directed version"), described by analogy with PLS.

A local search problem is the problem of finding a sink of a directed acyclic graph whose nodes are feasible solutions and whose arcs are improving local moves (Figure 29, left). The graph is generally exponentially large in the instance size; for maximum cut its nodes are all cuts. The three PLS algorithms make it possible to follow a path: the first gives a starting node and the third gives the next arc from a non-sink node.

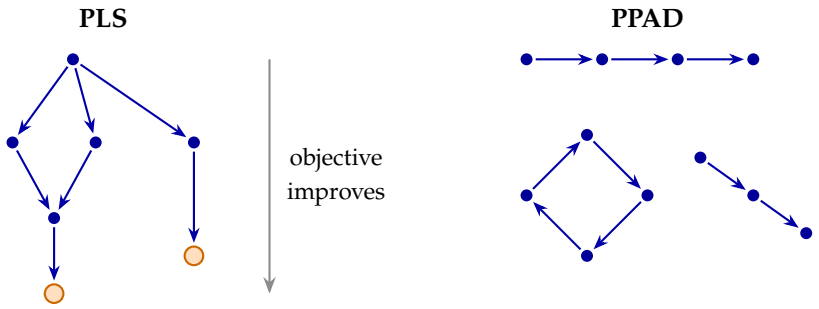


Figure 29: Left: a PLS problem as a directed acyclic graph of improving moves, ending at local optima (orange). Right: a PPAD problem as a graph with in- and out-degree at most 1, made of directed paths and cycles.

PPAD problems are those solvable by a particular naive directed path-following algorithm. Again nodes are "intermediate solutions" and arcs are "next moves", and the graph is exponentially large. Instead of being acyclic, a PPAD graph has in- and out-degree at most 1 and at least one source node; there is no objective function, and cycles are allowed (Figure 29, right). A PPAD problem is specified by three algorithms. The first gives an initial solution that is a source. The third either verifies that the current solution is a WITNESS, meaning a source or a sink other than the starting point, or returns the next solution. The second takes a solution and returns the previous one; it enforces in-degree at most 1, so naive path following always terminates: if the third algorithm says y follows x but the second does not agree that x precedes y, then x is treated as a sink, hence a witness. (In PLS the second algorithm plays the same honesty-enforcing role, via the objective

function.) Invoking the first algorithm once and then the third until the end of the path is reached is guaranteed to find a witness, but the path may be exponentially long; the question is whether a witness can be computed in polynomial time.

20.6 A canonical PPAD problem: Sperner's lemma

Consider a triangle (a 2-simplex) subdivided into small triangles. A LEGAL COLORING of the vertices colors the top corner red, the left corner green, and the right corner blue; a vertex on the boundary must have one of the two colors of the endpoints of its side, and interior vertices may have any of the three colors. A small triangle is TRICHROMATIC if all three colors appear at its corners. Figure 30 shows an example.

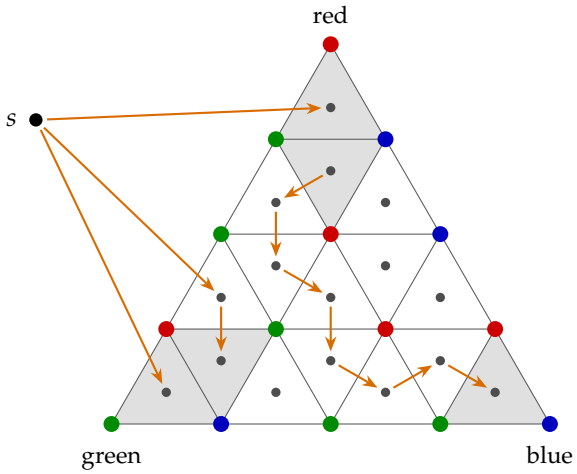


Figure 30: A legal coloring of a subdivided triangle with corners red (top), green (left), and blue (right). Arrows cross red-green sides. Starting from the source s, each of its three paths ends at a trichromatic triangle; the remaining path joins two trichromatic triangles. The five trichromatic triangles are shaded.

Theorem 20.2 [SPERNER'S LEMMA]. *For every legal coloring of a subdivided triangle, there is an odd number of trichromatic triangles, and hence at least one.*

Proof. Define a graph G with one node per small triangle plus a source node outside the simplex, and one edge for each red-green side of a small triangle: the edge joins the two triangles sharing that side, or the triangle and the source if the side lies on the boundary. Red-green sides on the boundary lie only on the left side of the big triangle, by legality.

- A trichromatic triangle has exactly one red-green side, so it has degree 1.
- A triangle with two red and one green corner, or two green and one red, has exactly two red-green sides, so it has degree 2; every other triangle has degree 0.
- The source has degree equal to the number of red-green segments on the left side. Walking down that side from red to green, the color changes between red and green an odd number

of times, so this degree is odd.

Every graph has an even number of odd-degree nodes. The source has odd degree, so the number of trichromatic triangles is odd. □

The proof is constructive: starting from the source, naive path following must end at a trichromatic triangle, as in Figure 30. So computing a trichromatic triangle of a legally colored subdivided triangle is a PPAD problem. Two details are omitted: the edges of G can be directed canonically, and a source of odd degree $2k+1$ is split into a source of out-degree 1 and k nodes of in- and out-degree 1. The same result and proof extend by induction to higher dimensions: every subdivided simplex in $\mathbb{R}^n$ whose vertices are legally colored with $n+1$ colors has an odd number of panchromatic subsimplices.

20.7 MNE and PPAD

There are two fundamental connections between computing an MNE and PPAD.

First connection: Sperner, Brouwer, Nash. Sperner's lemma is the combinatorial heart of Nash's proof that every finite game has an MNE, and its proof yields a path-following algorithm for computing an approximate MNE (in exponential time). The argument has two parts.

Theorem 20.3 [Brouwer's fixed-point theorem]. *Every continuous function $f : C \to C$ on a compact convex set $C \subseteq \mathbb{R}^n$ has a fixed point: a point $x \in C$ with $f(x) = x$.*

All the hypotheses (continuity, boundedness, closedness, convexity) are necessary.

Proof sketch for a triangle in $\mathbb{R}^2$. Let C be a triangle and $f : C \to C$ continuous. Write points in barycentric coordinates, $x = (x_{\text{top}}, x_{\text{left}}, x_{\text{right}})$ with nonnegative coordinates summing to 1, where a larger coordinate means closer to that corner. Subdivide C into small triangles and color each vertex x red if $f(x)_{\text{top}} < x_{\text{top}}$ ($f(x)$ is farther from the top corner than x), green if $f(x)_{\text{left}} < x_{\text{left}}$, and blue if $f(x)_{\text{right}} < x_{\text{right}}$, choosing either color when two conditions hold. If none holds then $f(x) = x$ and we are done. The coloring is legal: at a corner only its own coordinate can decrease, and on a side the coordinate of the opposite corner is 0 and cannot decrease. By Sperner's lemma there is a trichromatic triangle, whose corners are pulled in three different directions. Repeat with finer and finer subdivisions. By compactness, the centers of these trichromatic triangles have a subsequence converging to some $x^* \in C$, and the triangles' corners converge to x^* too. By continuity, every coordinate satisfies $f(x^*)_j \le x^*_j$, and since both coordinate vectors sum to 1, $f(x^*) = x^*$. □

For general compact convex sets: Sperner's lemma in higher dimensions gives the theorem for simplices, and every compact convex set of equal dimension is homeomorphic to a simplex, while homeomorphisms preserve fixed-point theorems.

Nash reduced the existence of MNE in finite games to Brouwer's theorem. Take a k-player game with strategy sets $S_1, \dots, S_k$ and payoff functions $\pi_1, \dots, \pi_k$, and let $C = \Delta_1 \times \cdots \times \Delta_k$, where Δ_i is the simplex of player i's mixed strategies. Define $f = (f_1, \dots, f_k) : C \to C$ by

$$f_i(\mathbf{x}) = \arg\max_{x_i' \in \Delta_i} g_i(x_i', \mathbf{x}), \qquad g_i(x_i', \mathbf{x}) = \underbrace{\mathbb{E}_{s_i \sim x_i', \mathbf{s}_{-i} \sim \mathbf{x}_{-i}}[\pi_i(\mathbf{s})]}_{\text{linear in } x_i'} - \underbrace{\|x_i' - x_i\|_2^2}_{\text{strictly convex}}.$$

Taking f_i to be a plain best response would not give a well-defined, let alone continuous, function, so the penalty term regularizes it, discouraging big changes to i's mixed strategy. Since g_i is strictly concave in x_i', f_i is well defined, and f is continuous. Every MNE $\mathbf{x}$ is a fixed point, since x_i maximizes the linear term and makes the penalty 0. Conversely, if $\mathbf{x}$ is not an MNE, some player i can increase its expected payoff by deviating from x_i to some x_i'. Then for $\epsilon > 0$,

$$g_i\big((1-\epsilon)x_i + \epsilon x_i', \mathbf{x}\big) - g_i(x_i, \mathbf{x}) = \epsilon\big(\pi_i(x_i', \mathbf{x}_{-i}) - \pi_i(\mathbf{x})\big) - \epsilon^2 \|x_i' - x_i\|_2^2 > 0$$

for sufficiently small ϵ, where π_i also denotes expected payoff; so $f_i(\mathbf{x}) \neq x_i$ and $\mathbf{x}$ is not a fixed point.

Second connection: the Lemke–Howson algorithm. It reduces computing an MNE of a bimatrix game to a path-following problem, much as the simplex method follows improving edges of the feasible region of a linear program. Unlike the simplex method it is not guided by an objective function; all known proofs of its convergence use parity arguments akin to Sperner's lemma, which shows that the problem lies in PPAD.

The two connections are incomparable. Lemke–Howson applies only to two players but shows that computing an *exact* MNE of a bimatrix game is in PPAD. The Sperner-based algorithm applies to any finite number of players but shows only that computing an *approximate* MNE is in PPAD; with three or more players, computing an exact MNE appears strictly harder than PPAD problems.

So PPAD is a subclass of TFNP that contains computing an MNE of a bimatrix game, and it is syntactic: the generic reason for membership is solvability by naive path following from a source in a directed graph with all in- and out-degrees at most 1. As such it admits complete problems, and it is the right class for evidence of intractability.

Theorem 20.4 [Chen–Deng–Teng; Daskalakis–Goldberg–Papadimitriou]. *Computing an MNE of a bimatrix game is a PPAD-complete problem.*

The proof is far too technical to describe here.

20.8 Discussion

One interpretation of Theorem 20.4, not without controversy, is that the intractability of Nash equilibria makes them unsuitable for general-purpose behavioral prediction: if no polynomial-time

algorithm can compute an MNE, strategic players should not be expected to find one quickly either. More generally, polynomial-time tractability of an equilibrium in a class of games can be used as a necessary condition for its predictive plausibility. This suggests focusing on tractable classes of games and equilibrium concepts, for example on properties shared by all correlated or coarse correlated equilibria.

After translating and scaling payoffs to lie in $[0, 1]$, an ϵ-APPROXIMATE MNE of a bimatrix game is a pair of mixed strategies in which neither player can increase its payoff by more than ϵ by a unilateral deviation. One can be computed in polynomial time for $\epsilon \approx \frac{1}{3}$, and in quasi-polynomial time for every constant $\epsilon > 0$. Whether polynomial time is possible for arbitrarily small constants ϵ is open, as is how the assumption PPAD $\not\subseteq$ P relates to other assumptions stronger than P $\neq$ NP, such as the existence of one-way functions.

Source. Tim Roughgarden, Algorithmic Game Theory, Lecture 20: Mixed Nash Equilibria and PPAD-Completeness.